

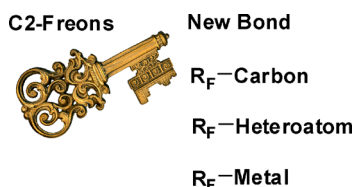
Polyfluorinated Ethanes as Versatile Fluorinated C2-Building Blocks for Organic Synthesis

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CONTENTS

1. Introduction	A	3.2.10. Generation of Perfluoroethyl Anions by Electrolysis	AP
2. Synthesis of Polyfluorinated Ethanes	B	3.2.11. Synthesis of Fluorinated Alkenes by β -Elimination Reaction	AQ
2.1. Halogenation of Ethylenes and Ethanes	B	3.3. Reaction of PFE Involving Carbocations	AQ
2.2. Isomerization Freons with Lewis Acids and Halogen Exchange Reaction	E	3.4. Reactions of PFE Involving Carbenes	AR
2.3. Nucleophilic Substitution	F	3.5. Reactions of PFE with Nucleophiles	AR
2.4. Synthesis of PFEs via Formation of Carbanions	G	3.5.1. Reactions with C-Nucleophiles	AT
2.5. Reduction of Fluorinated Ethanes (PFE)	G	3.5.2. Reactions with N-Nucleophiles	AV
3. Chemical Properties of Polyfluorinated Ethanes	H	3.5.3. Reactions with P-, As-, and Sb-Reagents	AW
3.1. Reactions Involving Free Radicals	H	3.5.4. Reactions with O-Nucleophiles	AX
3.1.1. Reactions with Alkenes	H	3.5.5. Reactions with S-Nucleophiles and Other Chalcogen Nucleophiles	AZ
3.1.2. Reactions with Alkynes	Q	3.6. Olefination Using PFE	BC
3.1.3. Reactions with Vinyl Ethers, Silyl Ethers, and Enolates	R	3.6.1. Wittig, Horner, and Fujita–Hiyama Olefination	BC
3.1.4. Reactions with Enamines and Ynamines	T	3.6.2. Catalytic Olefination Reaction (Nenajdenko–Shastin Olefination)	BD
3.1.5. Reactions with Aromatics and Heteroaromatics	U	3.7. Halogenation of Carbanions Using PFE	BF
3.1.6. Reactions with Fullerenes	W	3.8. Complexation of ICF ₂ CF ₂ I	BG
3.1.7. Other Radical Reactions Leading to C–C Bond Formation	W	3.9. Polyfluoroethyl Derivatives of Polyvalent Iodo Compounds	BG
3.1.8. Other Radical Reactions Leading to Formation of C–Heteroatom Bond	W	3.10. Reactions of PFE with HalOSO ₂ F, (FSO ₂ O) ₂ , CF ₃ SO ₃ Cl, and SO ₃	BK
3.2. Reactions of PFE, Involving C-2 Anionic Species as Intermediates	Z	4. Application of PFE	BK
3.2.1. Polyfluorinated C-2 Lithium Compounds	Z	5. Conclusions	BN
3.2.2. Fluorinated Vinyl Lithiums	AD	Author Information	BN
3.2.3. Polyfluorinated Ethyl Sodium and Potassium Compounds	AG	Corresponding Author	BN
3.2.4. Polyfluorinated Ethylmagnesium Compounds	AH	Notes	BN
3.2.5. Organozinc Compounds	AI	Biographies	BN
3.2.6. Perfluorinated Vinyl Zinc Derivatives	AK	Acknowledgments	BO
3.2.7. Organocopper Derivatives	AL	References	BO
3.2.8. Other Organometallics	AM		
3.2.9. Generation of Perfluorinated Anions Using Tetrakis(dimethylamino)ethylene (TDAE)	AP		

1. INTRODUCTION

Fluorinated compounds attracted remarkable interest in recent decades.¹ Unique complex of physicochemical and biological properties provided by incorporation of fluorine or perfluoroalkyl groups (as a rule CF₃) into a molecule of organic compound results in wide application of fluorinated compounds for construction of new materials, agrochemicals, and

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drugs. For example, about 20–25% of currently used drugs contain in the structure at least one fluorine atom or trifluoromethyl group (Figure 1).

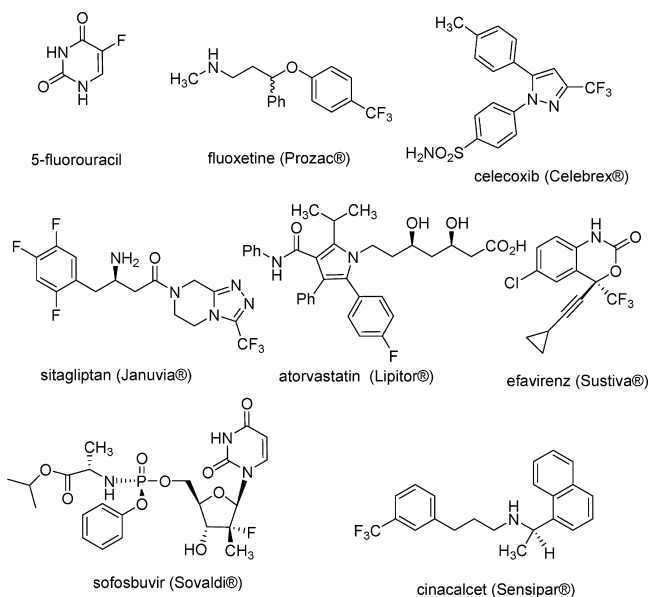


Figure 1. Some marketed drugs containing fluorine.

Biologically active compounds containing the pentafluoroethyl (C_2F_5) group are still more rare; for example, the angiotensin II receptor antagonist DuP 532 and antihypertensive K^+ channel opener KC-515 have been reported to date (Figure 2).² However, nowadays incorporation of trifluoroethyl

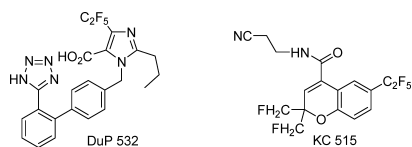


Figure 2. Bioactive molecules containing pentafluoroethyl fragment.

fragment CF_3CH_2 became a very popular tool in medicinal chemistry for new drug design (see section 4 of this Review). Therefore, the development of practical and reliable methods for incorporation of fluorinated C2 fragments into target structures is strongly desirable.³

The known methods for direct fluorination of organic compounds do not always allow the incorporation of the fluorine or a fluorinated group into the desired position of a molecule. Hence, a more flexible synthetic strategy based on the use of simple and available fluorine-containing compounds as building blocks plays an important role as a powerful supplement to direct fluorination methods.⁴

Utilization of new reagents and improved techniques for the selective introduction of fluorine or fluorinated fragments has become a major target in modern organofluorine chemistry. Polyfluoroethanes are readily available and cheap compounds having a big synthetic potential due to a broad variety of transformations in which these compounds can be involved. Nevertheless, there are only two small reviews regarding the chemistry of polyfluoroethanes.⁵ The aim of this Review is to attract the interest of a broad audience of chemists to these simple and very synthetically potent molecules. This Review is

divided into several parts, according to the reaction types. Methods for preparation of polyfluoroethanes are given separately. As a rule, only the chemistry of polyfluoroethanes having at least three fluorine atoms is discussed in this Review to reflect the special influence of fluorine atoms.

Fluorinated ethanes are a significant part of an important class of industrial chemicals, so-called “fluorocarbons”. The main type of compounds of this class is hydrofluorocarbons (HFC) and chlorofluorocarbons (CFC), which are widely used as refrigerants, foam blowing agents, and inert solvents. HFCs and CFCs are also known as “freons”. Bromofluorocarbons have the industrial name “halons”. There is a trivial nomenclature for freons, in which every compound has a specific three-digit numeric code (Table 1). The last digit shows the number of fluorine atoms. The first digit (from left to right) is the number of carbon atoms minus one; in the case of methane derivatives, this digit is not shown (the code shrinks to only two digits). The second last digit is a number of hydrogen atoms plus one (to avoid zero values in the case of fully halogenated molecules). This nomenclature can be used for halons by adding suffix B to the end of the code. The number of bromine atoms, replacing chlorine atoms, is placed to the right of suffix B. Suffixes a, b, or c show increasingly unsymmetrical isomers. More general nomenclature for halons uses five-digit codes, which includes numbers of C, F, Cl, Br, and I in exactly listed order.^{1,6}

Because of ozone layer depletion, some previously used fluorinated ethanes should have a restricted consumption and production after 2015 according to the Montreal protocol. The influence on ozone layer of fluorinated ethanes can be divided into three classes: dangerous, really leading to ozone layer depletion (class A), having small depletion effect (class B), and nondangerous (class C) (Table 1).

2. SYNTHESIS OF POLYFLUORINATED ETHANES

2.1. Halogenation of Ethylenes and Ethanes

An addition of elemental fluorine (or its surrogates), halogens (or interhalogens), and $HHal$ to ethene and its halogenated derivatives is the general pathway to the synthesis of fluorinated ethanes. A specific reactivity of some of these alkenes opens the possibility of radical, electrophilic, and even nucleophilic addition to the double bond. As a result, almost any type of C2 freons can be prepared using this methodology. This approach is one of the most useful and popular methods for the preparation of C2 freons because such alkenes are easily available industrial compounds.

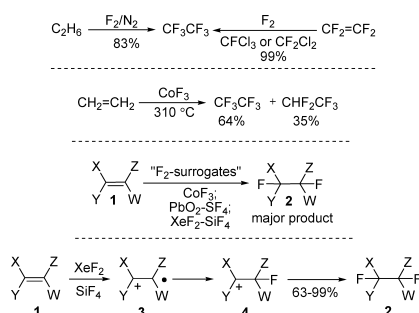
In contrast, ethane and halogenated ethanes are rarely used for the synthesis of polyfluorinated ethanes due to the low effectiveness of their fluorination, which usually proceeds nonselectively to give products of exhaustive fluorination and the C–C bond cleavage. A synthetic scope of a fluorination of ethane is restricted to the synthesis of hexafluoroethane. Thus, direct fluorination of ethane by molecular fluorine in jet flow reactor gives C_2F_6 in 83% yield (Scheme 1).⁷ An alternative approach is based on the reaction of addition of elemental fluorine to tetrafluoroethylene in inert liquid to afford C_2F_6 quantitatively.⁸ Fluorination by addition of F_2 was also reported for 1,2-dichloroethylene and trichloroethylene to give the corresponding PFEs in good yields (Table 2).⁹

However, it was demonstrated that some other reagents can also be applied for fluorination of alkenes as very useful surrogates of molecular fluorine capable of replacing this

Table 1. Polyfluoroethanes

code	chemical name	formula	class	code	chemical name	formula	class
R-111	pentachlorofluoroethane	C ₂ FCl ₅	A	R-132	dichlorodifluoroethane	C ₂ H ₂ F ₂ Cl ₂	B
R-112	1,1,2,2-tetrachloro-1,2-difluoroethane	C ₂ F ₂ Cl ₄	A	R-132a	1,1-dichloro-2,2-difluoroethane	C ₂ H ₂ F ₂ Cl ₂	B
R-112a	1,1,1,2-tetrachloro-2,2-difluoroethane	C ₂ F ₂ Cl ₄	A	R-132b	1,2-dichloro-1,1-difluoroethane	C ₂ H ₂ F ₂ Cl ₂	B
R-113	1,1,2-trichlorotrifluoroethane	C ₂ F ₃ Cl ₃	A	R-132c	1,1-dichloro-1,2-difluoroethane	C ₂ H ₂ F ₂ Cl ₂	B
R-113a	1,1,1-trichlorotrifluoroethane	C ₂ F ₃ Cl ₃	A	R-132bB2	1,2-dibromo-1,1-difluoroethane	C ₂ H ₂ Br ₂ F ₂	B
R-114	1,2-dichlorotetrafluoroethane	C ₂ F ₄ Cl ₂	A	R-133	1-chloro-1,2,2-trifluoroethane	C ₂ H ₂ F ₃ Cl	B
R-114a	1,1-dichlorotetrafluoroethane	C ₂ F ₄ Cl ₂	A	R-133a	1-chloro-2,2,2-trifluoroethane	C ₂ H ₂ F ₃ Cl	B
R-114B2	1,2-dibromotetrafluoroethane	C ₂ F ₄ Br ₂	A	R-133b	1-chloro-1,1,2-trifluoroethane	C ₂ H ₂ F ₃ Cl	B
R-115	chloropentafluoroethane	C ₂ F ₅ Cl	A	R-134	1,1,2,2-tetrafluoroethane	C ₂ H ₂ F ₄	C
R-116	hexafluoroethane	C ₂ F ₆	C	R-134a	1,1,1,2-tetrafluoroethane	C ₂ H ₂ F ₄	C
R-121	1,1,2,2-tetrachloro-1-fluoroethane	C ₂ HFCl ₄	B	R-141	1,2-dichloro-1-fluoroethane	C ₂ H ₂ FCF ₂	B
R-121a	1,1,1,2-tetrachloro-2-fluoroethane	C ₂ HFCl ₄	B	R-141B2	1,2-dibromo-1-fluoroethane	C ₂ H ₂ Br ₂ F	B
R-122	1,1,2-trichloro-2,2-difluoroethane	C ₂ HF ₂ Cl ₃	B	R-141a	1,1-dichloro-2-fluoroethane	C ₂ H ₂ FCF ₂	B
R-122a	1,1,2-trichloro-1,2-difluoroethane	C ₂ HF ₂ Cl ₃	B	R-141b	1,1-dichloro-1-fluoroethane	C ₂ H ₂ FCF ₂	B
R-122b	1,1,1-trichloro-2,2-difluoroethane	C ₂ HF ₂ Cl ₃	B	R-142	chlorodifluoroethane	C ₂ H ₂ F ₂ Cl	B
R-123	2,2-dichloro-1,1,1-trifluoroethane	C ₂ HF ₃ Cl ₂	B	R-142a	1-chloro-1,2-difluoroethane	C ₂ H ₂ F ₂ Cl	B
R-123a	1,2-dichloro-1,1,2-trifluoroethane	C ₂ HF ₃ Cl ₂	B	R-142b	1-chloro-1,1-difluoroethane	C ₂ H ₂ F ₂ Cl	B
R-123b	1,1-dichloro-1,2,2-trifluoroethane	C ₂ HF ₃ Cl ₂	B	R-143	1,1,2-trifluoroethane	C ₂ H ₂ F ₃	C
R-124	2-chloro-1,1,1,2-tetrafluoroethane	C ₂ HF ₄ Cl	B	R-143a	1,1,1-trifluoroethane	C ₂ H ₂ F ₃	C
R-124a	1-chloro-1,1,2,2-tetrafluoroethane	C ₂ HF ₄ Cl	B	R-151	chlorofluoroethane	C ₂ H ₄ ClF	C
R-125	pentafluoroethane	C ₂ HF ₅	C	R-151a	1-chloro-1-fluoroethane	C ₂ H ₄ ClF	C
R-131	1,1,2-trichloro-2-fluoroethane	C ₂ H ₂ FCF ₃	B	R-152	1,2-difluoroethane	C ₂ H ₄ F ₂	C
R-131a	1,1,2-trichloro-1-fluoroethane	C ₂ H ₂ FCF ₃	B	R-152a	1,1-difluoroethane	C ₂ H ₄ F ₂	C
R-131b	1,1,1-trichloro-2-fluoroethane	C ₂ H ₂ FCF ₃	B	R-161	fluoroethane	C ₂ H ₅ F	C

Scheme 1



dangerous reagent in laboratory use. Using CoF₃ as a fluorine source in the reaction with ethylene at elevated temperature led to a mixture of C₂F₆ and CF₂HCF₃ in almost quantitative total yield. Other halogenated ethenes also reacted easily with CoF₃ or MnF₃ under a range of conditions (25–125 °C) to form various fluorinated ethanes by this way.¹⁰ However, the reaction is not very selective, giving as a rule a mixture of fluorinated products (Table 2). A mixture of lead dioxide and sulfur tetrafluoride is another very nice reactive system for the selective addition of fluorine to the double bond of halogenated olefins, giving excellent results for the synthesis of various fluorinated ethanes (Table 2).¹¹ XeF₂–SiF₄ system (1.2 equiv of each) also provided high efficiency of fluorination of alkenes. In this case, fluorination includes an intermediate formation of radical cation **3** resulting from starting alkene **1** (via SET mechanism), then radical cation **3** is transformed to α -fluorinated carbocation **4**. Finally, the result of this reaction is the formation of vicinal difluoroadducts **2** (Scheme 1).¹²

The addition of halogens or interhalogens to tetrafluoroethylene is a very efficient synthetic path to fluorinated ethanes. Tetrafluoroethylene is an easily available industrial compound produced mainly for the preparation of Teflon. It should be

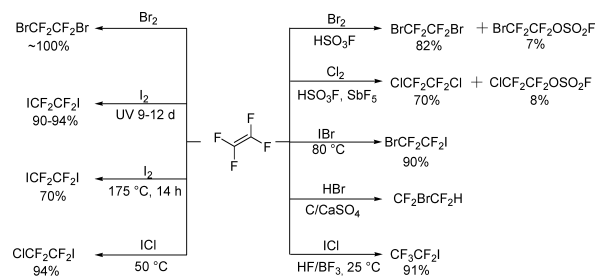
noted that because of low reactivity of the double bond toward electrophiles it demands additional activation in some cases. For example, the addition of bromine to tetrafluoroethylene was achieved even at low temperatures. Thus, tetrafluoroethylene and bromine were liquefied together and warmed slowly to room temperature to give BrCF₂CF₂Br quantitatively.¹³ Superacidic conditions were employed for the chlorine addition. A mixture of HSO₃F and SbF₅ catalyzed this reaction providing ClCF₂CF₂Cl in 70% yield. Similarly, BrCF₂CF₂Br was obtained in 82% yield using HSO₃F as a catalyst.¹⁴ XCF₂CF₂OSO₂F (X = Br, Cl) were obtained as the only byproducts in 7–8% yield. In the case of iodine, this method did not work. The addition of iodine to the double bond of tetrafluoroethylene took place at prolonged (9–12 days) UV-irradiation, giving the desired ICF₂CF₂I in high yield.¹⁵ Increasing the reaction temperature to 175 °C allowed one to dramatically reduce the duration of the addition, however decreasing the yield to 70%.¹⁶ The addition of iodine monochloride and iodine monobromide led to the corresponding tetrafluoroethanes BrCF₂CF₂I, ClCF₂CF₂I in high yields under neat conditions.¹⁷ In contrast, C₂F₅I was obtained in 91% yield by the reaction of ICl with tetrafluoroethylene employing an activation by the HF/BF₃ system. In this case, the intermediate iodonium-ion is attacked by excessive hydrogen fluoride instead of chloride-anion to form C₂F₅I.¹⁸ The addition of HBr was performed using carbon on CaSO₄ as a catalyst to give CF₂BrCF₂H (Scheme 2).¹⁹

CF₂=CFCl and CF₂=CFBr also provide broad possibilities for the synthesis of various fluoroethanes having heavy halogens in their structure. The addition of bromine to CF₂=CFCl proceeds at room temperature to give CF₂BrCFBrCl in almost quantitative yield.²⁰ The reaction with ICl at –5 °C in CH₂Cl₂ led regioselectively to ICF₂CFCl₂ with a small admixture of ClCF₂CFCF₂I. In contrast, the Fe-catalyzed (iron halides act as real catalysts of this reaction) addition of ICl at 50 °C led to

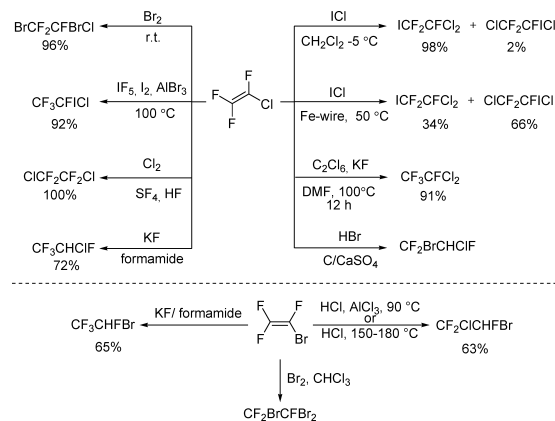
Table 2. Fluorination of Alkenes with Elemental Fluorine and Its Surrogates

alkene	ethane	yield, %	method
<i>cis</i> -CHCl=CHCl	CHFCICHFCI	70	F ₂
<i>trans</i> -CHCl=CHCl	CHFCICHFCI	73	F ₂
CCl ₂ =CHCl	CFCl ₂ CHFCI	42	F ₂
CH ₂ =CHCl	CH ₂ FCHClF + CHF ₂ CH ₂ F	28 + 13	CoF ₃
<i>cis</i> -CHCl=CHCl	CHFCICHFCI + CHF ₂ CHClF	53 + 12	CoF ₃
CH ₂ =CCl ₂	CF ₂ ClCHFCI	11	CoF ₃
CCl ₂ =CHCl	CF ₂ ClCHFCI + CFCl ₂ CHFCI	4 + 31	CoF ₃
CF ₂ =CH ₂	CF ₃ CH ₂ F + CF ₃ CHF ₂	54 + 15	CoF ₃
CF ₂ =CHBr	CF ₃ CHBrF	60	CoF ₃
CF ₂ =CFCl	CF ₃ CF ₂ Cl	60	PbO ₂ -SF ₄
CF ₂ =CFBr	CF ₃ CF ₂ Br	57	PbO ₂ -SF ₄
CF ₂ =CFI	CF ₃ CF ₂ I	65	PbO ₂ -SF ₄
CF ₂ =CCl ₂	CF ₃ CFCl ₂	88	PbO ₂ -SF ₄
CCl ₂ =CCl ₂	CFCl ₂ CFCl ₂	77	PbO ₂ -SF ₄
CCl ₂ =CClF	CFCl ₂ CF ₂ Cl	77	PbO ₂ -SF ₄
CF ₂ =CHF	CF ₃ CF ₂ H	40	PbO ₂ -SF ₄
CCl ₂ =CHCl	CFCl ₂ CHFCI	85	PbO ₂ -SF ₄
CHF=CFCl	CHF ₂ CF ₂ Cl	96	PbO ₂ -SF ₄
CF ₂ =CHI	CF ₃ CHFI	53	PbO ₂ -SF ₄
CF ₂ =CH ₂	CF ₃ CH ₂ F	28	PbO ₂ -SF ₄
CH ₂ =CCl ₂	CFCl ₂ CFH ₂	59	PbO ₂ -SF ₄
CH ₂ =CHBr	CH ₂ FCHBrF	20	PbO ₂ -SF ₄
<i>cis</i> -CHCl=CHCl	CHFCICHFCI	69 DL + 21 meso	PbO ₂ -SF ₄
<i>trans</i> -CHCl=CHCl	CHFCICHFCI	54 DL + 26 meso	PbO ₂ -SF ₄
CH ₂ =CH ₂	CH ₂ FCH ₂ F + CHF ₂ CH ₃ + CHF ₂ CH ₂ F	63 + 25 + 9	XeF ₂ -SiF ₄
CHF=CH ₂	CHF ₂ CH ₂ F	93	XeF ₂ -SiF ₄
CH ₂ =CHBr	CH ₂ FCHBrF + CHFBrCHF ₂	21 + 53	XeF ₂ -SiF ₄
CF ₂ =CH ₂	CF ₃ CH ₂ F	94	XeF ₂ -SiF ₄
CF ₂ =CHF	CF ₃ CHF ₂	99	XeF ₂ -SiF ₄
CF ₂ =CFCl	CF ₃ CF ₂ Cl	93	XeF ₂ -SiF ₄

34/66 mixture of the above-mentioned products.²¹ The opposite regioisomer distribution can be explained by catalytic action of formed iron halide behaving as a Lewis acid to promote isomerization of primary products (see also Schemes 8 and 9). CF₃CFCl₂ was prepared by chlorofluorination of CF₂=CFCl with KF/CCl₃CCl₃ mixture in DMF at 100 °C in 91% yield.²² Otherwise, chlorofluorination of CF₂=CFCl using SF₄-HF-Cl₂ system afforded quantitatively ClCF₂CF₂Cl, which arose from initially formed addition product

Scheme 2

ClCF₂CFCl₂ by Cl-F exchange reaction.²³ Fluoroiodination of CF₂=CFCl by iodine and IF₅ in the presence of AlBr₃ at 100 °C gave CF₃CFICl in high yield.²⁴ Nucleophilic addition of HF to chlorotrifluoroethylene using KF/formamide system has been used for the preparation of CF₃CHClF in 72% yield.²⁵ CF₂BrCHClF has been prepared by carbon-catalyzed addition of HBr to chlorotrifluoroethylene (Scheme 3).¹⁹

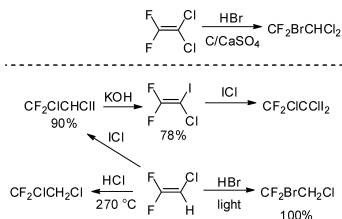
Scheme 3

Several syntheses of bromo-substituted polyfluoroethanes are based on reactions of CF₂=CFBr. For example, nucleophilic addition to CF₂=CFBr was carried out using KF/formamide system to give CF₃CFBrH.²⁶ Under electrophilic conditions, addition of HCl (activation with aluminum chloride or under elevated temperature) led to CF₂ClCFHBr.²⁷ Reaction with bromine in chloroform gave CF₂BrCFBr₂ (Scheme 3).²⁸ The regiochemistry of the addition of unsymmetrical electrophiles to CF₂=CFCl and CF₂=CFBr can be explained by an interplay of electronic and steric factors; the formation of CF₃-derivatives is a highly favored process (for example, see formation of CF₃CFICl, CF₃CFCl₂, CF₃CHClF, CF₃CFBrH).

Similarly, the radical addition of HBr, HCl, and ICl to other halogenated 1,1-difluoroethylenes opens a simple and effective route to various fluorinated ethanes. As a rule, almost any type of target freons can be prepared from the corresponding alkene. For instance, using this approach, CF₂ClCCl₂ and CF₂ClCHClCl were synthesized by the addition of ICl to CF₂=CClI and CF₂=CClIH. The addition of HBr or HCl to CF₂=CClIH and CF₂=CCl₂ afforded CF₂BrCHCl₂, CF₂ClCH₂Cl, CF₂BrCH₂Cl correspondingly (Scheme 4).²⁹

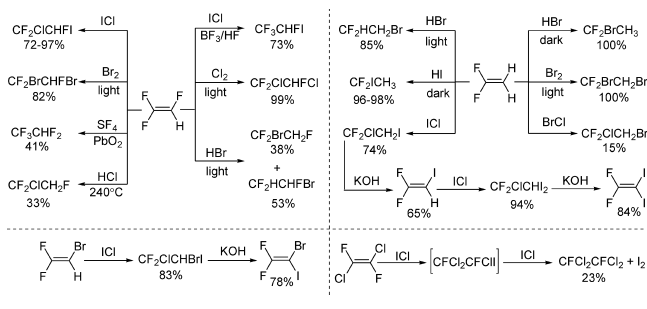
1,1-Difluoroethylene³⁰ and trifluoroethylene³¹ are also industrially available monomers, the reactions (radical and electrophilic) with halogens, interhalogens, and hydrogen halides of which have been studied very thoroughly. It should be noted that 1,1-difluoroethylene is a highly reactive alkene for

Scheme 4



electrophilic addition as well as for radical reactions, whereas fluorinated alkenes having three or four halogens are almost inactive toward electrophiles without additional activation. It has been determined that the rate of radical addition is also higher for the reaction with 1,1-difluoroethylene. The row of reactivity toward radical addition for fluorinated alkenes decreases in the order $\text{CF}_2=\text{CH}_2 > \text{CF}_2=\text{CF}_2 > \text{CF}_2=\text{CHCl} > \text{CF}_2=\text{CFCl}$.³² For example, $\text{CF}_2=\text{CFH}$ does not react with HBr in the dark and gives the corresponding adducts CF_2HCFHBr and $\text{CF}_2\text{BrCFH}_2$ only by light activation (radical conditions). Alternatively, 1,1-difluoroethylene gives selectively the adduct with HBr (opposite regioselectivity) under electrophilic conditions in 100% yield. In this case, the regiochemistry is controlled by intermediate carbocation CH_3CF_2^+ formed selectively due to mesomeric stabilization provided by fluorines (Scheme 5).

Scheme 5

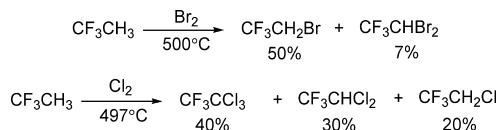


The reaction of various fluorinated alkenes with iodine monochloride was studied in detail. It was found that the reaction proceeds regioselectively to give fluorinated ethanes having one or two iodine atoms in the molecule in high yield. This approach was used for the synthesis of some iodinated alkenes (by the subsequent elimination with KOH) opening broad possibilities for the synthesis of various fluorinated ethanes and ethylenes. Unsaturated derivatives prepared by a base-induced elimination of HX from freons were used as valuable fluorinated building blocks for the preparation of fluorinated organometallics and fluorinated butadienes.³³ As a result, the addition to fluorinated alkenes is a very flexible and useful methodology, giving the possibility for the synthesis of almost any type of fluorinated ethanes from readily available and cheap starting materials (Scheme 5).

The radical chlorination and bromination of some partially fluorinated ethanes can be used for the preparation of halogenated derivatives having hydrogen atoms in their structure. For example, the reaction of 1,1,1-trifluoroethane with chlorine and bromine under elevated temperature gave the corresponding halogenated trifluoroethanes. However, while in the case of bromination the selectivity was rather high, it was difficult to control the chlorination of CF_3CH_3 . As a result, a

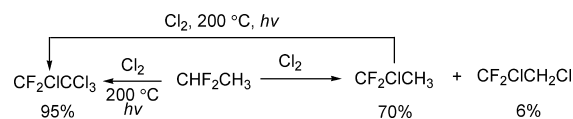
mixture of mono-, di-, and trichloride ($\text{CF}_3\text{CH}_2\text{Cl}$, CF_3CHCl_2 , CF_3CCl_3) was formed (Scheme 6).³⁴

Scheme 6



Similarly, radical chlorination of 1,1-difluoroethane gave a mixture of monochloride $\text{CF}_2\text{CH}_2\text{Cl}$ and dichloride CF_2CHCl_2 in 70% and 6% yield, respectively. It should be noted that $\text{CF}_2\text{HCH}_2\text{Cl}$ is not formed under these conditions. The observed selectivity of the reaction indicates higher stability of CH_3CF_2 radical as compared to isomeric CF_2HCH_2 radical.³⁵ The exhaustive halogenation of 1,1-difluoroethane takes place to give the corresponding tetrachloride $\text{CF}_2\text{ClCCl}_3$ in 74% yield using excess of chlorine at 200 °C (Scheme 7).³⁶

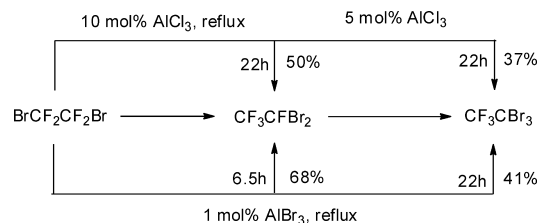
Scheme 7



2.2. Isomerization Freons with Lewis Acids and Halogen Exchange Reaction

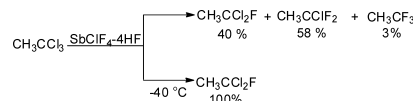
Lewis acid-induced isomerization of fluorinated ethanes is another general method for their preparation. Usually such rearrangements lead to the concentration of fluorine atoms on one carbon atom, which is more thermodynamically favorable.^{1c} Using this approach, both CF_3CBr_3 and CF_3CFBr_2 were obtained efficiently via the isomerization of $\text{BrCF}_2\text{CF}_2\text{Br}$ under the corresponding reaction conditions. AlCl_3 ³⁷ as well as AlBr_3 ³⁸ were used as catalysts for the isomerization. It should be noted that AlBr_3 was shown to be more effective, providing higher yields by a lesser amount of the catalyst (Scheme 8).

Scheme 8



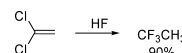
Similarly, CF_3CCl_3 , CF_3CFCl_2 , and $\text{CF}_2\text{ClCCl}_3$ were obtained from industrial freon $\text{CFCl}_2\text{CF}_2\text{Cl}$ (Scheme 9). CF_3CCl_3 was isolated in 45–50% yield when isomerization was catalyzed by AlCl_3 under reflux.³⁹ Addition of CS_2 to the reaction mixture led to the formation of $\text{CF}_2\text{ClCCl}_3$ instead of CF_3CCl_3 .⁴⁰ Use of $\gamma\text{-Al}_2\text{O}_3$ fluorinated by SF_4 or $\beta\text{-AlF}_3$ at elevated temperature gave a mixture of CF_3CCl_3 and CF_3CFCl_2 in maximal 62% and 31% yields, respectively.⁴¹ The isomerization of $\text{BrCF}_2\text{CFClBr}$ by AlCl_3 catalysis at room temperature gave $\text{CF}_3\text{CClBr}_2$ in high yield (Scheme 9).³⁷

Scheme 12



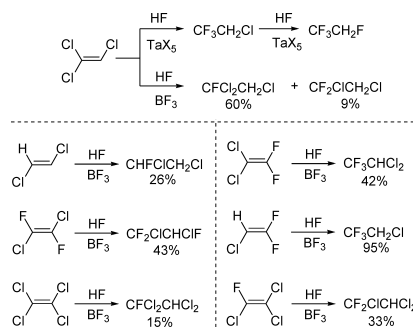
There are also very valuable methods based on the combination of addition to double bond and halogen exchange. Such methods are based on reactions with hydrogen fluoride and its combination with Lewis acids. For example, the reaction of 1,1-dichloroethylene with hydrogen fluoride gives directly 1,1,1-trifluoroethane by the addition of HF–halogen exchange sequence (Scheme 13).³⁴

Scheme 13



Depending on the reaction conditions, the hydrogen fluoride/boron trifluoride system converts various chlorinated ethenes to the corresponding fluorinated ethanes in good yields. Tantalum halides have also been used as Lewis acid for an activation of HF during the formation of $\text{CF}_3\text{CH}_2\text{F}$ and $\text{CF}_3\text{CH}_2\text{Cl}$ (Scheme 14).⁴⁶

Scheme 14

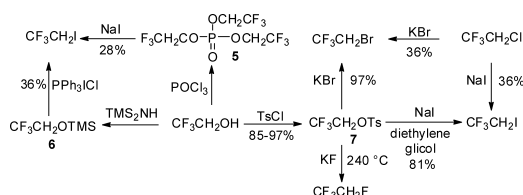


2.3. Nucleophilic Substitution

Nucleophilic substitution can also be used for the preparation of various fluorinated ethanes; for example, target molecules were prepared starting from trifluoroethanol derivatives or using the interconversion of freons. Thus, 1,1,1-trifluoroethyl tosylate **7**, fluorinated phosphonate **5**, and silyl ether **6** can be prepared from trifluoroethanol. The subsequent reaction with halides opens a simple laboratory approach to trifluoromethylated halomethanes.⁴⁷ $\text{CF}_3\text{CH}_2\text{I}$ has also been prepared from phosphate **5** by reaction with sodium iodide.⁴⁸ Another laboratory synthesis of $\text{CF}_3\text{CH}_2\text{I}$ is based on the transformation of trifluoroethanol into trimethylsilyl ether **6** followed by reaction with Ph_3PICl to form target compound in 36% yield (Scheme 15).⁴⁹ Alternatively, $\text{CF}_3\text{CH}_2\text{Br}$ and $\text{CF}_3\text{CH}_2\text{I}$ can be prepared by the reaction of $\text{CF}_3\text{CH}_2\text{Cl}$ with KBr or KI correspondingly, but unfortunately in lower yields (Scheme 15).⁵⁰

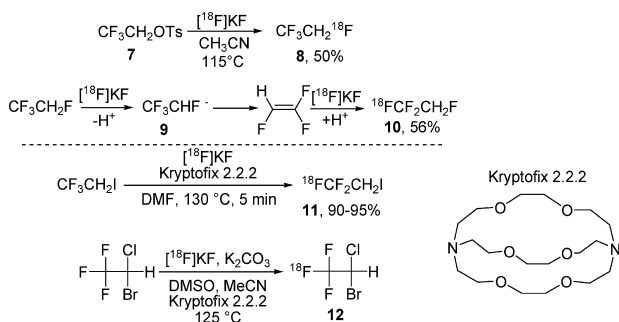
Isomeric ^{18}F labeled 1,1,1,2-tetrafluoroethanes **8** and **10** were prepared by the reaction of ^{18}F -fluoride with 2,2,2-trifluoroethyl tosylate (**7**) or 1,1,1,2-tetrafluoroethane. The first reaction proceeds as a normal $\text{S}_{\text{N}}2$ reaction (the presence of kryptand allows one to obtain the target compound in higher yield at lower temperature). The reaction with 1,1,1,2-tetrafluoroethane

Scheme 15



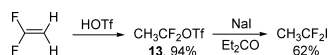
is more complex. The first step is F^- promoted deprotonation leading to trifluoroethylene through intermediate formation of anion **9**. ^{18}F labeled at the 1-position 1,1,1,2-tetrafluoroethane (**10**) was formed after subsequent nucleophilic addition of ^{18}F -fluoride and protonation.⁵¹ Labeling CF_3CH_2I with ^{18}F via ^{18}F – ^{19}F exchange proceeded with high efficiency (90–95%) using similar reaction conditions.⁵² ^{18}F – ^{19}F exchange in trifluoromethyl group was achieved in the case of reaction with Halothane ($CF_3CHBrCl$) to form **12** (Scheme 16).⁵³

Scheme 16



The reaction of 1,1-difluoroethylene with triflic acid (CF_3SO_3H) opens a simple route to a very interesting reagent, 1,1-difluoroethyl triflate (**13**). It was found that this triflate is a rather reactive alkylating agent; for example, 1,1-difluoro-1-iodoethane was prepared by the reaction with sodium iodide in 62% yield (Scheme 17).⁵⁴

Scheme 17

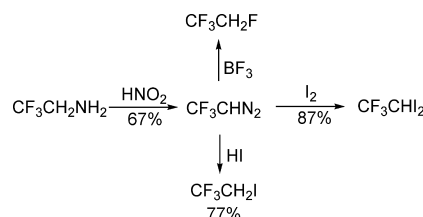


Some rare examples of the preparation of fluorinated ethanes from trifluoromethyldiazomethane were published. The target diazocompound can be prepared quite easily by the reaction of commercially available 2,2,2-trifluoroethylamine with nitrous acid. Rather stable trifluoromethyldiazomethane (due to electron-withdrawing properties of CF_3 -group) is formed in 67% yield. For example, it can be distilled at 13 °C without degradation.⁵⁵ The subsequent reaction of trifluoromethyldiazomethane with iodine gave 1,1,1-trifluoro-2,2-diiodoethane in 87% yield; alternatively, reactions with HI and HF (formed by BF_3 hydrolysis) allowed the preparation of 2,2,2-trifluoroethyl iodide⁵⁶ (77% yield) and 1,1,1,2-tetrafluoroethane (Scheme 18).⁵⁷

2.4. Synthesis of PFEs via Formation of Carbanions

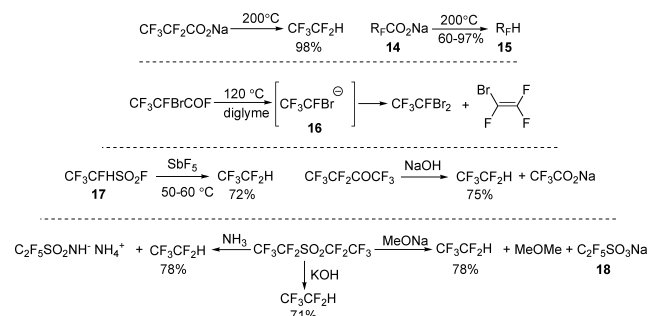
Pyrolysis of some salts of polyfluorinated acids proceeds similarly to that with nonfluorinated analogues, allowing the preparation of fluorinated alkanes via decarboxylation and intermediate formation of fluorinated carbanions followed by

Scheme 18



protonation. For example, pentafluoroethane and higher fluorinated homologues **15** were prepared by this method in 98% yield by heating sodium pentafluoropropionate or higher carboxylates **14** in ethylene glycol at 200 °C.⁵⁸ Pyrolytic decarboxylation of 2-bromotetrafluoropropionyl fluoride in dry diglyme at 120 °C in the presence of anhydrous sodium carbonate gave a mixture of 1,1-dibromotetrafluoroethane and trifluorobromoethylene via postulated formation of carbanion intermediate **16**.⁵⁹ Fluorinated sulfonic acid derivatives can be also used for the preparation of fluorinated ethanes. For example, pentafluoroethane was prepared from the corresponding sulfonyl fluoride **17**; however, in this case, the reaction demands the use of SbF_5 .⁶⁰ Pentafluoroethane was similarly prepared by the haloform cleavage of perfluorinated butanone-2 and bis(pentafluoroethyl)sulfone by NaOH, KOH, sodium methoxide, or ammonia.⁶¹ The formation of dimethyl ether and sulfonate **18** can be explained by reaction of intermediate methyl pentafluoroethylsulfonate with a second molecule of sodium methoxide. It should be noted that in the case of unsymmetrical polyfluorobutanone-2 the reaction proceeds selectively to give pentafluoroethane through the formation of more stable anion (Scheme 19).

Scheme 19

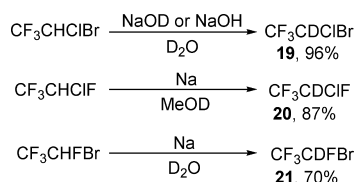


The proton–deuterium exchange in basic media was used for the synthesis of deuterium-labeled polyfluoroethanes. Because of the electronegative properties of halogen atoms, the acidity of protons in these compounds is quite high and the exchange can be performed under relatively mild conditions. Thus, exposure of $CF_3CHClBr$, CF_3CHClF , CF_3CHFBr in MeOD or D_2O in the presence of NaOD or NaOH at ambient temperature affords deuterated polyfluoroethanes **19**–**20** (analogues of Halothane and terflurane) in high to quantitative yields (Scheme 20).^{25b,62}

2.5. Reduction of Fluorinated Ethanes (PFE)

A partial reduction is a very useful method for the synthesis of various polyfluorinated ethanes (Table 3). Hydrogenation using various metal catalysts (both heterogeneous and homogeneous), low-valent phosphorus and sulfur salts, trialkyl silanes, ammonium formate, oxalic acid, and zinc were used for the

Scheme 20



reduction. Usually this transformation proceeds in such a way that only one halogen is removed to form C–H bond. C–F bond is never cleaved; C–Br bond can be reduced highly selectively in the presence of C–Cl bond. Some examples of reduction of PFE are divided by nature of reducing agent (Table 3).

In the case of the reduction using ammonium formate, the reaction proceeds via a radical mechanism, which starts from the oxidation of formate with persulfate to give carbon dioxide radical anion. This radical anion reacts with PFE **22** to form the radical anion **23**, which eliminates halide giving the perfluorinated radical **24**. Finally, the reaction of **24** with formate leads to the final product **15** and carbon dioxide radical anion, continuing the radical process (Scheme 21, Table 3).

Unusual transformation was found by Gu and co-workers. Heating the fluoroalkyl halides **22** under weakly basic conditions in DMF gave the corresponding hydrogenolysis products **15**. The mechanism of the reaction has not been clarified, but it was proposed that perfluoroalkyl radicals could be involved in the transformation.⁷⁹ Most probably DMF is the reductant in this case (Scheme 22).

3. CHEMICAL PROPERTIES OF POLYFLUORINATED ETHANES

3.1. Reactions Involving Free Radicals

A formation of a radical-type intermediate is very typical for the initial reaction of polyfluorinated ethanes. For example, all reactions of polyfluorinated ethanes with multiple C–C bonds occur as radical addition, and they are overviewed in this part of the Review. These reactions are arranged according to the type of the reaction initiation. Thus, application of a transition metal catalysis, sodium dithionite, UV-irradiation, heating, and radical initiators will be considered in detail. Also, reactions of some organometallic compounds, generated from polyfluorinated ethanes by means of radical reactions, will be discussed.

3.1.1. Reactions with Alkenes. **3.1.1.1. Catalysis by Transition Metal Derivatives.** Copper halides (CuCl, CuI, etc.) are employed very often for the initiation of radical additions of polyfluorinated ethanes. The first examples were given by Burton in 1970, who examined the reaction in detail. It was found that a series of fluorinated ethanes react smoothly with 1-octene under catalysis by copper chloride (CuCl)–ethanolamine redox system in refluxing *tert*-butanol. As a result, a variety of 1:1 adducts were prepared in good yields.^{37a} It was discovered that ethanolamine as a ligand gave higher yields as compared to other amines. However, 2-propylamine showed better results in the case of reaction with 1-hexene.⁸⁰ As a proof of this principle, it was shown that Cu(II) compounds can also be used as catalysts.^{37a} The first step of the reaction is the transformation of fluoroethane to radical **24** by the reaction with Cu(I). The subsequent addition of **24** to alkene **1** gives radical intermediate **26**. The reaction of **26** with CuCl^{•+} (radical-cation) gives the adduct **25** and regenerates copper

cation.^{37a} Later, Van Der Puy examined a set of various alkenes in the reaction using a CuCl–CuCl₂ × 2H₂O–ethanolamine catalytic system.⁸¹ It was demonstrated that both internal and terminal double bonds can be involved in the reaction. An attack of fluorinated radicals occurred only from the less substituted side of the double bond despite its electron properties. Addition to limonene afforded regioselectively the adduct with the exocyclic double bond, the formation of which is explained by its higher activity as compared to the endocyclic one. Alkenes bearing functional groups can also be used as starting materials for the addition of fluorinated ethanes. Thus, some derivatives of carboxylic acids,⁸² esters,^{82–84} and phosphonates⁸⁵ were prepared in high yields using this approach (Scheme 23, Table 4).

In the case of reaction of CF₃CCl₃ with conjugated 1,3-dienes **27**, regular 1,4-addition takes place to give an isomeric mixture of alkenes **28** in moderate to high yields. The reaction proceeds stereoselectively to form *E*-isomers predominantly (Scheme 24).⁸⁶

The molecule of ICF₂CF₂I has two highly reactive centers, which are able to form radicals under Cu(I) catalysis. As a result, the formation of bis-adducts with alkenes is possible. Thus, the reaction with excess of ethylene gave diiodide **29** in high yields under both CuCl and CuI catalysis. Less active BrCF₂CF₂Br did not reveal such reactivity (Scheme 25).¹⁵

An unusual reaction was found by Mitani.⁸⁷ The interaction of CF₃CCl₃ and CF₂BrCFBrCl with α -bromoallylsilane **1a** led to an elimination of bromine, accompanied by migration of a double bond. Vinylsilanes **30a** and **30b** were isolated in high yield in both cases (Scheme 26).

Another very effective copper-based catalytic system is the Cu(II) acetate–hydrazine hydrate redox system, which allows the preparation of polyfluoroalkyl-substituted derivatives **25** in good to high yields (Scheme 27).⁸⁸

Metallic copper was also used as a catalyst. The postulated⁸⁹ mechanism includes the formation of perfluoroethylcopper, the decay of which results in formation of copper and C₂F₅-radical. An addition to an alkene leads to the corresponding radical **26** (see Scheme 23), the reaction of which with C₂F₅I affords iodide **25** and C₂F₅-radical. In such a way, the interaction of C₂F₅I and cyclohexene gives the pentafluoroethylated derivative of iodocyclohexane **25a** as a 1:2 mixture of isomers in low yield. More effectively, this transformation proceeds in the case of CF₃CHCl₂. The copper-catalyzed addition of CF₃CHCl₂ to styrene leads to fluorinated phenylbutane **25b** in 97% yield (Scheme 28).⁹⁰

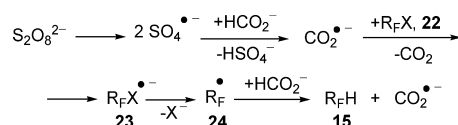
Several catalytic systems were proposed on the basis of iron and its mixtures with other metals. High efficiency was revealed by dichlorobis(π -cyclopentadienyl)titanium(IV) Cp₂TiCl₂/Fe redox couple. The addition of PFE to alkenes promoted by catalytic amounts of Cp₂TiCl₂ (2 mol %) and Fe powder (30 mol %) gave 1:1 adducts **25** in excellent yields⁹¹ (Table 5). A radical chain mechanism was postulated to occur in the case of this system. The first step is reduction of Ti(IV) into Ti(III) by iron. Next, low valent titanium species react with R_FX to give the perfluoroalkyl radical **24**, which initiates the radical chain process. To prove the first step of the mechanism, some facts should be noted. It was found that the reactions did not take place when both components of the couple (Cp₂TiCl₂ or Fe) were used separately. At the same time, TiCl₃ can initiate the reaction. Reduction of Cp₂TiCl₂ into greenish Cp₂Ti(III)Cl was also observed (Scheme 29, Table 5).

Table 3. Reduction of PFEs in Different Conditions

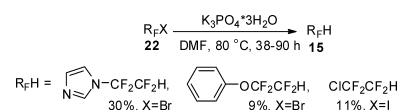
PFE	conditions	products	yield, %	ref
CF ₃ CCl ₃	(1) Zn, cat. CuCl–DMF, rt, 1 h; (2) H ₃ O ⁺	CF ₃ CCl ₂ H	99	63
CF ₃ CCl ₃	H ₂ (6 equiv), 20% Ni/SiO ₂ , 450 °C, 8 kPa	CF ₃ CCl=CClCF ₃	85	64
		CF ₃ CH=CClCF ₃	3	
		CF ₃ CH=CHCF ₃	1	
		CF ₃ CCl ₂ H	4	
CF ₃ CFCl ₂	H ₂ (1/3 equiv), Pd/C, 180 °C	CF ₃ CFH ₂	62	65
		CF ₃ CH ₃	13	
		(CF ₃ CFCl ₂ + CF ₂ ClCF ₂ Cl)	16	
CF ₃ CFCl ₂ /CF ₂ ClCF ₂ Cl 96:4	H ₂ (1/3.5 equiv), Pd/C, 170 °C	CF ₃ CFH ₂	76	65
		CF ₃ CH ₃	12	
		CF ₂ HCF ₂ H	10	
CF ₂ ClCFCl ₂	H ₂ , NiO/Cr ₂ O ₃ , 400 °C	CF ₂ =CHF	7	66
		CF ₂ =CClF	67	
C ₂ F ₅ Cl	H ₂ , Pd/AlF ₃	C ₂ F ₅ H	98	67
CF ₃ CBr ₃	NaH ₂ PO ₂ , AcONa–Pt/C, AcOH, 30 °C, 2 h	CF ₃ CHBr ₂	88	68
CF ₃ CBr ₃	NaH ₂ PO ₂ , AcONa–Pt/C, AcOH, 40 °C, 6 h	CF ₃ CH ₂ Br	90	68
CF ₃ CBr ₂ Cl	NaH ₂ PO ₂ , AcONa–Pt/C, AcOH, 30 °C, 2 h	CF ₃ CHBrCl	86	68
CF ₃ CBr ₂ Cl	NaH ₂ PO ₂ , AcONa–Pt/C, AcOH, 40 °C, 6 h	CF ₃ CH ₂ Cl	95	68
CF ₃ CFBr ₂	NaH ₂ PO ₂ , AcONa–Pt/C, AcOH	CF ₃ CHBrF	86	68
CF ₃ CCl ₃	NaH ₂ PO ₂ , AcONa–Pt/C, AcOH	CF ₃ CHCl ₂	92	68
CF ₃ CBr ₂ Cl	Na ₂ SO ₃ , 1.5 M NaOH, H ₂ O, 65–72 °C	CF ₃ CHClBr	80–90	69
CF ₃ CCl ₃	H ₂ (8 atm), 100 °C, THF, RhCl(PPh ₃) ₃	CF ₃ CHCl ₂	95	70
CF ₃ CCl ₃	H ₂ , RhCl(PPh ₃) ₃ , benzene, 4.5 h	CF ₃ CHCl ₂	95	71
CF ₃ CCl ₃	H ₂ (8 atm), 100 °C, THF, RuCl(NO)(PPh ₃) ₂	CF ₃ CHCl ₂	96	70
CF ₃ CFCl ₂	H ₂ , Pd/Al ₂ O ₃ , 101 °C	CF ₃ CH ₂ F	85	72
		CF ₃ CHFCl	9	
		CF ₃ CH ₃	2	
CF ₃ CCl ₃	HCO ₂ NH ₄ (1.5 equiv), (NH ₄) ₂ S ₂ O ₈ (0.25 equiv), DMF, 30–40 °C, 1–5 h	CF ₃ CHCl ₂	82	73
CF ₃ CCl ₃	HCO ₂ NH ₄ (2.5 equiv), (NH ₄) ₂ S ₂ O ₈ (0.25 equiv)	CF ₃ CH ₂ Cl	77	73
CF ₃ CFBr ₂	HCO ₂ NH ₄ (2.8 equiv), (NH ₄) ₂ S ₂ O ₈ (0.3 equiv)	CF ₃ CH ₂ F	87	73
CF ₃ CBr ₂ Cl	HCO ₂ NH ₄ (2.8 equiv), (NH ₄) ₂ S ₂ O ₈ (0.3 equiv)	CF ₃ CH ₂ Cl	65	73
ClCF ₂ CFCl ₂	HCO ₂ NH ₄ (1.5 equiv), (NH ₄) ₂ S ₂ O ₈ (0.3 equiv)	ClCF ₂ CFHCl	80	73
ClCF ₂ CFCl ₂	HCO ₂ NH ₄ (2.6 equiv), (NH ₄) ₂ S ₂ O ₈ (0.3 equiv)	HCF ₂ CFHCl	60	73
BrCF ₂ CFBrCl	HCO ₂ NH ₄ (2.8 equiv), (NH ₄) ₂ S ₂ O ₈ (0.3 equiv)	HCF ₂ CFHCl	63	73
CF ₃ CFBr ₂	(HCO ₂) ₂ , Na ₂ S ₂ O ₈ , NaOH, THF, 80 °C	CF ₃ CH ₂ F	95	74
CF ₃ CCl ₃	H ₂ , Pd/Fe	CF ₃ CHCl ₂	70	75
CF ₃ CFBr ₂	NaHSO ₃ , morpholine, THF	CF ₃ CHFBr	82	76
CFCl ₂ CFCl ₂	thiourea dioxide, Et ₃ N, MeCN, 14 h	CFCl ₂ CFHCl	90	76
CF ₃ CBr ₂ Cl	NaH ₂ PO ₂ , Fe ₂ (SO ₄) ₃ , MeOH	CF ₃ CHBrCl	82	74
CF ₃ CFCl ₂	HSiEt ₃ (2 equiv), (PhCO) ₂ O ₂ (0.05 equiv), 80 °C, 10 h	CF ₃ CFHCl	88	77
		CF ₃ CH ₂ F	12	
ClCF ₂ CFCl ₂	HSiEt ₃ (2 equiv), (PhCO) ₂ O ₂ (0.04 equiv), 80 °C, 10 h	ClCF ₂ CFHCl	90	77
		HCF ₂ CFCl ₂	2	
		ClCF ₂ CFH ₂	1	
		HCF ₂ CFHCl	7	
ClCF ₂ CFCl ₂	PS ^a (2 equiv), (PhCO) ₂ O ₂ (0.04 equiv), 80 °C, 10 h	ClCF ₂ CFHCl	97	77
		HCF ₂ CFCl ₂	2	
		HCF ₂ CFHCl	1	
ClCF ₂ CFCl ₂	PhSiMe ₂ H (2 equiv), (PhCO) ₂ O ₂ (0.04 equiv), 80 °C, 10 h	ClCF ₂ CFHCl	97	77
		HCF ₂ CFCl ₂	3	
CF ₂ ClCF ₂ I	EtOH, pyrogallol, irradiation	CF ₂ ClCF ₂ H	47	47

^aPS: poly(methylhydrosiloxane), [–(H)Si(Me)O–]_n.

Scheme 21



Scheme 22



Scheme 23

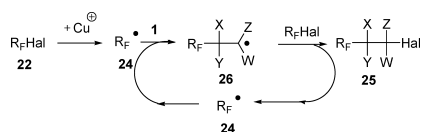
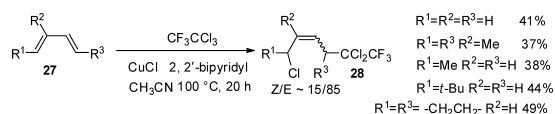


Table 4. Cu-Catalyzed Additions of Fluorinated Ethanes to Alkenes

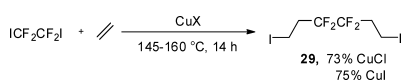
R _F Hal	Alkene	Product	Yield, %	ref.
BrCF ₂ CF ₂ Br			77 ^a	37
BrCF ₂ CFBrCl			95 ^a	37
ClCF ₂ CFCl ₂			62 ^a	37
CF ₃ CClBr ₂			98 ^a	37
CF ₃ CBBr ₂			74 ^a	37
CF ₃ CCl ₂			83 ^a	37
ClCF ₂ CFCl			73 ^b	37
CF ₃ CCl ₂			55 ^b	31
CF ₃ CCl ₂			62 ^b	31
CF ₃ CCl ₂			70 ^b	31
CF ₃ CCl ₂			63/82 ^c	30, 31
CF ₃ CCl ₂			40 ^b	31
CF ₃ CCl ₂			58 ^b	31
CF ₃ CCl ₂			65 ^b	31
CF ₃ CCl ₂			40 ^d	32
CF ₃ CCl ₂			83 ^a	33
CF ₃ CCl ₂			83 ^a	32
CF ₃ CCl ₂			57 ^d	32, 34
CF ₃ CCl ₂			70 ^e	35
CF ₃ CCl ₂			70 ^e	35

^aCuCl, ethanolamine, *t*-BuOH, reflux. ^bCuCl–CuCl₂ × 2H₂O (1:1), ethanolamine, *t*-BuOH, reflux. ^cConditions (a) or CuCl, 2-propylamine, reflux. ^dCuCl, CH₃CN, 140 °C. ^eCuCl, CH₃CN, 130 °C.

Scheme 24

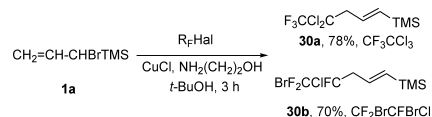


Scheme 25

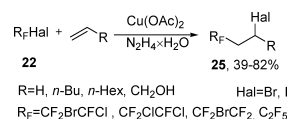


It also turned out that in the case of some dienes cyclizations are possible (formation of five- and six-membered cycles). Thus, diene **1b** was transformed easily into tetrahydrofuran derivatives **31a** and **31b** in almost quantitative yields (Scheme

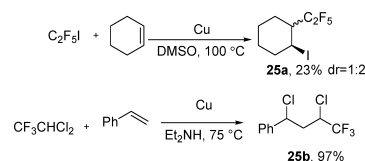
Scheme 26



Scheme 27



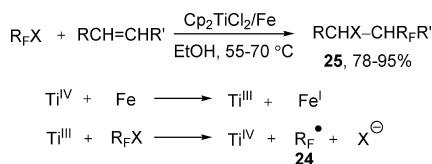
Scheme 28

Table 5. Reaction of PFEs with Alkenes Catalyzed with Cp₂TiCl₂/Fe

R _F	X	R	R'	yield, %	ref
BrCF ₂ CF ₂	I	<i>n</i> -C ₆ H ₁₃	H	96	91
BrCF ₂ CF ₂	I	<i>n</i> -C ₆ H ₁₃	H	91 ^a	91
BrCF ₂ CF ₂	I	<i>n</i> -C ₆ H ₁₃	H	93	92
BrCF ₂ CF ₂	I	CH ₂ OH	H	96	91
BrCF ₂ CF ₂	I	-(CH ₂) ₄ -	H	92	91
BrCF ₂ CF ₂	Br	<i>n</i> -C ₆ H ₁₃	H	95	91
BrCF ₂ CF ₂	Br	<i>n</i> -C ₆ H ₁₃	H	0 ^a	91
BrCF ₂ CF ₂	Br	<i>n</i> -C ₅ H ₁₁	H	93	91
BrCF ₂ CF ₂	Br	CH ₂ =CH(CH ₂) ₂	H	78	91
BrCF ₂ CF ₂	Br	-(CH ₂) ₄ -	H	92	91
ClCF ₂ CF ₂	I	CH ₂ =CH(CH ₂) ₄	H	90	93
ClCF ₂ CF ₂	I	CH ₂ =CH(CH ₂) ₄	H	85 ^b	93
BrCF ₂ CF ₂	I	CH ₂ =CH(CH ₂) ₄	H	91	93
ClCF ₂ CFCl	I	CH ₂ =CH(CH ₂) ₄	H	81	93

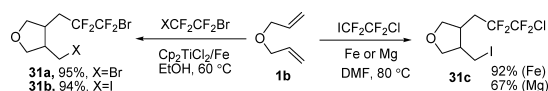
^aAbsence of Cp₂TiCl₂. ^bIn THF.

Scheme 29



30).⁹¹ Another example of the iron-catalyzed addition of PFE to alkenes is the reaction of ICF₂CF₂Cl with **1b**. This reaction

Scheme 30

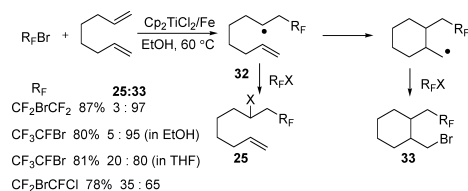


was performed in DMF in the presence of catalytic amounts of Fe powder to give the cyclic adduct **31c** in 92% yield.⁹⁴ Using Mg instead of Fe was less effective to afford **31c** in lower yield.

A more complicated situation was observed if a six-membered ring could be formed. Depending on PFE nature,

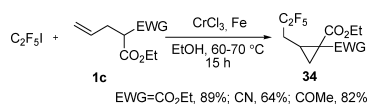
the reaction led to open-chain adducts or their mixtures with cyclohexane derivatives. The reaction of octa-1,7-diene with iodides $\text{BrCF}_2\text{CF}_2\text{I}$, $\text{ClCF}_2\text{CF}_2\text{I}$, $\text{ClCF}_2\text{CFClI}$ gave open-chain adducts **25** exclusively (Table 5), while the reaction with bromides $\text{BrCF}_2\text{CF}_2\text{Br}$, CF_3CFBr_2 , $\text{BrCF}_2\text{CFClBr}$ afforded mixtures of open-chain adducts **25** and cyclohexane derivatives **33**. Apparently, relatively weak C–I bond in iodides favors fast reaction of these PFEs with intermediate **32** to give iodide **25** without cyclization. In the case of bromides, the rate of cyclization is faster than reaction of **25** with $\text{R}_\text{F}\text{Br}$, which makes the cyclization the predominant process (Scheme 31).^{91,93}

Scheme 31



A versatile approach to electrophilic cyclopropanes was elaborated using the CrCl_3 –Fe system. The best results were observed when 20 mol % of $\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$ and 1.5 equiv of iron powder were used. Under these conditions, the addition of $\text{C}_2\text{F}_5\text{I}$ to alkenes **1c** led to highly substituted cyclopropanes **34** in high yields. The mechanism of the transformation was not discussed. However, it is reasonable to propose that after the standard radical addition the final step of the sequence is a base-induced intramolecular nucleophilic substitution of the intermediate secondary alkyl iodide to form a cyclopropane ring (Scheme 32).⁹⁵

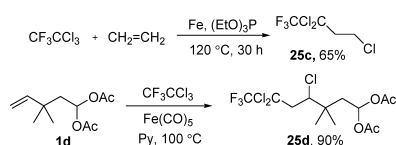
Scheme 32



Superior results were achieved when iron triethylphosphite was used as a catalytic system for the addition of CF_3CCl_3 to ethylene. Performed without a solvent in the presence of 2.5 mol % of Fe and 1.5 mol % of $\text{P}(\text{OEt})_3$, the addition gave adduct **25c** in 65% yield.⁹⁶ As compared to other metal-catalyzed additions of PFE to alkenes, this method is advanced in terms of cost, selectivity, and rate. Solvent-free procedure is also an advantage. It should be pointed out that the procedure was successfully scaled up allowing one to prepare more than 300 lb of **25c** per batch (Scheme 33). Iron complexes were also employed successfully. The addition of CF_3CCl_3 to acetal **1d** afforded adduct **25d** in 90% yield using $\text{Fe}(\text{CO})_5$ as an initiator of the reaction (Scheme 33).⁹⁷

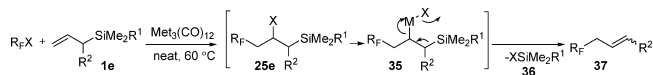
An interesting tandem reaction was discovered by Fuchikami.⁹⁸ Catalyzed by $\text{Fe}_3(\text{CO})_{12}$ or $\text{Ru}_3(\text{CO})_{12}$, addition

Scheme 33



of PFE to allylsilanes **1e** gave alkenes **37** in good yields through a cascade transformation. In contrast, lower yields were obtained if the reaction was carried out in the presence of AIBN or under UV-irradiation. The addition of PFE initiates the sequence to give saturated silanes **25e**. Next, an oxidative addition of transition metal complex led to organometallic intermediate **35**, which gave the alkene **37** by an elimination of halotrialkylsilane **36** (Scheme 34, Table 6).

Scheme 34

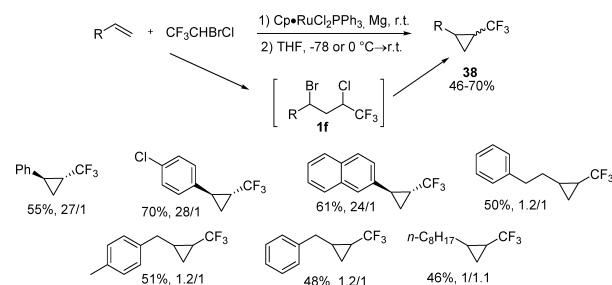
Table 6. Addition of PFE to Allylsilanes **1e**

R_F	X	R^1	R^2	cat. (mol %)	yield of 37 , %
BrCF_2CF_2	I	Me	H	$\text{Fe}_3(\text{CO})_{12}$ (1.3)	85
ClCF_2CFCl	I	Me	H	$\text{Fe}_3(\text{CO})_{12}$ (1.3)	85
ClCF_2CFH	I	Me	H	$\text{Fe}_3(\text{CO})_{12}$ (1.3)	75
BrCF_2CFCl	Br	Me	H	$\text{Ru}_3(\text{CO})_{12}$ (0.3)	59
ClCF_2CFCl	I	Me	CH_2CF_3	$\text{Fe}_3(\text{CO})_{12}$ (2.0)	85
ClCF_2CFCl	I	H	Me	$\text{Ru}_3(\text{CO})_{12}$ (0.6)	82
ClCF_2CFCl	I	Me	H	$h\nu^a$	65
ClCF_2CFCl	I	Me	H	AIBN (12) ^b	40

^aIrradiated with 400 W high-pressure lamp at 0 °C. ^bReaction was performed at 80 °C.

A versatile approach to CF_3 -cyclopropanes was elaborated using reaction of Halothane with terminal alkenes. The Ru-catalyzed addition of Halothane to alkenes in the presence of magnesium led to intermediate compounds **1f**, which undergo cyclization into CF_3 -cyclopropanes **38** after evaporation of excessive Halothane and addition of THF to the reaction mixture.⁹⁹ The reaction proceeds highly stereoselectively to give *trans*-isomers in the case of phenylethenes, while alkylethenes formed a mixture of *trans/cis*-isomers near 1/1 (Scheme 35).

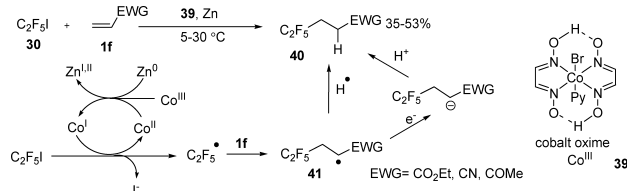
Scheme 35



Several other metal derivatives can catalyze the addition as well. Promoted by a cobalt oxime **39**–Zn redox system, the reaction of $\text{C}_2\text{F}_5\text{I}$ with such electron-deficient alkenes **1f** as ethyl acrylate, acrylonitrile, and methyl vinyl ketone afforded fluorinated compounds **40** in moderate to good yields (35–53%). The reduction of **39** by Zn gave low valent Co-complex, which initiates a radical chain reaction. The addition of perfluoroethyl radical to alkene **1f** leads to radical intermediate **41**. The formation of product **40** is a result of the reaction of radical **41** with hydrogen source or reduction with Zn followed by protonation.¹⁰⁰ It should be noted that this example is a rare

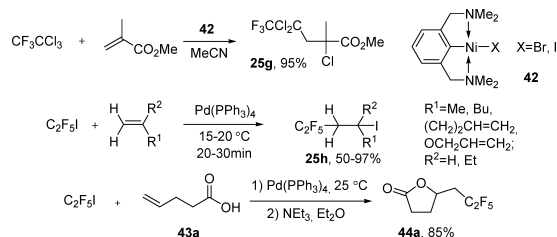
type of PFE reactivity, which formally resulted from reduction of C–Hal bond (Scheme 36).

Scheme 36



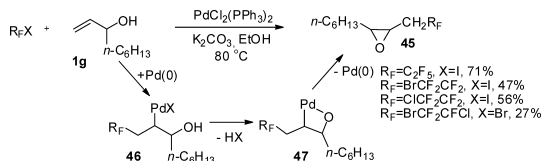
Application of other catalysts in this reaction was also reported. For example, the reaction of methyl methacrylate with CF_3CCl_3 catalyzed by organonickel(II) complexes **42** gave fluorinated ester **25g** quantitatively.¹⁰¹ A number of fluorinated alkyl iodides **25h** were synthesized under $\text{Pd}(\text{PPh}_3)_4$ catalysis. SET mechanism was postulated.¹⁰² In the case of unsaturated acid **43a**, the reaction gave lactone **44a** in 85% yield via a base-catalyzed cyclization of intermediate addition adduct **25** (Scheme 37).¹⁰³

Scheme 37



Polyfluoroalkyl-substituted oxiranes **45** were synthesized by $\text{PdCl}_2(\text{PPh}_3)_2$ -catalyzed addition of PFEs to allyl alcohol **1g**. The addition to the double bond leads to the organopalladium derivative **46**, which transforms into palladia-oxacyclobutane intermediate **47**, followed by reductive elimination to give the corresponding partially fluorinated oxiranes **45** (Scheme 38).¹⁰⁴

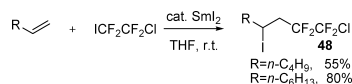
Scheme 38



Samarium(II) iodide was also used as a radical chain initiator for PFE additions. The reaction of $\text{ICF}_2\text{CF}_2\text{Cl}$ with alkenes led to the corresponding adducts **48** in moderate to high yields. SET mechanism was postulated for perfluoroalkyl radical formation (Scheme 39).¹⁰⁵

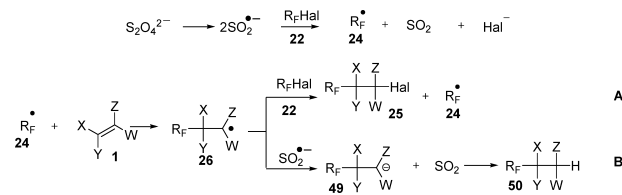
3.1.1.2. Catalysis by Sodium Dithionite and Related Species. A significant number of publications dealt with PFEs additions to multiple bonds catalyzed by sodium dithionite.¹⁰⁶

Scheme 39



Being a strong reducing agent, $\text{Na}_2\text{S}_2\text{O}_4$ is a good electron donor, which can be easily involved in electron transfer processes. As a result, $\text{Na}_2\text{S}_2\text{O}_4$ was widely used for an initiation of a radical chain reaction for PFEs additions to multiple bonds. The initiation of perfluoroethyl radical is shown in Scheme 40.

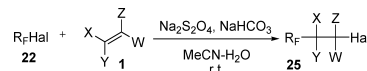
Scheme 40



Dissociation of dithionite leads to SO_2 radical anion, which reacts with PFE **22** to give radical **24**. The addition of **24** to alkene **1** gives rise to radical intermediate **26**. Next, depending on particular conditions, a formation of the normal adduct **25** (pathway A) or the reduced product **50** (pathway B) is observed. Thus, the reaction of the radical intermediate **26** with PFE **22** leads to product **25** having C–Hal bond in the structure and radical **24**, which continues the radical chain process. Alternatively, SET from SO_2 radical anion to radical **24** gives anion **49**, which transforms into reduced product **50** by protonation. Reactions initiated by Na_2SO_3 , NaHSO_3 , $\text{Na}_2\text{S}_2\text{O}_3$, $\text{Na}_2\text{S}_2\text{O}_5$, $\text{Na}_2\text{S}_2\text{O}_6$, and Zn-SO_2 couple can be rationalized by means of similar steps (Scheme 40).

In most cases, the reaction is performed in $\text{MeCN-H}_2\text{O}$ mixture, using NaHCO_3 as a base. Rarely are other solvents used ($\text{DMF-H}_2\text{O}$ or $\text{DMSO-H}_2\text{O}$ mixtures). Thus, the dithionite-catalyzed addition of PFEs to alkenes **1** has broad synthetic possibilities and can be used for alkenes bearing various functional groups at the double bond (alkyl, ester, hydroxyl) to afford the corresponding derivatives of type **25** in good yields (Scheme 41, Table 7).

Scheme 41



The reaction of alkenes with PFEs is quite sterically sensitive. Thus, the terminal double bond of diene **1h** reacted with $\text{ICF}_2\text{CF}_2\text{Cl}$ selectively to give adduct **25i** in the presence of the

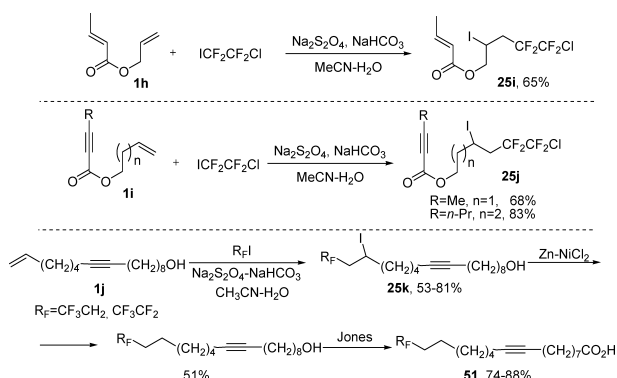
Table 7. Reaction of PFEs with Alkenes Giving Adducts 25

Product	Yield, % /ref.	Product	Yield, % /ref.
	not given ¹⁰⁷		68 ¹⁰⁸
	84 ¹⁰⁹		84 ¹¹⁰
	70 ¹¹¹		78 ¹¹²
	51 ¹¹³		56 ¹¹⁴
	25 ¹¹⁵		70 ¹¹⁶
	55 ¹¹⁵		47–61 ¹¹⁶

^a Na_2SO_3 in $\text{DMF-H}_2\text{O}$ 40–60 °C. ^b $\text{Na}_2\text{S}_2\text{O}_5$ in $\text{DMF-H}_2\text{O}$.

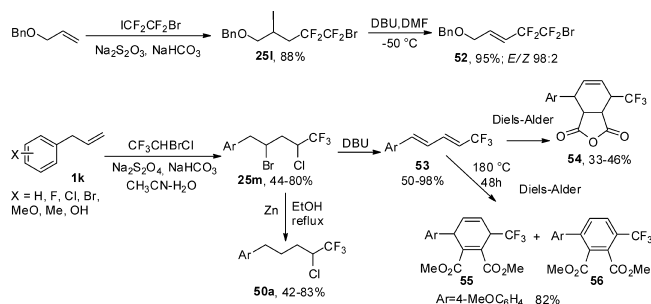
internal double bond.¹¹⁷ Similarly, the addition of PFE to a double bond is faster than that to a triple bond. It was found that, initiated by $\text{Na}_2\text{S}_2\text{O}_4$, reaction of unconjugated eneynes **1i**, **1j** with iodo-substituted PFEs gave chemoselectively fluorinated acetylenes **25j**, **25k**.^{117,118} This radical addition was used as a key step for the synthesis of novel stearic acid (9-octadecynoic acid) analogues **51**. The corresponding products **25k**, thus obtained, were deiodinated with Zn–NiCl₂ system. Finally, Jones oxidation provided the target fluorine-containing fatty acids **51**. It was demonstrated that the interfacial stability of the monolayers formed from these acids at the air–water interface increased dramatically with increasing fluorine content in the fatty acid molecule. From the viewpoint of the design of fluorinated fatty acid analogues, monolayer stability and/or surface activity can be sufficiently improved as long as hydrogen atoms at the terminal position of the hydrophobic segment are partially replaced by fluorine, and the replacement of at least five hydrogen atoms by fluorine is required to exhibit the stabilization of monolayers (Scheme 42).

Scheme 42



Products of radical addition of PFEs to alkenes have in the structure polyfluorinated fragment as an EWG and a halogen atom at β -position. Therefore, a base-induced elimination opens a simple way to alkenes having perfluoroethyl group at the double bond. For example, this sequence (radical addition–elimination) was used in the reaction of allylbenzyl ether with $\text{ICF}_2\text{CF}_2\text{Br}$ to form iodoalkane **25l**, which was transformed into fluorinated alkene **52** by treatment with DBU in DMF (Scheme 43).¹¹⁹ Similarly, some trifluoromethylated dienes were prepared. The addition of CF_3CHBrCl to the double bond of allylbenzenes **1k** in $\text{MeCN}/\text{H}_2\text{O}$ resulted in formation of 1-(2-bromo-4-chloro-5,5,5-trifluoropentyl)benzenes **25m** as a mixture of diastereoisomers in 3:1 ratio. The radical addition products were formed in moderate to high yields. Also, small

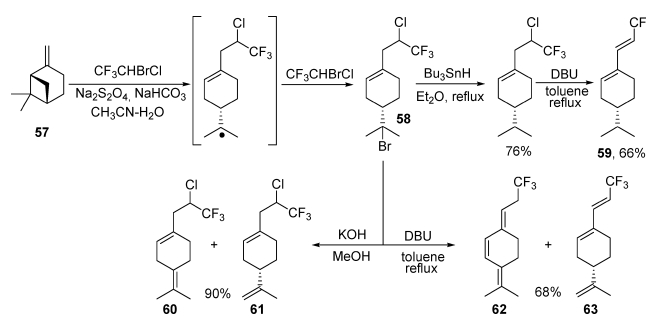
Scheme 43



amounts of reduced products **50a** were observed. The reduction of adducts **25m** with zinc in refluxing ethanol allows selective reduction of C–Br bond to form chlorides **50a**. The treatment of adducts **25m** with DBU in refluxing hexanes resulted in double dehydrohalogenation affording trifluoromethylated dienes **53** in good yields. These dienes can be used in Diels–Alder reaction using maleic anhydride and dimethyl acethylenedicarboxylate. The corresponding adducts **54–56** were isolated in moderate to good yields (Scheme 43).¹²⁰

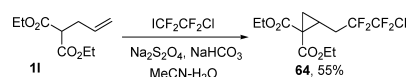
The exocyclic double bond of β -pinene (**57**) reacts with 1-bromo-1-chloro-2,2,2-trifluoroethane in the presence of sodium dithionite. As a result, rearrangement of the pinene skeleton takes place, converting it to menthan derivative **58** formed as a 1:1 mixture of diastereoisomers. Trifluoromethylated dienes **60**, **61** and trienes **62**, **63** can be obtained by dehydrobromination of **58** with various bases (pyridine, KOH, DBU). A reduction of bromide **58** with Bu_3SnH followed by dehydrochlorination with DBU afforded conjugated diene **59** in 50% overall yield. All of these transformations proceeded with retention of the configuration at the carbon atom C-4, and final compound **59** exhibited high optical purity (Scheme 44).¹²¹

Scheme 44



A very important part of this type of chemistry is the possibility to synthesize various carbo- and heterocyclic compounds (three-, five-, and six-membered) starting from the radical addition reaction of PFEs. For example, the reaction of allyl malonate **11** with polyfluoroalkyl iodide gave at the first step of the sequence the corresponding addition product, which afforded cyclopropane derivative **64** via an intramolecular nucleophilic substitution (but not elimination) by treatment with a base (Scheme 45).¹²²

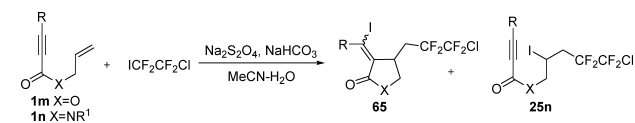
Scheme 45



If a formation of five-membered ring is possible, addition of PFEs to enynes occurs as tandem perfluoroalkylation-cyclization to give the cyclic products. Thus, the reaction of acyclic alkynoates **1m** (X = O) with ICF₂CF₂Cl afforded an isomeric mixture of fluorinated α -(iodoalkylidene)- γ -butyrolactone derivatives **65** in high yields.^{117,123} Polyfluoroalkylated 3-alkylidene-2(3*H*)-pyrrolidones **65** were prepared similarly by the reaction of *N*-allyl alkynamides **1n** (X = NR¹).¹¹⁷ In the latter case, admixture of acyclic products **25n** was obtained in noticeable amounts (Scheme 46, Table 8).

In the case of pent-4-en-1-yl amine derivatives **1o**, the addition of $\text{ClCF}_2\text{CF}_2\text{I}$ is very sensitive to the nature of the

Scheme 46



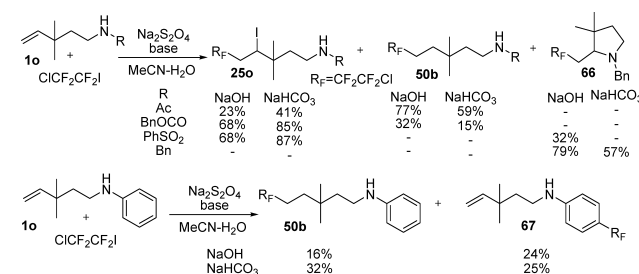
substituent adjacent to the amino group. Thus, **1o** protected with acyl- and benzyloxycarbonyl groups reacted with $\text{ClCF}_2\text{CF}_2\text{I}$ to give a mixture of the addition products **25o** and its reduced derivatives **50b**. The ratio **25o**/**50b** depends on the base used, giving higher yields of the addition product **25o** in the case of weakly basic NaHCO_3 . In the case of the amine **1o** protected with the phenylsulfonyl group, an admixture of the cyclization product **66** appeared, while reduced derivative **50b** was not formed. This pyrrolidine derivative **66** was the only product in the case of *N*-benzyl-substituted **1o**. At last, the perfluoroalkylation of *N*-phenyl-substituted amine **1o** led to a mixture of reduced derivative **50b** and compound **67**, which is the product of the aromatic substitution (see also the corresponding part of this Review) (Scheme 47).¹²⁴

If the addition of PFE to unsaturated acids is performed, a nucleophilic cyclization of the intermediate iodide into lactones is observed. Thus, a number of polyfluoroalkyl- γ -lactones **44** was synthesized from PFE and 4-pentenoic acids **43** in good yields (Scheme 48, Table 9). In the case of α -amino pentenoic acid amides **43b**, the reaction with $\text{ClCF}_2\text{CF}_2\text{I}$ led to racemic lactones **44b** and **44c** (yields were not given).¹²⁵ In the case of conjugated *E*-isomer of dienoic acid **68**, having unfavorable configuration for the cyclization, the reaction with $\text{ICF}_2\text{CF}_2\text{Cl}$ gave halide-free 1,4-adduct **69** with the *E*-configuration instead of the corresponding γ -lactone (Scheme 48).¹²⁶

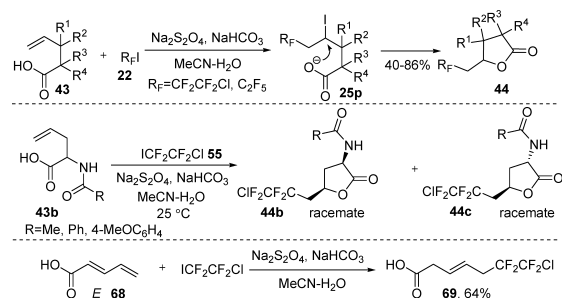
Similarly, 4-pentenamides **70a** gave γ -butyrolactones **44d**, **44e** in good yields by the reaction with $\text{ICF}_2\text{CF}_2\text{Cl}$. The expected pyrrolidinone **71** was not detected in the reaction mixture. The formation of γ -butyrolactones **44d,e** can be explained by cyclization of intermediate adduct **75** via hydrolysis of amide and an intramolecular nucleophilic substitution of iodide with carboxylate (Scheme 49).¹²⁹

Tandem reaction of perfluoroalkylation-lactonization with closure of six-membered ring was also described. In contrast to 4-pentenoic acids, 5-hexenoic acids **76a** afforded the corresponding δ -valerolactones **77** with admixture of acyclic 7,7,8,8,8-pentafluorooct-4-enoic acids **78** formed by competitive elimination of hydrogen iodide from intermediate iodine-substituted acids.¹³⁰ In the case of 2-allyl-4-pentenoic acids **76b**, the addition of $\text{ICF}_2\text{CF}_2\text{Cl}$ starts a cascade transformation, leading to the lactone **79** as a final product. At the first step, the radical addition followed by cyclization leads to cyclopentane derivative **80**. The next step is a base-induced lactonization, giving the final bicyclic valerolactone **79**. This free radical addition-lactonization tandem process is highly sterically sensitive. Thus, in the case of proton adjacent to the α -position at the carboxylic group of **76b**, the reaction gives only lactone **79** in 82% yield. The reaction of **76b** with more bulky ethyl

Scheme 47



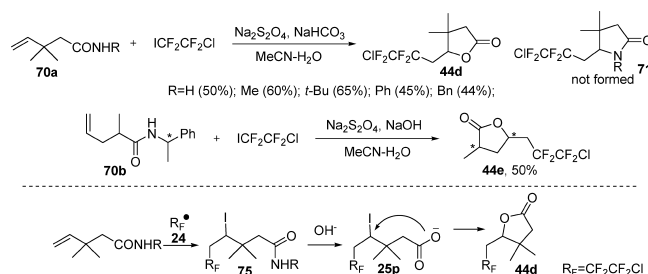
Scheme 48

Table 9. Synthesis of Polyfluoroalkyl- γ -lactones

Product	Yield, %/ref	Product	Yield, %/ref
$\text{ClF}_2\text{CF}_2\text{C}-\gamma\text{-lactone}$	81 ^[107]	$\text{ClF}_2\text{CF}_2\text{C}-\gamma\text{-lactone}$	85 ^[107,127]
$\text{ClF}_2\text{CF}_2\text{C}-\gamma\text{-lactone}$	40 ^[91]	$\text{ClF}_2\text{CF}_2\text{C}-\gamma\text{-lactone}$	70 ^[107,127]
$\text{ClF}_2\text{CF}_2\text{C}-\gamma\text{-lactone}$	86 ^[107,111,128]	$\text{C}_2\text{F}_5-\gamma\text{-lactone}$	81 ^[127]

^a NaHSO_3 in $\text{DMF}-\text{H}_2\text{O}$ 50–60 °C. ^b $\text{Na}_2\text{S}_2\text{O}_6$ in $\text{MeCN}-\text{H}_2\text{O}$.

Scheme 49



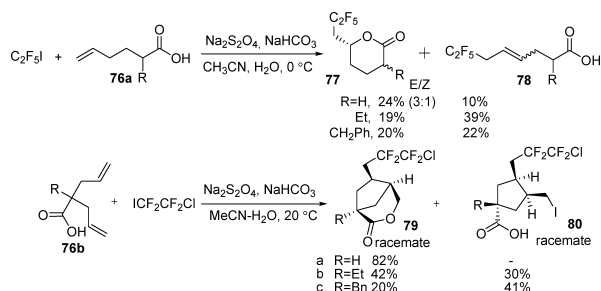
group gives a noticeable admixture of cyclopentane derivative **80**, which dominates in the case of **76b** bearing benzyl group (Scheme 50).¹³¹

Examples of formation of reduced product **50** (pathway B, Scheme 40) are less presented in the literature as compared to normal adduct **25** (pathway A). Thus, the reaction of 1-octene with some PFEs gave fluorinated alkanes **50c** in good yields.

Table 8. Reaction of **1m** and **1n** with $\text{ICF}_2\text{CF}_2\text{Cl}$

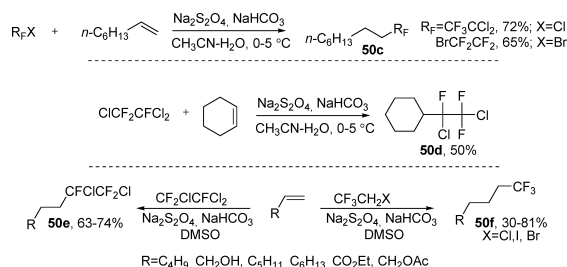
X	R	yield of 65 , %	E/Z	yield of 25n , %	X	R	yield of 65 , %	E/Z	yield of 25n , %
O	H	47	60:40		NBn	Me	50	97:3	35
O	Me	83	97:3		NBn	<i>n</i> -C ₃ H ₇	41	97:3	23
O	<i>n</i> -C ₈ H ₁₇	60	97:3		NAc	H	46	52:48	
NMe	Me	53	97:3	19	NBz	Me	70	97:3	

Scheme 50



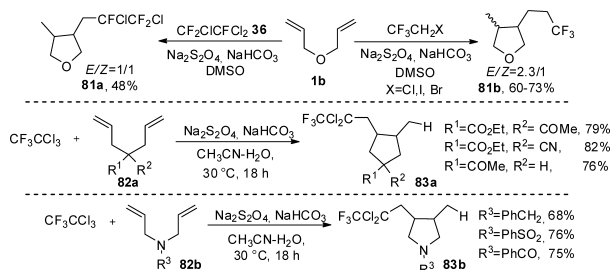
Similarly, cyclohexene afforded cyclic derivative **50d**.¹¹⁴ Using heating at 75–80 °C in DMSO, a number of reactions of the addition of PFE to alkenes were carried out to afford the corresponding polyfluoroalkylated adducts **50e**, **50f** in moderate to high yields (Scheme 51).¹³²

Scheme 51



Fluorinated derivatives of tetrahydrofuran **81a**, **81b** were prepared by an intramolecular radical cyclization.^{132,133} In the case of 1,6-dienes **82a** and **82b**, the addition of CF_3CCl_3 initiated by $\text{Na}_2\text{S}_2\text{O}_4$ led to five-membered ring closure accompanied by reduction to give polyfluoroalkylated cyclic derivatives **83a** and **83b** in high yields (Scheme 52).¹³⁴

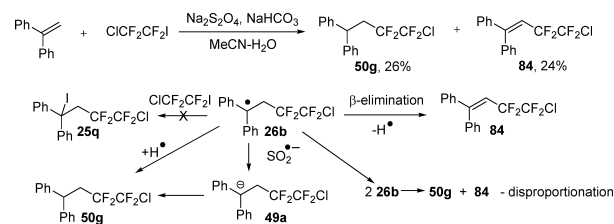
Scheme 52



In the case of the addition to sterically hindered α -phenylstyrene, a mixture of reduced product **50g** and styrene **84** was obtained. At first glance, the formation of both products can be easily explained in terms of previously mentioned pathways A and B (Scheme 40). Indeed, realization of pathway B can result in the formation of **50g**, as well as realization of pathway A, followed by HI elimination from “normal adduct” **25q**, can afford alkene **84**. However, it is not true. Thus, if the reaction is carried out in MeCN–D₂O, no formation of the corresponding deuterium containing product is observed, which means that the formation of the anion **49a** does not take place or is “less important”, literally citing the authors.¹³⁵ The reaction of radical **26b** with $\text{ClCF}_2\text{CF}_2\text{I}$ is also impossible due to sterical hindrance of α -phenylstyrene. Most probable

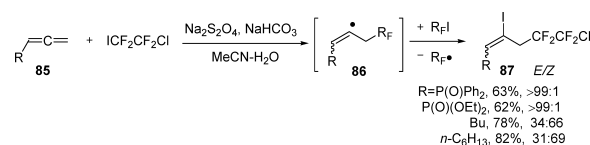
explanations of the reaction results are the following. Radical **26b** captures hydrogen-radical from solvent to form reduced product **50g**, while β -elimination of hydrogen-radical from **26b** leads to alkene **84**. Alternatively, **50g** and **84** can be a result of radical **26b** disproportionation, which is in good agreement with their almost equal ratio in the reaction mixture (Scheme 53).

Scheme 53



Various allenes **85** were also involved successfully into the reaction with some PFE in the presence of sodium dithionite. The reaction proceeded regio- and stereoselectively via the formation of the vinyl radical **86** to introduce a perfluoroalkyl group into the terminal position of allenes. As a result, iodoalkenes **87** were formed in high yields (Scheme 54).¹³⁶

Scheme 54



3.1.1.3. The Initiation by Heating or Irradiation. In contrast to the above-mentioned methods of the reaction, initiation, heating, UV, or X-ray irradiation were rarely used, which resulted from some distinct drawbacks of these methods. Indeed, under these conditions, reaction often gives mixtures of telomers or oligomers in lower yields in comparison with other methods. Prolonged reaction time, noticeable or severe tarring, elevated temperatures, and UV-irradiation itself produce some additional inconveniences. In some cases, however, high yields of the adducts were achieved to provide an alternative pathway for their preparation (Table 10).

Being heated, $\text{ICF}_2\text{CF}_2\text{I}$ can be partially transformed into oligomers of general formula $\text{I}(\text{CF}_2\text{CF}_2)_n\text{I}$; oligomers with $n = 2$ and $n = 3$ are dominating ones in the reaction mixture.^{146,147} The reaction proceeds through an elimination of iodine to give tetrafluoroethylene, which reacts with $\text{ICF}_2\text{CF}_2\text{I}$ to result in the oligomerization. In contrast, heating or UV-irradiation of $\text{BrCF}_2\text{CF}_2\text{Br}$ led only to decomposition to form tetrafluoroethylene by two sequential eliminations of bromine-radical (Scheme 55).¹⁴⁸

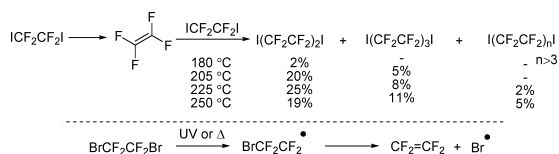
Thermally or photochemically induced reaction of $\text{CH}_3\text{CF}_2\text{H}$, CH_3CF_3 with hexafluoropropene resulted in a CH-insertion to form radical addition products, partially fluorinated pentanes **88**, in good yield. It should be noted that 1,1-difluoroethane forms radical **24a** stabilizing with C–F bonds in contrast to 1,1,1-trifluoroethane, forming 2,2,2-trifluoroethyl radical **24b** (Scheme 56).¹⁴⁹

3.1.1.4. Initiation by Radicals. Another way to initiate the addition of PFE is application of radical initiators such as AIBN, peroxides, as well as the rarely used $\text{O}_2\text{--BR}_3$ ($\text{R} = \text{Alk}$) system.

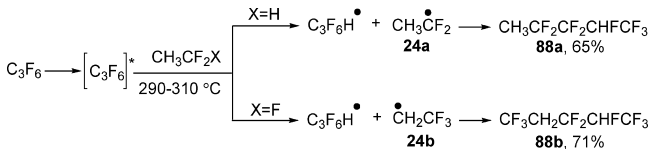
Table 11. Addition of C₂F₅I to Alkenes Catalyzed by Radical Particles

R ₁ Hal	alkene	Conditions	products	Yield, %	ref.
C ₂ F ₅ I	C ₆ F ₇ CH=CH ₂	(<i>t</i> -BuO) ₂ , D-100°, 140 °C, 3 h	$\begin{array}{c} \text{C}_6\text{F}_{17} \\ \\ \text{---CH}_2\text{---CH}_2\text{---C}_2\text{F}_5 \\ \\ \text{C}_6\text{F}_{17} \end{array}$ $\left\{ \begin{array}{c} \text{C}_6\text{F}_{17} \\ \\ \text{---CH}_2\text{---CH}_2\text{---C}_2\text{F}_5 \\ \\ \text{---CH}_2\text{---CH}_2\text{---C}_2\text{F}_5 \\ \\ \text{---CH}_2\text{---CH}_2\text{---C}_2\text{F}_5 \end{array} \right\}_n$	25 17	[1]
C ₂ F ₅ I	AcOCH=CH ₂	AIBN, 70 °C, 10 h	$\begin{array}{c} \text{AcO} \\ \\ \text{---CH}_2\text{---CH}_2\text{---C}_2\text{F}_5 \\ \\ \text{---CH}_2\text{---CH}_2\text{---C}_2\text{F}_5 \end{array}$	83	[1]
C ₂ F ₅ I	MeO ₂ CCH=CH ₂	AIBN, 70 °C, 24 h	$\begin{array}{c} \text{MeO}_2\text{C} \\ \\ \text{---CH}_2\text{---CH}_2\text{---C}_2\text{F}_5 \\ \\ \text{---CH}_2\text{---CH}_2\text{---C}_2\text{F}_5 \end{array}$	80	[1]
C ₂ F ₅ I		AIBN, 4 h, 80 °C	$\begin{array}{c} \text{C}_2\text{F}_5 \\ \\ \text{---[Bicyclo[2.2.1]heptane]} \end{array}$	99	[1]
C ₂ F ₅ I	CH ₂ =CH(CH ₂) ₃ CO ₂ H	1) AIBN ^a , r.t. 2) Zn, HCl-H ₂ O, 70 °C	C ₂ F ₅ (CH ₂) ₃ CO ₂ H	48	[1]
C ₂ F ₅ I		(<i>t</i> -BuO) ₂ , 130 °C	$\begin{array}{c} \text{---[Cyclohexane ring]} \end{array}$ $\begin{array}{c} \text{---[Cyclohexane ring]} \end{array}$	21 18	[1]
BrCF ₂ CF ₂ Br	CH ₂ =CH ₂	(PhCO) ₂ O ₂ , 100 °C	$\begin{array}{c} \text{BrCF}_2\text{CF}_2\text{CH}_2\text{CH}_2\text{Br} \\ \text{BrCF}_2\text{CF}_2\text{CH}_2\text{CH}_2\text{Br} \end{array}$	32 20	[1]
CF ₃ CCl ₂ I	<i>n</i> -C ₈ H ₁₁ CH=CH ₂	(<i>t</i> -BuO) ₂	<i>n</i> -C ₈ H ₁₁ CH ₂ CH ₂ CCl ₂ CF ₃	Not given	[1]
CF ₃ CFBr ₂	CH ₂ =CH ₂	(<i>t</i> -BuO) ₂ , 120 °C	CF ₃ (CFBrCH ₂) ₂ Br	26	[1]
CF ₃ CFBr ₂		(<i>t</i> -BuO) ₂ , 120 °C	$\begin{array}{c} \text{Br} \\ \\ \text{---CFBrCF}_3 \end{array}$	69	[1]
CF ₃ CFBr ₂		(<i>t</i> -BuO) ₂ , 120 °C	$\begin{array}{c} \text{Br} \\ \\ \text{---CFBrCF}_3 \end{array}$	30	[1]
CF ₃ CCl ₂ I	HO ₂ CCH=CH ₂	(PhCO) ₂ O ₂ , 80 °C ^a	$\begin{array}{c} \text{HO}_2\text{C} \\ \\ \text{---CH}_2\text{---CH}_2\text{---CCl}_2\text{CF}_3 \\ \\ \text{Cl} \end{array}$	78	[1]
ICF ₂ CF ₂ I	CH ₂ =CH ₂	(<i>t</i> -BuO) ₂ , 135 °C	[(CH ₂) ₂ CF ₂ CF ₂ (CH ₂) ₂] _n	52	[1]
ICF ₂ CF ₂ I	AcOCH=CH ₂	AIBN, 80-85 °C	$\begin{array}{c} \text{AcO} \\ \\ \text{---CH}_2\text{---CH}_2\text{---CF}_2\text{CF}_2\text{I} \end{array}$	12	[1]
CH ₃ CF ₂ I	C ₄ H ₉ CH=CH ₂	AIBN, 80-85 °C	$\begin{array}{c} \text{C}_4\text{H}_9 \\ \\ \text{---CH}_2\text{---CH}_2\text{---CF}_2\text{CH}_3 \\ \\ \text{---CH}_2\text{---CH}_2\text{---CF}_2\text{CH}_3 \end{array}$	82	[5]
CH ₃ CF ₂ I	CF ₂ =CF ₂	(PhCO) ₂ O ₂ , 80 °C	$\begin{array}{c} \text{CH}_3(\text{CF}_2)_n\text{I} \\ \text{CH}_3(\text{CF}_2)_n\text{I} \\ \text{CH}_3(\text{CF}_2)_n\text{I} \\ \text{CH}_3(\text{CF}_2)_n\text{I} \\ \text{CH}_3(\text{CF}_2)_n\text{I} \\ \text{CH}_3(\text{CF}_2)_n\text{I} \end{array}$	48 23 9 4 3	[8]
C ₂ F ₅ I	CH ₂ =CF ₂	TBPP ^b , 75 °C	$\begin{array}{c} \text{CF}_3\text{CF}_2\text{CH}_2\text{CF}_2\text{I} \\ \text{CF}_3\text{CF}_2\text{CH}_2\text{CF}_2\text{I} \\ \text{CF}_3\text{CF}_2\text{CH}_2\text{CF}_2\text{I} \\ \text{CF}_3\text{CF}_2\text{CH}_2\text{CF}_2\text{I} \\ \text{CF}_3\text{CF}_2\text{CH}_2\text{CF}_2\text{I} \\ \text{CF}_3\text{CF}_2\text{CH}_2\text{CF}_2\text{I} \end{array}$	16 49 18 9 5	[1]

Scheme 56



Scheme 56



As an exclusion, reaction of diethyl maleate with $\text{C}_2\text{F}_5\text{I}$ afforded reduced product **50h** instead of iodide. This observation was explained by a reaction of radical intermediate **26c** with “hydrogen source” in the reaction media (Scheme 58). The nature of the hydrogen source was not clarified.¹⁶¹

^aD-100: perfluorobutyltetrahydrofuran 80% and perfluoropropyltetrahydropyran 20% with bp 100 °C. ^b2,2'-Azobis(2,4-dimethyl)-valeronitrile. ^cSolid-state synthesis using Wang resin ((Advanced ChemTech, U.S.); 100–200 mesh; substitution 0.8 mmol/g; cross-linking, 1% DVB). ^d*tert*-Butylperoxypivalate.

CCOC(=O)/C=C/C(=O)OCC
 $\xrightarrow[\text{AIBN, 220 } ^\circ\text{C}]{\text{C}_2\text{F}_5\text{I}}$
CCOC(=O)C(C(=O)OCC)C(F)(F)F

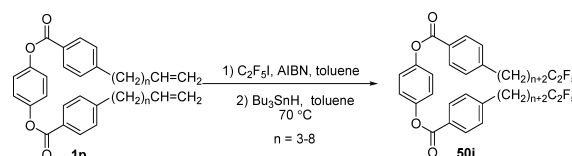
 50h, 40%

The reaction proceeds via a radical intermediate **26c**, which is a 1,1,1-trifluoro-2,2-dimethyl-2-(diethoxycarbonylmethyl)ethyl radical.

Double addition $\text{C}_2\text{F}_5\text{I}$ to 1, ω -diene **1p** was described. Reaction of $\text{C}_2\text{F}_5\text{I}$ with **1p** followed by treatment with *t*- Bu_3SnH gave the fluorinated derivative **50i**, which was found to possess liquid crystalline properties. The yields of the products were not given (Scheme 59).¹⁶²

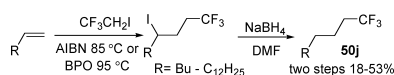
AIBN or benzyl peroxide-catalyzed addition of $\text{CF}_3\text{CH}_2\text{I}$ to terminal alkenes followed by sodium borohydride reduction

 $(\text{t-BuO})_2 \longrightarrow 2 \text{ t-BuO}^\bullet + \text{O}_2$	 $\text{NC-C(CH}_3)_2\text{-N-C(CH}_3)_2\text{-CN} \longrightarrow 2 \text{ } ^\bullet\text{C(CH}_3)_2\text{CN} + \text{N}_2$
 $\text{Ph-CO-O-O-CO-Ph} \longrightarrow 2 \text{ Ph-CO-O}^\bullet \longrightarrow 2 \text{ Ph}^\bullet + 2 \text{ CO}_2$	 $\text{R}_\text{f}\text{Hal} + \text{R}^\bullet \longrightarrow \text{R}_\text{f}^\bullet + \text{RHal}$



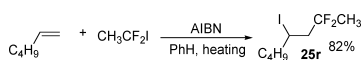
represents a very useful two-step protocol for the preparation of various alkanes **50j** having trifluoromethyl group at the end of the chain (Scheme 60).¹⁶³

Scheme 60



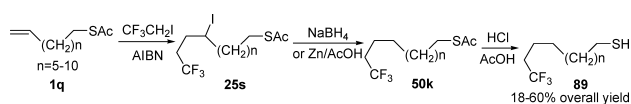
AIBN-catalyzed radical addition of 1,1-difluoro-1-iodoethane to hexene-1 confirmed additionally the efficiency of this protocol. The corresponding adduct **25r** was prepared in high yield. The reactivity of the formed radical is in agreement with order of reactivity for alkyl radicals $\text{RCH}_2\text{CH}_2 < \text{RCH}_2\text{CF}_2 < \text{CF}_3$ (Scheme 61).¹⁶⁴

Scheme 61



CF_3 -substituted alkanethiols **89** were synthesized to study self-assembly on gold surface. The AIBN-catalyzed radical addition of 1,1,1-trifluoroethyl iodide to ω -olefins **1q** that are functionalized at the α -terminus with a thioacetate group was a key step of the reaction sequence. Subsequent reduction of **25s** and acidic hydrolysis of **50k** gave target ω -trifluoromethylated thiols **89** in low to good overall yields (Scheme 62).¹⁶⁵

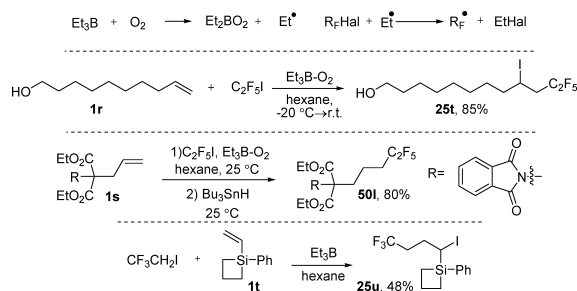
Scheme 62



In contrast to the activation of the reaction by peroxides or AIBN, giving radicals by thermal decomposition, ethyl-radicals are formed using the $\text{O}_2\text{--Et}_3\text{B}$ system at rather low temperature. As a result, addition can be carried out under milder conditions, which is an advantage of that system. For example, the addition of $\text{C}_2\text{F}_5\text{I}$ to long-chain alkene **1r** was performed in -20°C to room temperature range to give derivative **25t** in 85% yield.¹⁶⁶ In a similar manner, the addition of $\text{C}_2\text{F}_5\text{I}$ to alkene **1s** followed by reduction with $t\text{-Bu}_3\text{SnH}$ led to fluorinated malonate **50l** in high yield.¹⁶⁷ The reaction of vinylsilane **1t** with $\text{CF}_3\text{CH}_2\text{I}$ afforded derivative **25u** in 48% yield without destruction of the four-membered cycle (Scheme 63).¹⁶⁸

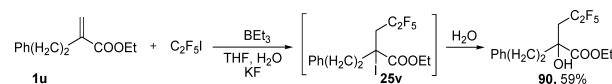
The reaction of acrylic ester derivative **1u** with $\text{C}_2\text{F}_5\text{I}$, performed in $\text{THF--H}_2\text{O}$ mixture in the presence of KF ,

Scheme 63



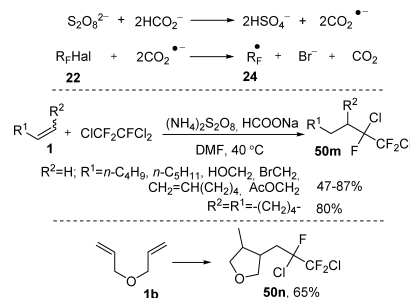
afforded alcohol **90** in good yield. That formation is a result of nucleophilic substitution of iodide (activated with ester moiety in α -position) by water in intermediate adduct **25v** (Scheme 64).¹⁶⁹

Scheme 64



Another redox system used for initiation of radical addition of freons is a mixture of $(\text{NH}_4)_2\text{S}_2\text{O}_8$ and HCO_2Na . Oxidation of formate anion by persulfate led to CO_2 radical-anion, which reacts with perfluoroethylhalide **22** to give perfluoroethyl radical **24** via SET mechanism. The addition of perfluoroethyl radical **24** to alkene **1** gave the corresponding radical, which captured hydrogen from the reaction media to afford the reduced product (see Scheme 40, pathway B). Using this redox system, various fluorinated derivatives **50m** were prepared in high yields under mild conditions (Scheme 65). In case of bisallyl ether **1b**, cyclic tetrahydrofuran derivative **50n** was obtained.¹⁷⁰

Scheme 65



3.1.2. Reactions with Alkynes. Alkynes were also involved into the reaction with PFEs (generally, iodo-substituted ethanes were used). Reaction can be initiated by using various redox systems (peroxides, AIBN, heating, and UV-irradiation) to give iodoalkenes bearing perfluoroethyl group at the formed double bond in high yields. Usually, this reaction proceeds low selectively to form a variable *E/Z*-mixture of the addition products; however, in some cases very good stereoselectivity was achieved (Table 12).

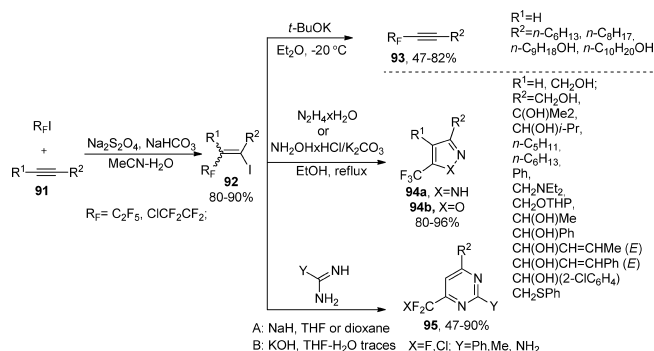
The reaction of alkynes initiated with sodium dithionite gave fluorinated iodoalkenes **92** in high yields as a mixture of *Z/E* isomers in 1:1 ratio. In most cases, obtained products were used for further transformations into polyfluoroalkylated alkynes or heterocycles. Thus, using this approach, acetylenes **93** were prepared in good yields by addition of $\text{C}_2\text{F}_5\text{I}$ and $\text{ClCF}_2\text{CF}_2\text{I}$ to terminal acetylenes **91** ($\text{R}^1 = \text{H}$) followed by treatment with $t\text{-BuOK}$.¹⁷⁸ Similarly, 3-(trifluoromethyl)pyrazoles **94a**, 5-(trifluoromethyl)isoxazoles **94b**, and 4-(trifluoromethyl)-pyrimidines **95** were synthesized in high yields by the subsequent reaction with hydrazine, hydroxylamine, amidines, and guanidine correspondingly (Scheme 66).¹⁷⁹

Initiated by sodium dithionite, the addition of $\text{C}_2\text{F}_5\text{I}$ to propargyl alcohol afforded quantitatively an isomeric mixture of alkenes **92a**, which was used as a starting material for efficient laboratory scale preparation of 4,4,5,5,5-pentafluoropentane-1-thiol **96** (Scheme 67). The latter is an important side-chain

Table 12. Addition of PFE to Alkynes

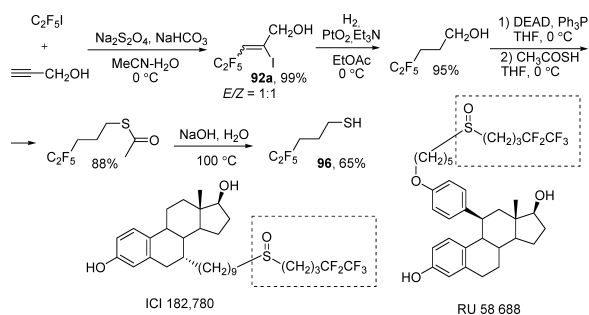
R ₂ Hal	Acetylene	Conditions	Product	E/Z	Yield, %	ref.
BrCF ₂ CF ₃ I	Ph—C≡C—	Cp ₂ TiCl ₂ /Fe EtOH, 55–70 °C		50:50	75	171
ClCF ₂ CF ₃ I	Ph—C≡C—	Cp ₂ TiCl ₂ /Fe EtOH, 55–70 °C		50:50	72	171
ClCF ₂ CF ₃ I	<i>n</i> -Bu—C≡C—	Cat. SmI ₂ , THF, r.t.		14:86	76	172
C ₂ F ₅ I	<i>n</i> -C ₈ H ₁₇ —C≡C—	Zn (0.1 eq.) CF ₃ CO ₂ H (0.2 eq.) CH ₂ Cl ₂ , 25 °C		95:5	85	173
C ₂ F ₅ I	F ₃ C—C≡C—	220 °C, 35 MPa		-	67	174
C ₂ F ₅ I	H—C≡C—H	UV or 220–260 °C		Not given	57–72 0.5	175
C ₂ F ₅ I	HO(CH ₂) ₈ C≡CH	AIBN, 80 °C, 20 h		92:8	82	176
ICF ₂ CF ₂ I	TMS—C≡C—	<i>t</i> -(BuO) ₂ , 120 °C		Not given	Not given	177

Scheme 66



moiety of antitumor cancer agents compounds RU 58 688 and ICI 182,780,¹⁸⁰ which revealed high antitumor potency.¹⁸¹

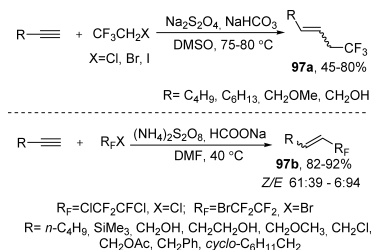
Scheme 67



Similarly to the reaction with alkenes, in some cases the addition of PFEs to alkynes affords reduced products instead of halides. Thus, the reaction of 1,1,1-trifluoro-2-chloro-, 2-bromo-, and 2-iodoethanes with terminal acetylenes yields the corresponding fluorinated alkenes **97a** as a mixture of diastereomers.¹³² For example, in the case of (NH₄)₂S₂O₈–HCO₂Na redox system, the reaction afforded various fluorinated alkenes **97b** in high yields as mixtures of *E/Z* isomers (Scheme 68).^{170,182}

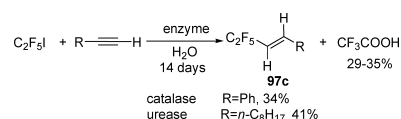
In contrast to other methods, the addition of pentafluoroethyl iodide catalyzed by enzymes (catalase and urease) gave 100% stereoselectively *E*-isomer of C₂F₅-alkenes **97c** in moderate yields. The reaction was performed in the presence of catalase or urease in water. The formation of alkenes **97c** was

Scheme 68



accompanied by formation of trifluoroacetic acid. It was mentioned that electron transfer from enzyme to C₂F₅I took place to initiate the addition of C₂F₅I and a reduction of the C–I bond (Scheme 69).¹⁸³

Scheme 69



3.1.3. Reactions with Vinyl Ethers, Silyl Ethers, and Enolates. The reaction of trimethylsilyl enolates **98** with PFEs is a very valuable approach to α -perfluoroethylated aldehydes and ketones **99**. Both cyclic and open-chain silyl ethers were successfully involved in the transformation to form target carbonyls in good to high yields. The reaction proceeds through the intermediate formation of the radical, produced by the addition of polyfluorinated radical to the double bond of silyl ether **98** followed by desilylation (Scheme 70, Table 13).

Scheme 70

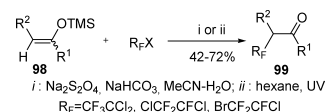


Table 13. Reaction of Silyl Enolates with PFEs

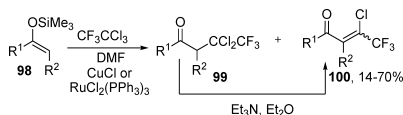
X	Product	Yield, %	ref.	X	Product	Yield, %	ref.
I		78 ^a	184	Cl		48	185
Cl		72	185	Br		42	185
Cl		70	185	I		52	186
Cl		61	185	Br		45–60 ^a	187
Br		40–60 ^a	187				

^aNa₂S₂O₄, NaHCO₃, MeCN–H₂O. All other reactions were performed in hexane under UV-irradiation.

Various catalytic systems can initiate this transformation of trimethylsilyl enolates **98**. In the case of metal-catalyzed reaction with CF₃CCl₂, a mixture of CCl₂CF₃-substituted ketones **99** and α,β -unsaturated CF₃-ketones **100** was formed. The reaction was performed in DMF using a large excess of CF₃CCl₂ (about 20 equiv), equimolar amount of CuCl, and 4 Å molecular sieves for trapping the liberated HCl. Subsequent treatment with triethylamine or DBN resulted in access to α,β -

unsaturated CF₃-ketones **100** in moderate to good yields for aromatic enol ethers and lower yields for aliphatic ones (Scheme 71, Table 14).¹⁸⁸ Later, the same authors examined

Scheme 71

Table 14. Synthesis of CF₃-Ketones **99** by the Reaction of Trimethylsilyl Enols **98** with CF₃CCl₃

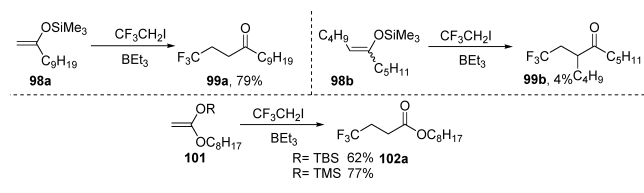
R ¹	R ²	Yield of, % (Z/E)		Cyclic silyl enolates	Yield of, % (Z/E)	
		CuCl ¹⁸⁸	RuCl ₂ (PPh ₃) ₃ ¹⁸⁹		CuCl ¹⁸⁸	RuCl ₂ (PPh ₃) ₃ ¹⁸⁹
Ph	H	52 (90:10)	75 (92:8)	OTMS		
4-MeOC ₆ H ₄	H	44 (92:8)	-	OTMS	49 (30:70)	60 (27:73)
4-MeC ₆ H ₄	H	49 (93:7)	-	OTMS		
4-ClC ₆ H ₄	H	55 (93:7)	-	OTMS		
Ph ^a	Ph	14 (12:88)	30 (13:87)	OTMS	21 (12:88)	37 (24:76)
2-furyl	H	70 (97:3)	-	OTMS		
2-pyridyl	H	0	25 (99:1)	OTMS		
PhCH=CH	H	59 (99:1)	-	OTMS		
<i>t</i> -Bu	H	15 (88:12)	53 (99:1)	OTMS	0	31 (81:19)
CH ₂ (CH ₃) ₂ ^a	H	15 (79:21)	-	OTMS		
<i>n</i> -BuO	H	-	20 (95:5)	OTMS	32 (58:42)	60 (60:40)

^aInitial alkene has *Z*-configuration.

RuCl₂(PPh₃)₃ as catalyst for this reaction. Despite a lower amount of the catalyst used (5 mol %), higher yields of desired products **100** were obtained. Moreover, 2-pyridyl and cyclohexenyl enol ethers were involved in the reaction to provide satisfactory yields of α,β -unsaturated CF₃-ketones, although nothing was isolated in the case of CuCl catalysis. It should be also mentioned that, in the case of ruthenium catalysis, intermediate CCl₂CF₃-ketones **99** were formed without admixture of CF₃-enones **100**.¹⁸⁹

A number of silyl enolates and silyl ketene acetals were trifluoroethylated with CF₃CH₂I in the presence of triethylborane. In the case of silyl enolates **98a**, **98b**, the reaction requires prolonged reaction time (15–20 h) to form target products **99a**, **99b** in variable yields (Scheme 72).¹⁹⁰ Contrary, silyl ketene acetals **101** gave high yields of ω -trifluoromethylated esters **102a** during shorter reaction time (up to 1 h).

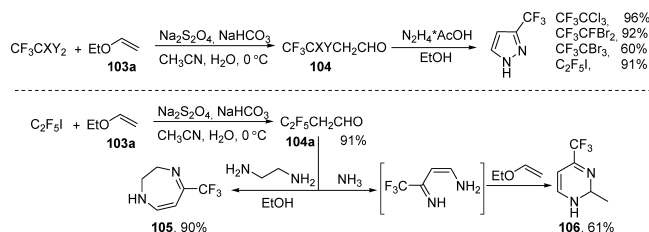
Scheme 72



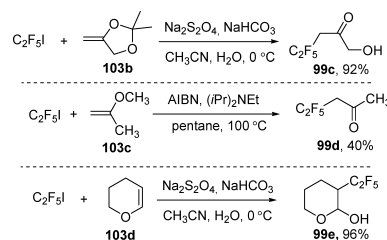
In a similar manner, the addition of PFEs to vinyl ethers provides a versatile approach to α -fluoroalkyl substituted carbonyl compounds or their synthetic equivalents. Thus, the perfluoroalkylation of ethyl vinyl ether (**103a**) gave a fistful of fluorinated aldehydes **104**, which were successfully transformed into α -CF₃-pyrazole,¹⁹¹ 2,3-dihydro-1*H*-1,4-diazepine (**105**),¹⁹² and 1,2-dihydro-2-methylpyrimidine (**106**)¹⁹³ by treatment with the corresponding binucleophiles (Scheme 73).

The addition of C₂F₅I to vinyl ethers **103b** and **103c** resulted in the formation of hydroxy ketone **99c**¹⁹⁴ and 4,4,5,5,5-pentafluoropenta-2-one (**99d**), respectively. The reaction of C₂F₅I with 3,4-dihydro-2*H*-pyran (**103d**) led to cyclic hemiacetal **99e** in almost quantitative yield (Scheme 74).¹⁹⁴

Scheme 73

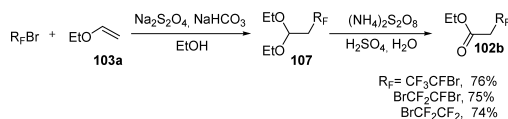


Scheme 74



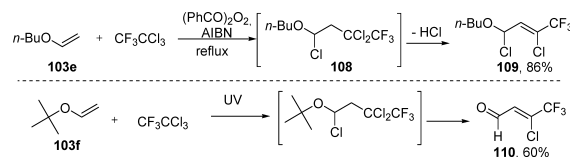
A number of fluorinated butyric acid esters **102b** were prepared in good yields using that type of addition. In the first step, the reaction of PFEs with ethyl vinyl ether **103a** led to acetals **107**, which were subsequently transformed into esters **102b** by persulfate oxidation (Scheme 75).¹⁹⁵

Scheme 75



The reaction of *n*-butyl vinyl ether (**103e**) with CF₃CCl₃ afforded stereoselectively fluorinated allyl ether **109** in high yield. Formed as an intermediate, ether **108** eliminates HCl to give alkene **109** as *Z*-isomer only.¹⁹⁶ In contrast, *tert*-butyl vinyl ether (**103f**) reacted with CF₃CCl₃ to result in unsaturated aldehyde **110**. In this case, additional elimination of *t*-BuCl took place (Scheme 76).¹⁹⁷

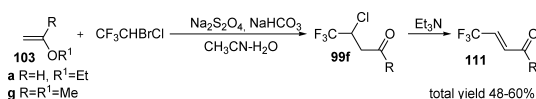
Scheme 76



The reaction of Halothane with ethyl vinyl ether (**103a**) and 2-methoxypropene (**103g**) in the presence of sodium dithionite opens an efficient and simple route to synthetically attractive α,β -unsaturated carbonyl compounds **111** bearing trifluoromethyl group at β -position. These products are important fluorinated building blocks for a preparation of fluorinated heterocycles. Radical addition of Halothane gives intermediate β -chlorosubstituted carbonyl compound **99f**. Subsequent treatment with triethylamine allows one to prepare target trifluoromethylated α,β -unsaturated carbonyl compounds **111** in good total yields (Scheme 77).¹⁹⁸

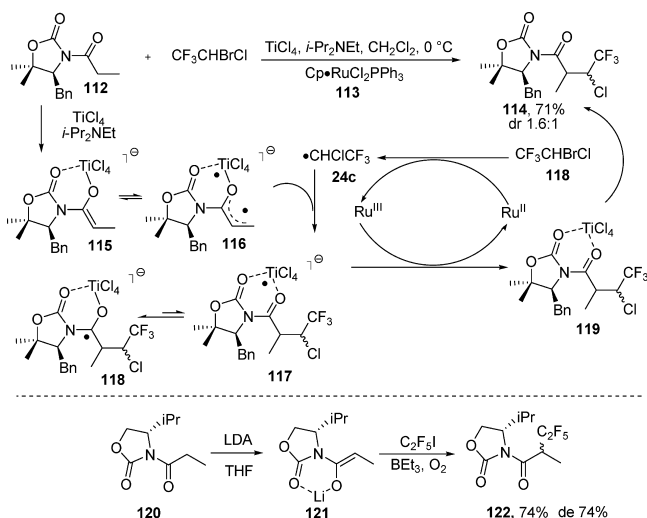
Enolates can also react with PFEs. Thus, amide **112** reacts with Halothane in the presence of TiCl₄ (Lewis acid), *i*-Pr₂NEt

Scheme 77



(base), and Ru(II) complex **113** (radical initiator) to form trifluoromethyl derivative **114** as a mixture of diastereomers. In the first step, deprotonation of **112** followed by enolization and coordination with TiCl_4 affords titanium enolate **115**. Enolate **115** exists in equilibrium with biradical form **116** due to valence tautomerism. Biradical **116** reacts with trifluoromethylchloromethyl radical **24c** formed from Halothane by Ru(II)-catalyzed single electron transfer (SET). Next, “titanium” radical **117**, existing with its valence isomer **118** (carbon “radical”), participates in SET with Ru(III) complex to give complex **119** and regenerated Ru(II) form of radical initiator. Decomposition of the complex **119** leads to the final product **114**.¹⁹⁹ Generated by deprotonation of **120** with LDA, enolate **121** forms the fluorinated amide **122** by treatment with $\text{C}_2\text{F}_5\text{I}$ under assistance of $\text{BET}_3\text{-O}_2$ radical initiating system (Scheme 78).²⁰⁰

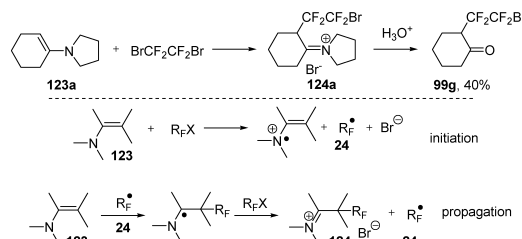
Scheme 78



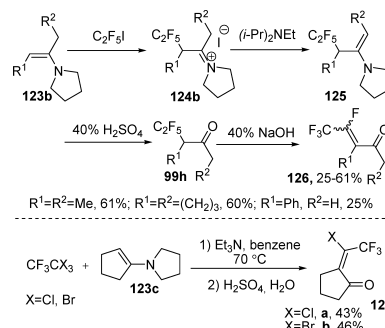
3.1.4. Reactions with Enamines and Ynamines. Being rather strong electron donors, enamines react with PFEs without additional activators. Electron transfer from enamine **123** to PFE leads to the formation of perfluoroalkyl radical **24** (via the intermediate radical anion), which starts the radical chain process to give radical **125**. The interaction of **125** with PFE affords iminium salt **124**, which is converted into ketone by acidic hydrolysis. Thus, fluorinated cyclohexanone **99g** was prepared in moderate yield starting from enamine **123a** and $\text{BrCF}_2\text{CF}_2\text{Br}$ (Scheme 79).²⁰¹

Contrary, polyfluoroalkylated enamines **125** can be obtained by basic workup of the intermediately formed iminium salts **124b**. Subsequent treatment of these enamines **125** with diluted sulfuric acid afforded the C_2F_5 -substituted ketones **99h**, which were converted into α,β -unsaturated β -trifluoromethyl enones **126** in moderate to good yields by a treatment with sodium hydroxide.²⁰² Similarly, in the case of CF_3CCl_3 and CF_3CBr_3 , the reaction with enamine **123c** in the presence of Et_3N gave α,β -unsaturated trifluoromethyl ketones **127a**, **127b** after acidic workup (Scheme 80).²⁰³

Scheme 79



Scheme 80



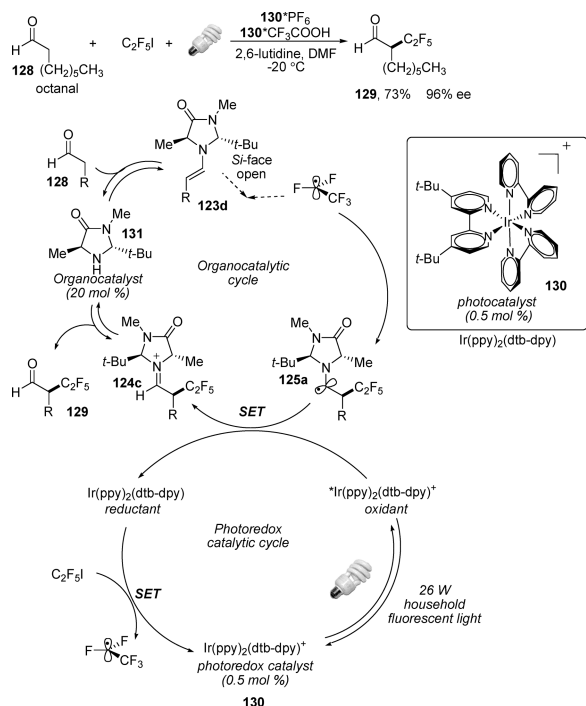
Enantioselective perfluoroalkylation of aldehydes via photoredox organocatalysis was reported by MacMillan. Using the combination of chiral imidazolidinone **131** as an amine catalyst and iridium complex **130** as a photocatalyst, $\alpha\text{-C}_2\text{F}_5$ -substituted aldehyde **129** was obtained in 73% yield and 96% ee. The reaction was performed in DMF under irradiation with household fluorescent 26 W lamp light via the mechanism including organocatalytic and photoredox catalytic cycles. As a first step, octanal (**128**) forms enamine **123d**, which reacts with perfluoroethyl radical to give radical particle **125a**, transforming into iminium salt **124c** by SET process (organocatalytic cycle). Hydrolysis of iminium salt **124c** gave aldehyde **129**. Perfluoroethyl radical is formed by SET from the iridium catalyst (photoredox catalytic cycle). The mechanism is shown below (Scheme 81). The reaction is general and permits also preparation of α -trifluoromethylated derivatives using trifluoromethyl iodide.²⁰⁴

An interesting $\text{Pd}(\text{PPh}_3)_4$ -catalyzed reaction of $\text{ClCF}_2\text{CF}_2\text{I}$ with triethylamine afforded $\text{CF}_2\text{CF}_2\text{Cl}$ -enamine **132** in moderate yield. The reaction proceeds through the formation of enamine **123e** by oxidation of NEt_3 with $\text{ClCF}_2\text{CF}_2\text{I}$. The subsequent reaction of **123e** with $\text{ClCF}_2\text{CF}_2\text{I}$ gave β -polyfluoroalkylated enamine **132** in moderate yield (Scheme 82).²⁰⁵

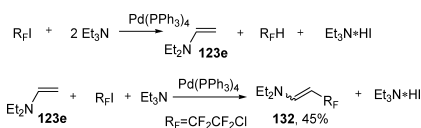
Enamines derived from methylene heterocyclic compounds **133** reacted with $\text{C}_2\text{F}_5\text{I}$ under irradiation to open an efficient access to trifluoromethylated cyanine dyes. The reaction with pentafluoroethyl iodide using 2 equiv of **133** and NaClO_4 gave β -perfluoroalkylimidocarbocyanine perchlorates **135** in moderate yields. The mechanism was not investigated in detail. However, it was proposed that the reaction occurs via an addition–elimination sequence, leading to intermediate **134**, which reacts with **133** and NaClO_4 to form compounds **135**. Similarly, starting from vinilogenous enamine **137** (prepared in situ from quinolinium salt **136**), quinoline derived dye **138** was obtained (Scheme 83).²⁰⁶

The reaction of ynamines **139** with PFEs led to polyfluoroethyl-substituted haloenamine **140** under UV-irradi-

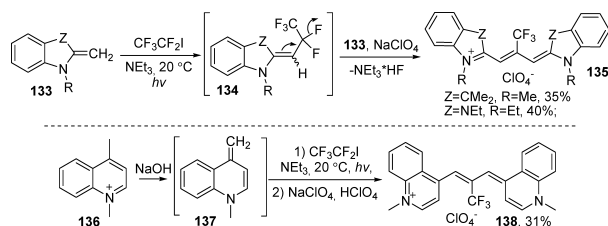
Scheme 81



Scheme 82

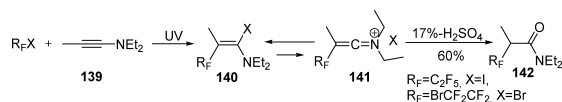


Scheme 83



ation. Because of isomerization of **140** into vinylidene iminium salts **141**, haloenamines **140** can be converted into polyfluoroethyl-substituted amides **142** in acidic media (Scheme 84).^{201,202}

Scheme 84



3.1.5. Reactions with Aromatics and Heteroaromatics.

Perfluoroethyl radicals are rather reactive toward electron-rich aromatic compounds. Thus, reaction of aniline with C_2F_5I , initiated by Zn/SO_2 couple or $Na_2S_2O_4$, led to polyfluoroethylated aniline as a mixture of *ortho*- and *para*-isomers **146**, **147** in a 1:1 ratio (Scheme 85, Table 15). The first step of the reaction is the formation of perfluoroethyl radical via SET mechanism. Next, the addition of the perfluoroethyl radical gives

Scheme 85

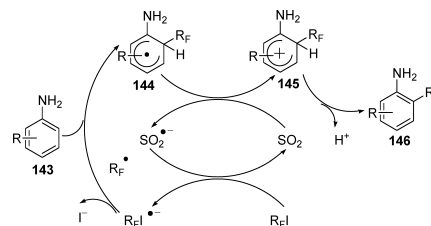


Table 15. Perfluoroalkylations of Anilines

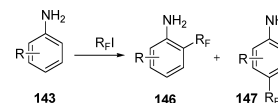
Aniline	R_FI	Conditions	Yield of 146 , %	Yield of 147 , %	ref
	C_2F_5I	$Zn, SO_2, DMF, 2\text{-methylpyridine}, Na_2S_2O_4, 65^{\circ}C$	7	7	207, 208
	C_2F_5I	$Na_2S_2O_4, Na_2HPO_4, DMF-H_2O, 65^{\circ}C$	21	16 ^a	208
	C_2F_5I	$Na_2S_2O_4, NaHCO_3, Bu_4N^+HSO_4^-, THF-H_2O, r.t., 2.5 h$	-	40	209
	C_2F_5I	$NaHSO_3, Zn, NaHCO_3, THF-H_2O, r.t., 3 h$	41	8	209
	C_2F_5I	$Na_2S_2O_4, NaHCO_3, Bu_4N^+HSO_4^-, t\text{-BuOMe-H}_2O, r.t., 8 h$	-	87	209
	C_2F_5I	$Na_2S_2O_4, NaHCO_3, Bu_4N^+HSO_4^-, THF-H_2O, r.t., 8 h$	-	94	209
	C_2F_5I	$Na_2S_2O_4, NaHCO_3, Bu_4N^+HSO_4^-, EtOAc-H_2O, r.t., 15 h$	18	-	209
	$ClCF_2CF_2I$	5–10 mol % $Ni(PPh_3)_4$, Dioxane, $80^{\circ}C, 6 h$	22	65	210

^a2,6-Bis(pentafluoroethyl)aniline (2%) and 2,4-bis(pentafluoroethyl)aniline (3%) were also isolated.

intermediate **144** (or the corresponding *para*-isomer), which transfers electron to SO_2 to form the cation **145** (σ -complex). Deprotonation leads to the final polyfluoroethylated aniline **146** (or the corresponding *para*-isomer **147**).

Using this reaction, a set of various anilines was perfluoroalkylated to afford the corresponding derivatives **146**, **147** in moderate to high yields. In addition to already mentioned initiating systems Zn/SO_2 and $Na_2S_2O_4$, $Zn/NaHSO_3$ and $Ni(PPh_3)_4$ were also used as promoters (Scheme 86, Table 15). It should be noted that yields are usually low due

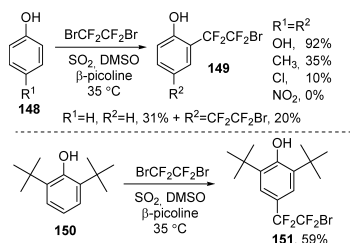
Scheme 86



to side reactions; however, when the amino group was blocked by substituents in *ortho*-positions, pentafluoroethylation proceeded very efficiently.

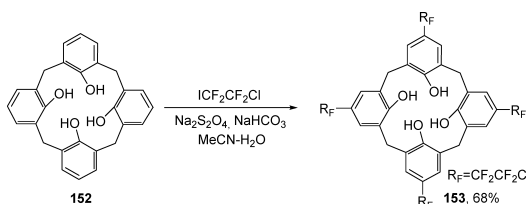
Polyfluoroalkylation of *para*-substituted phenols **148** by $BrCF_2CF_2Br$, activated by sulfur dioxide, afforded the corresponding *ortho*- CF_2CF_2Br derivatives **149**. In the case of phenol, the reaction led to a mixture of *ortho*- and *para*-isomers.²¹¹ Polyfluoroalkylation of 2,6-di-*tert*-butylphenol **150** resulted in the formation of highly substituted derivative **151** in 59% yield (Scheme 87).²¹²

Scheme 87



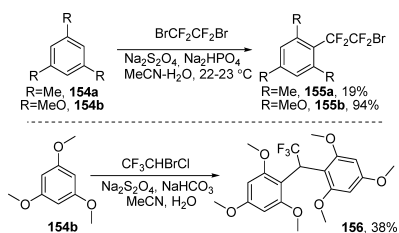
Aromatic rings of calix[4]arenes were also involved into the reaction with perfluoroethyl radicals. The corresponding tetrasubstituted *p*-perfluoroethylcalix[4]arene **153** was prepared in 68% yield by the reaction of calix[4]arene **152** with $\text{ClCF}_2\text{CF}_2\text{I}$ in the presence of sodium dithionite (Scheme 88).²¹³

Scheme 88



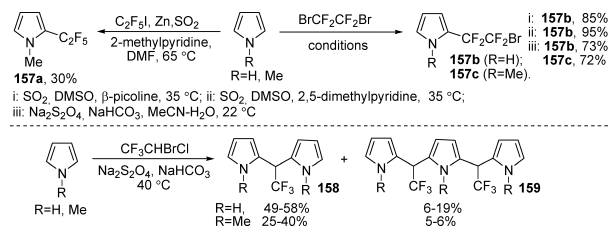
Mesitylene (**154a**) and trimethoxybenzene (**154b**) can also be perfluoroethylated using sodium dithionite as initiator. Thus, derivatives of 1,3,5-trimethyl- and trimethoxybenzenes **155** were prepared by the reaction of **154** with $\text{BrCF}_2\text{CF}_2\text{Br}$. However, in the case of trimethoxybenzene, the yield of the target product was much higher, which indicates that only very electron-rich aromatics can participate in this reaction.²¹⁴ In a similar condition, the reaction of 1-bromo-1-chloro-2,2,2-trifluoroethane with **154a** gave in 38% yield the corresponding diaryl-substituted trifluoroethane **156**, which is the result of the radical reaction of intermediate product of trimethoxybenzene alkylation with another molecule of **154a** (Scheme 89).²¹⁵

Scheme 89



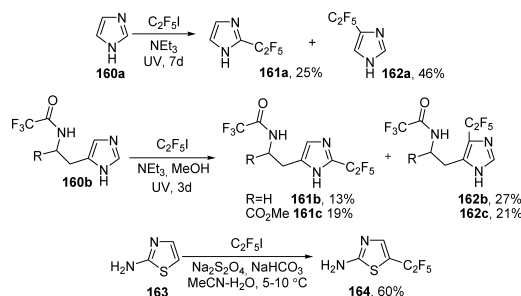
Heterocyclic compounds were also involved in perfluoroalkylation with PFEs. $\text{C}_2\text{F}_5\text{I}$ reacted with *N*-methylpyrrole to give 2- C_2F_5 -substituted pyrrole **157a** in 30% yield.²⁰⁸ Similarly, 2-(bromotetrafluoroethyl)pyrroles **157b**, **157c** were prepared by the reaction of pyrrole and 1-methylpyrrole with $\text{BrCF}_2\text{CF}_2\text{Br}$.^{211,214,216} Sodium dithionite-initiated reaction of pyrrole and 1-methylpyrrole with 1-bromo-1-chloro-2,2,2-trifluoroethane provided trifluoromethylated dipyrromethanes **158** as the main products in 40–58% isolated yields. Small amounts of tripyrroles **159** were also formed in these conditions (Scheme 90).²¹⁵

Scheme 90



Imidazole (**160a**) was smoothly perfluoroethylated with perfluoroethyl iodide to afford a mixture of 2- and 4-substituted isomers **161a**, **162a** in good yield.²¹⁷ Trifluoroacetamides of histamine and histidine **160b,c** gave similarly mixtures of 2- and 4-alkylated derivatives **161b**, **162b** and **161c**, **162c** in moderate yields.²¹⁸ In contrast, the reaction of aminothiazole **163** with $\text{C}_2\text{F}_5\text{I}$ gave only one pentafluoroethylated derivative **164** in which the C_2F_5 group located in 5-position in good yield and regioselectively.²¹⁹ The mechanism of the transformations includes radical addition as a key step. It should be noted that in all of these cases only C-alkylation took place (Scheme 91).

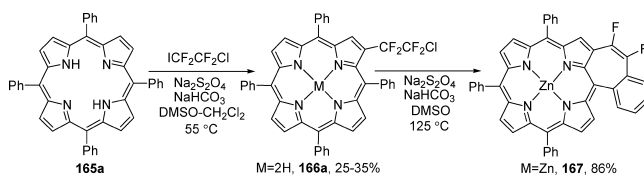
Scheme 91



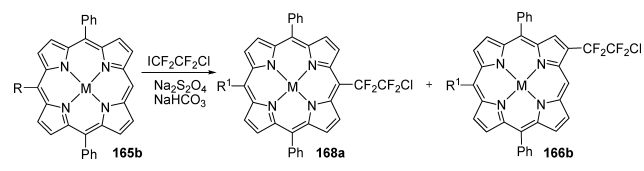
Porphyrins are another important type of macroheterocycles, which can be perfluoroalkylated to obtain electron-deficient porphyrin derivatives. In the case of tetrasubstituted porphyrin **165a**, the reaction proceeds in 25–35% yields²²⁰ to give $\text{C}_2\text{F}_4\text{Cl}$ -derivative **166a**.²²¹ The radical substitution is directed to β -position of a pyrrole ring of porphyrin if all meso-positions are substituted; however, in the case of meso-unsubstituted porphyrins, the reaction is directed as a rule to free meso-position. Zn(II) porphyrin derivative **166a** undergoes intramolecular radical cyclization at elevated temperature to form seven-membered condensed difluoride **167** in high yield (Scheme 92).²²¹

Bis(phenyl)-substituted porphyrins **165b** reacted with $\text{ICF}_2\text{CF}_2\text{Cl}$ to give polyfluoroethylated derivative **168a** or **166b** depending on temperature, solvents, and the ratio of reagents. Using that approach, porphyrins with two perfluoroalkyl groups were also obtained (Scheme 93, Table 16).²²¹

Scheme 92



Scheme 93

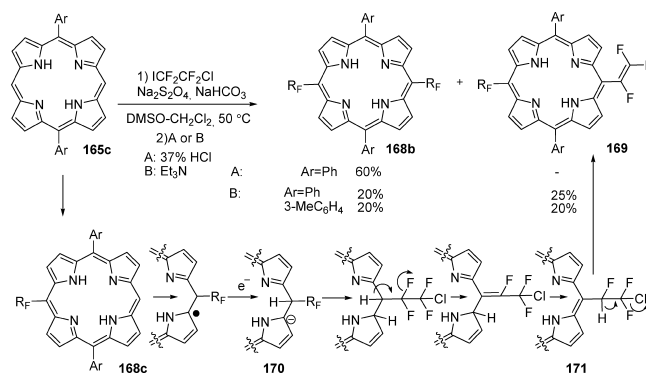


Unusual fluoroalkenylation during the radical reaction of PFEs with porphyrins was reported. The reaction of porphyrins **165c** (having two free meso-positions) with $\text{ICF}_2\text{CF}_2\text{Cl}$ followed by treatment with Et_3N gave a mixture of bis(chlorotetrafluoroethyl) derivative **168b** and the corresponding porphyrin derived trifluoroethylene **169** in good total yield. If acidic workup was used, trifluoroethylene **169** was not formed and derivative **168b** was isolated exclusively. The mechanism of trifluoroethylene **169** formation is depicted in Scheme 94.²²⁴ It includes reduction of monofluoroalkylated porphyrin **168c** to form carbanion intermediate **170**. Subsequent protonation–deprotonation sequence resulted in elimination and double bond shift to give finally **171** as a precursor for the formation of **169** (Scheme 94).

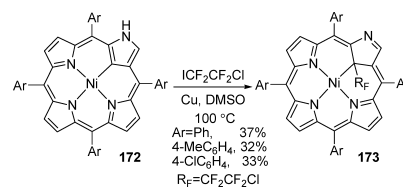
Ni(II) complexes of *N*-confused porphyrins **172** gave unusual fluorinated derivatives **173** in moderate yields by the reaction with $\text{ICF}_2\text{CF}_2\text{Cl}$ in the presence of copper (Scheme 95).²²⁵ In this case, the reaction resulted in selective formation of products having the fluoroalkyl group attached to the inner 21-carbon rather than to one of the peripheral carbon atoms because all meso-positions were occupied with aryl groups. No regioisomeric products were found. This regioselectivity gave information about possible participation of porphyrin coordinated nickel in this radical reaction.

3.1.6. Reactions with Fullerenes. Being a so-called “radical sponge”, fullerenes were efficiently involved in the reaction with PFEs. Thus, high-temperature reactions of C_{60} , C_{70} , and higher fullerenes (C_{76} – C_{96}) with $\text{C}_2\text{F}_5\text{I}$ afforded mixtures of broad spectra of isomeric $\text{C}_m(\text{C}_2\text{F}_5)_n$ adducts with n up to 12. Using HPLC separation of these mixtures, a manifold of pure adducts was isolated and characterized by MALDI, NMR, and Raman spectroscopy as well as X-ray analysis. In addition to mentioned adducts, epoxide $\text{C}_{60}(\text{C}_2\text{F}_5)_4\text{O}$ **174a** was isolated as an admixture. The formation of **174a** was the result of participation of oxygen or any oxygen-containing impurities in the reaction. Thermally induced reaction of $\text{ICF}_2\text{CF}_2\text{I}$ with fullerenes was also investigated. It was found that the addition of biradicals C_2F_4 and C_4F_8 , thermally generated from $\text{ICF}_2\text{CF}_2\text{I}$, led to a number of fullerene derivatives $\text{C}_{60}(\text{C}_2\text{F}_4)_n$ and $\text{C}_{70}(\text{C}_2\text{F}_4)_m$, where n and m ranged from 1 to 18 and from 1 to 14 correspondingly. On longer reaction times, C_{60} was nearly quantitatively consumed, and the fullerene-based yield was 90–95%. Compositions of the reaction mixtures and

Scheme 94



Scheme 95



formulas of isolated products are given in Table 17 and Figures 3 and 4.^{226–233}

3.1.7. Other Radical Reactions Leading to C–C Bond Formation. Bromotetrafluoroethyl radical, generated from $\text{BrCF}_2\text{CF}_2\text{Br}$ under UV-irradiation, reacted with [1.1.1]-propellane (**177**), containing very weak C–C bond (estimated energy as 59–65 kcal/mol), to give a mixture of adducts **178a**, **178b**, **178c** in the ratio 11:2:1 in 80% total yield (Scheme 96).²³⁴ Similarly, perfluoroethyl radical, generated from $\text{C}_2\text{F}_5\text{I}$, reacts with cyanogen (dicyane) to form a mixture of perfluorobutane and iodocyanide in 60% yield.²³⁵ The reaction of $\text{ICF}_2\text{CF}_2\text{I}$ with diazomethane led to insertion of methylene fragments into C–I bonds resulting in the synthesis of 3,3,4,4-tetrafluoro-1,6-diiodohexane **179** in 20% yield.²³⁶

Initiated by $\text{Na}_2\text{S}_2\text{O}_4$, the addition of $\text{Cl}(\text{CF}_2)_2\text{I}$ to arylisocyanides **180** afforded a series of fluorinated imidoyl iodides **181** in high yields.²³⁷ Surprisingly, as compared to their CF_3 -analogues, imidoyls **181** are stable enough to be stored and handled at room temperature (Scheme 97).

3.1.8. Other Radical Reactions Leading to Formation of C–Heteroatom Bond. $\text{C}_2\text{F}_5\text{I}$ reacts with tellurium and selenium in the presence of equimolar amounts of copper to give bis(perfluoroethyl)selenium **182a** or bis(perfluoroethyl)tellurium **182b** in high yields with admixtures of diselenium **183a** or ditelluride **183b** correspondingly (Scheme 98).²³⁸

The reaction of $\text{ICF}_2\text{CF}_2\text{I}$ with S_2F_{10} (extremely toxic compound) on heating in the presence of tetrafluoroethylene (added to prevent $\text{ICF}_2\text{CF}_2\text{I}$ thermal decomposition) gave SF_5 -

Table 16. Reactions of Porphyrins with PFEs

M	R	R ¹	conditions	168a/166b	total yield, %	ref
Zn(II)	H	H	DMSO, THF, 35 °C	1/0	40	222
Zn(II)	H	H	DMSO, CH_2Cl_2 , 40 °C, 1.1 equiv of 165b /1.1 equiv of $\text{Na}_2\text{S}_2\text{O}_4$ /1 h	4/1	50	223
Zn(II)	H	H	DMSO, CH_2Cl_2 , 40 °C, 3 equiv of 165b /3 equiv of $\text{Na}_2\text{S}_2\text{O}_4$ /5 h	1/1	40	223
Zn(II)	H	H	DMSO, CH_2Cl_2 , 40 °C, 1.1 equiv of 165b /1.1 equiv of $\text{Na}_2\text{S}_2\text{O}_4$ /2 h	3/1	60	223
Zn(II)	H	$\text{CF}_2\text{CF}_2\text{Cl}$	DMSO, 40 °C, 3 equiv of 165b /3 equiv of $\text{Na}_2\text{S}_2\text{O}_4$ /1 h	1/1	40	223
Zn(II)	$\text{CF}_2\text{CF}_2\text{Cl}$	$\text{CF}_2\text{CF}_2\text{Cl}$	DMSO, 40 °C, 1 equiv of 165b /1 equiv of $\text{Na}_2\text{S}_2\text{O}_4$ /1 h	1/1	80	223

Table 17. Reaction of Fullerenes with PFEs

PFE	substrate	temp, °C	compositions of reaction mixtures and isolated products	ref
C ₂ F ₅ I	C ₆₀	430 ^a	C ₆₀ (C ₂ F ₅) _{4–10}	226
C ₂ F ₅ I	C ₆₀	380–440	C ₆₀ (C ₂ F ₅) ₄ O (174a) C ₆₀ (C ₂ F ₅) ₆ (one isomer) (174b) C ₆₀ (C ₂ F ₅) ₈ (five isomers) (174d–g) C ₆₀ (C ₂ F ₅) ₁₀ (two isomers 174j, 174k)	227
C ₂ F ₅ I	C ₆₀	400–430	C ₆₀ (C ₂ F ₅) ₆ (174c) C ₆₀ (C ₂ F ₅) ₈ (174j)	228
C ₂ F ₅ I	C ₇₀	360	C ₇₀ (C ₂ F ₅) ₈ mostly C ₇₀ (C ₂ F ₅) ₁₀ (seven isomers 175a–g) C ₇₀ (C ₂ F ₅) ₁₂	229
C ₂ F ₅ I	C ₇₀	360	C ₇₀ (C ₂ F ₅) ₁₀ (nine isomers 175a–i)	230
C ₂ F ₅ I	a mixture C ₇₆ –C ₉₆ and small amounts of C ₆₀ and C ₇₀	250	C ₆₀ (C ₂ F ₅) _n C ₇₀ (C ₂ F ₅) _m C ₇₈ (C ₂ F ₅) ₁₀ (two isomers 176a,b)	231
C ₂ F ₄ I ₂	C ₆₀	400–450	C ₆₀ (C ₂ F ₄) _n , n = 1–18 C ₆₀ (C ₄ F ₈) ₂ (two isomers 174l,m)	232
C ₂ F ₄ I ₂	C ₇₀	400–450	C ₆₀ (C ₄ F ₈) ₆ (174n) C ₇₀ (C ₂ F ₄) _m , m = 1–14 C ₇₀ (C ₂ F ₄) ₂ (175j)	232
C ₂ F ₄ I ₂	C ₇₀	380–420	C ₇₀ (C ₂ F ₄) _m , m = 1–16 C ₇₀ (C ₂ F ₄) (175l) C ₇₀ (C ₄ F ₈) (175m) C ₇₀ (C ₄ F ₈) ₂ (175k)	233

^aThe reaction was carried out using Cu powder as an promoter.

substituted tetrafluoroethyl iodide **184** in 49% yield as a major product. An admixture of SF₅-derived perfluorobutyl iodide **185** was also formed.²³⁹ Similarly, the reaction of ICF₂CF₂I with disulfides **188** afforded fluorinated thioethers **187** in good yields at high pressure (Scheme 99).^{239a}

The reaction of ClCF₂CFCl₂ with diphenyl disulfide **186b** was also performed under Na₂S₂O₄ activation. Sulfide **188** formed via radical reaction was isolated in 52% yield.²⁴⁰ The pentafluoroethylation of homocystine (**186d**), performed by C₂F₅I in the presence of sodium in liquid ammonia, gave pentafluoromethionine analogue **189** in good yield (Scheme 100).²⁴¹ Similar reaction of diselenide **190** afforded C₂F₅-selenide **191**. Initial formation of perfluoroethyl radical was achieved by the reaction with sodium hydroxymethanesulfinate (rongalite).²⁴²

The photochemical oxidation of 1,1,1-trichloro-2,2,2-trifluoroethane was performed by molecular oxygen to give trifluoroacetic acid. A number of catalysts were tested in the reaction. Fe/C, Cu/C, Zn/C, Sn/C, AlF₃, Fe/Al₂O₃, C, ZrO₂, and other systems were investigated. The best results were obtained under Fe/C (20 wt %) catalysis. The reaction proceeds via intermediate formation of trifluoroacetyl chloride, which is hydrolyzed into trifluoroacetic acid. The mechanism of the transformation is depicted in Scheme 101 and includes as a key step the formation of CF₃CCl₂ radical.²⁴³

Similar photochemical oxidation of CF₃CCl₂H leads to trifluoroacetyl chloride in quantitative yield.²⁴⁴ Depending on

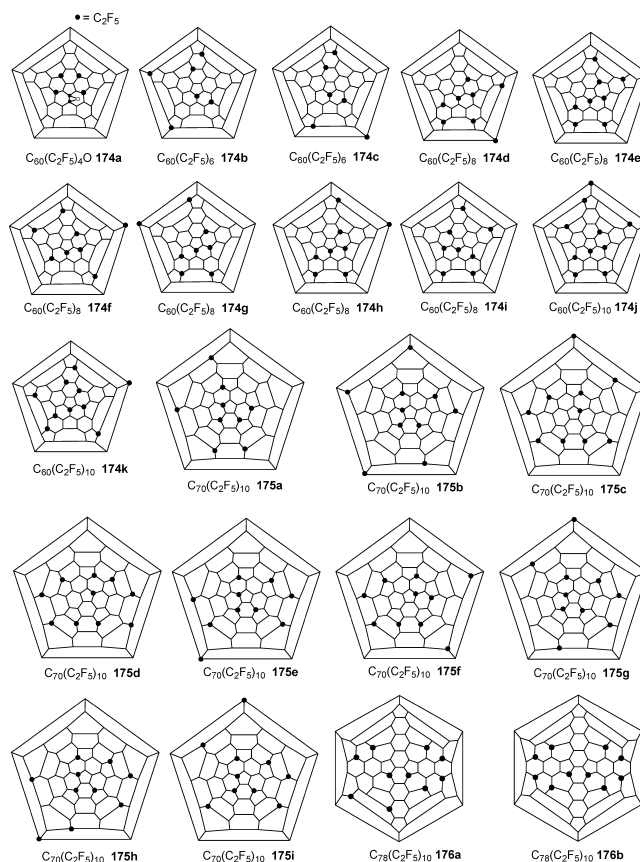


Figure 3. Shlegel diagrams of fullerene derivatives.

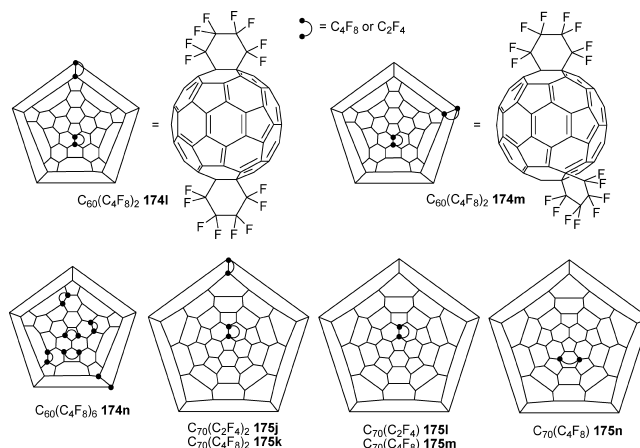
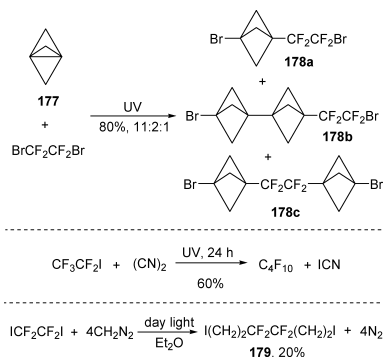


Figure 4. Condensed fullerene derivatives.

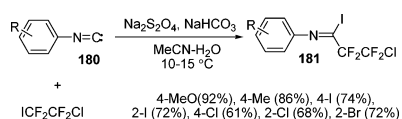
the reaction conditions, the oxidation of CF₃CClH₂ can afford trifluoroacetyl chloride or trifluoroacetic acid. Thus, thermal oxidation in the presence of catalytic amounts of chlorine leads to trifluoroacetic acid. In photochemical conditions, a mixture of trifluoroacetyl chloride with trifluoroacetic acid is formed.²⁴⁵ Photochemical oxidation in the presence of steam gives trifluoroacetic acid (Scheme 102).^{243a}

A convenient method for the preparation of various polyfluoroacetic acids was elaborated by Hu. It was found that hydrolysis of PFE into polyfluoroacetic acids can be carried out under mild conditions in the presence of redox system-(NH₄)₂S₂O₈/HCO₂Na in DMF. The mechanism of the transformation includes generation of polyfluoroalkyl radical

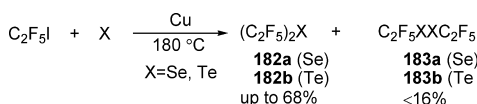
Scheme 96



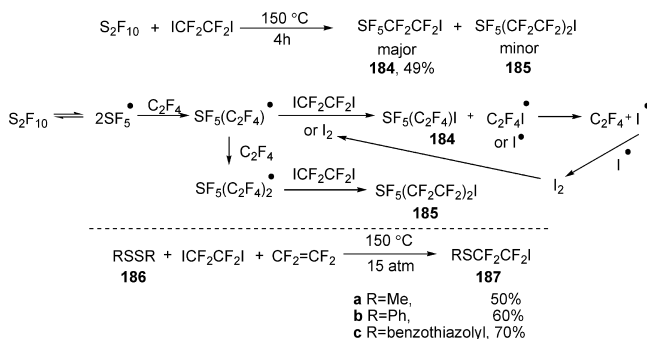
Scheme 97



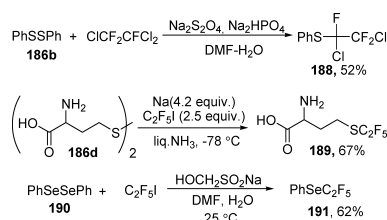
Scheme 98



Scheme 99

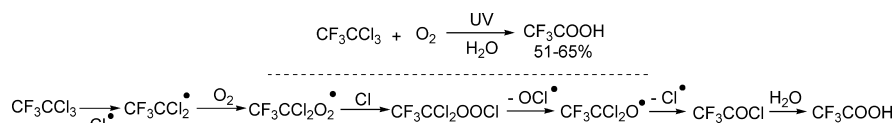


Scheme 100

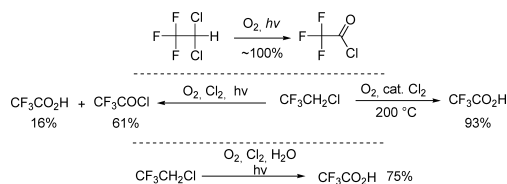


24 followed by reaction with oxygen to form hydroperoxide **192**, which transforms into polyfluoroacetyl halide by the extrusion of HOX. The last step is a reaction of

Scheme 101

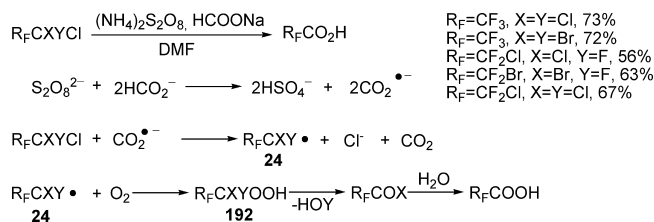


Scheme 102



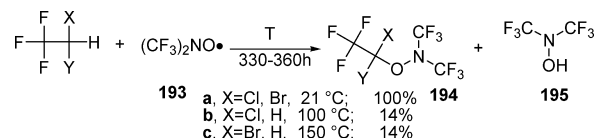
polyfluoroacetyl halide with water to form the corresponding polyfluoroacetic acids (Scheme 103).²⁴⁶

Scheme 103



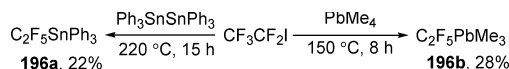
Reaction of some polyfluoroethanes with bis(trifluoromethyl)nitroxide **193** was investigated by Banks.²⁴⁷ It was found that this reaction is very sensitive to the nature of PFE. For instance, in the case of Halothane, the reaction proceeds at room temperature to give fluorinated ether **194a** and bis(trifluoromethyl)hydroxylamine **195** quantitatively. At elevated temperature, less reactive CF₃CH₂Cl and CF₃CH₂Br gave products of *O*-alkylation **194b,c** in low yield (Scheme 104).

Scheme 104



Perfluoroethyl derivatives of tin **196a** and lead **196b** were prepared in low yields by the radical reaction of C₂F₅I with Ph₃SnSnPh₃ and PbMe₄ correspondingly (Scheme 105).²⁴⁸

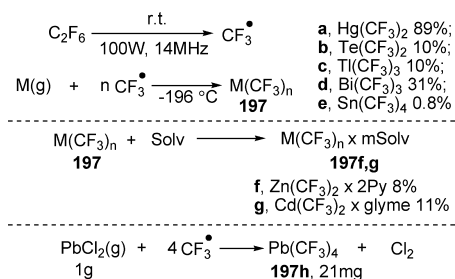
Scheme 105



Using metal vapor synthesis (MVP) technique, an interesting synthesis was carried out by Lagow. Generated in a radio frequency glow discharge, free trifluoromethyl radical generated from hexafluoroethane was co-condensed with metal vapor on a coldfinger at −196 °C to give σ-bonded metal trifluoroalkyl compounds **197**.²⁴⁹ In the case of cadmium and zinc, reaction products **197** were treated with glyme and pyridine to give stable complexes **197f,g**.²⁵⁰ As a proof of principle, inorganic

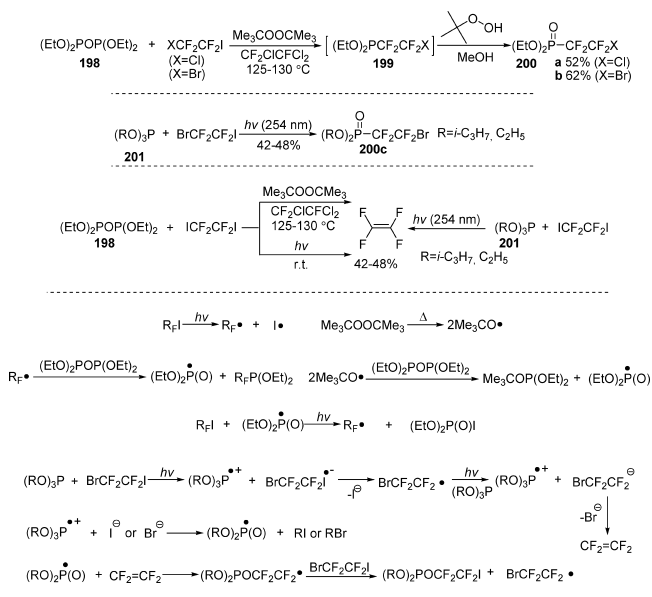
salts were also involved in the transformation. Thus, lead(II) chloride afforded tetrakis(trifluoromethyl)lead **197h** in trace amounts (Scheme 106).²⁵¹

Scheme 106



The reaction of BrCF₂CF₂I and ClCF₂CF₂I with (EtO)₂POP(OEt)₂ **198** in the presence of di-*tert*-butyl peroxide gave phosphonites **199**, which were easily oxidized into phosphonates **200a,b**. Alternatively to the thermally induced radical reaction, a photochemical pathway was also elaborated. The reaction of BrCF₂CF₂I with trialkyl phosphites **201** afforded phosphonates **200c** in 42–48% yields. In contrast, reaction of BrCF₂CF₂I with (RO)₃P **201** gave tetrafluoroethylene only under heating or with (EtO)₂PONa (without radical initiators or UV-irradiation). Reactions of ICF₂CF₂I with both tetraethylpyrophosphate **198** and trialkyl phosphites **201** gave tetrafluoroethylene as well.²⁵² A possible mechanism of the reactions includes formation of radical particles and can be rationalized as shown in Scheme 107.

Scheme 107



3.2. Reactions of PFE, Involving C-2 Anionic Species as Intermediates

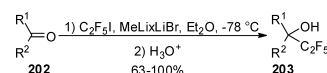
3.2.1. Polyfluorinated C-2 Lithium Compounds.

3.2.1.1. Formation of C–C Bond. Fluorinated organometallic compounds were shown to be very versatile reagents for organic synthesis. At the end of last century, some good general reviews dealing with their synthesis and reactivity were published.²⁵³ This type of reactivity is a very significant part of the PFE chemistry; therefore, we decided to include the

most important features of organometallic chemistry of PFE in this Review, to show the whole synthetic scope of fluorinated ethanes and rapid development of this field. A lot of new data appeared during the last years. Lithium, magnesium, zinc, and copper derivatives were mostly dealt with, although reactions involving mercury, germanium, lead, gold, cobalt, iridium, rhodium, platinum, and palladium species were also described.

The treatment of C₂F₅I with methyl lithium at –78 °C gives rise to perfluoroethyl lithium, which is moderately stable only at low temperatures, and one should use 5–8-fold excess of that reagent. The addition of perfluoroethyl lithium to carbonyl compounds **202** leads to the corresponding fluorinated alcohols **203** in high yields (Scheme 108, Table 18).

Scheme 108

Table 18. Reaction of C₂F₅Li with Carbonyl Compounds

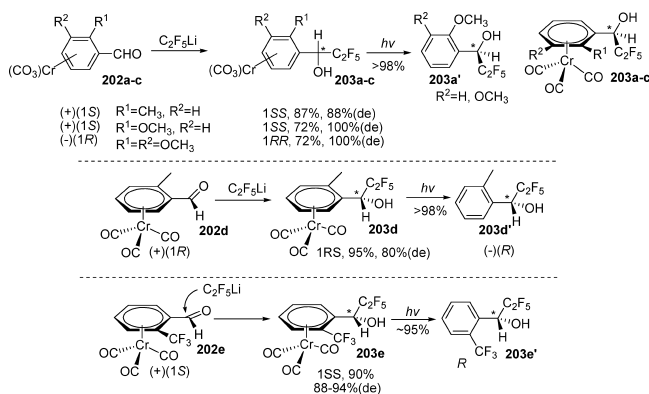
Product	Yield, %	Ref.	Product	Yield, %	Ref.	Product	Yield, %	Ref.
	91	254		100	254		84	254
	88	254		81	254		81	255
	98	256, 257, 258		93	254		88	259
	73	256, 258		56	260		63	259
	78	256, 258		100	261		81	262
	86	254		81	263		91(Ph) 52(Bn)	264

Stereoselective synthesis of C₂F₅-substituted carbinols by the addition of perfluoroethyl lithium was performed using Cr(CO)₃ complexes of some benzaldehydes **202a–e**. The treatment of optically pure complexes **202a–e** with Li/Na alloy and C₂F₅I gave the corresponding carbinols **203a–e** in good yields and high to excellent diastereoselectivity, which is a result of nucleophilic attack of pentafluoroethyl lithium to the aldehyde moiety from the upper side of **202a–e** (the other side is blocked by bulky Cr(CO)₃ fragment). The irradiation of the latter products **203a,d,e** in Et₂O/CHCl₃ led to pure enantiomers of alcohols **203a',d',e'** in 95–98% yield (Scheme 109).^{258,265}

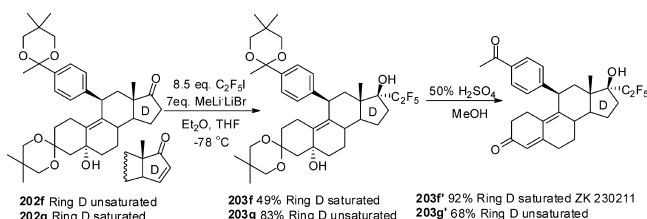
The addition of perfluoroethyl lithium to steroidal ketones **202f,g** afforded diastereoselectively alcohols **203f,g** (α-face addition), which gave ketones **203f',g'** after acidic treatment (Scheme 110). Ketone **203f'** (compound ZK 230211) is a highly potent progesterone receptor antagonist, combining high antiprogesterogenic activity with little or no other endocrinological effects. The pharmacological properties of ZK 230211 may prove useful in the treatment of endometriosis, leiomyomas, breast cancer, and in hormone replacement therapy.²⁶⁶

In a similar manner, steroidal compounds **203h** and **203i** were obtained using C₂F₅Li addition to ketones **202h** and **202i**. Again, the highly diastereoselective addition of the organolithium compound was observed. The prepared compounds

Scheme 109

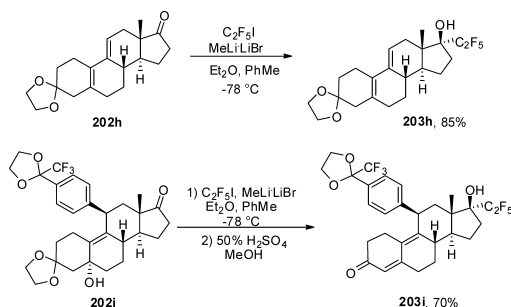


Scheme 110



were found to be partial agonists (mesoprogesterins) with significant applications for antiproliferative and antiovarian treatment strategies in gynecological therapy such as uterine fibroids, endometriosis, and heavy menstrual bleeding (Scheme 111).²⁶⁷

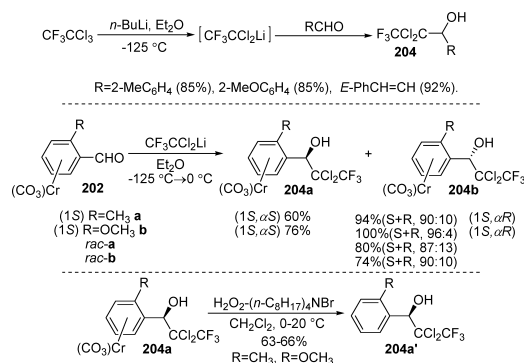
Scheme 111



CF₃CCl₂Li is a very unstable organolithium compound (much more unstable than pentafluoroethyl lithium), which can be handled only at temperatures below −95 °C. Only a few examples of addition to a carbonyl group using CF₃CCl₂Li were reported. Fluorinated alcohols **204** were prepared by the reaction of this organolithium compound with aldehydes. As in the case of C₂F₅Li, addition to chiral Cr(CO)₃-ArCHO complexes **202** proceeded stereoselectively to give mostly only one diastereomer **204a**, which was easily transformed into enantiomerically pure alcohol **204a'** by the oxidation with hydrogen peroxide (Scheme 112).²⁶⁸

Alternatively, perfluorinated ethyllithiums can be prepared by the deprotonation of appropriate PFEs. For example, C₂F₅Li can be obtained by the direct lithiation of pentafluoroethane by *n*-butyl lithium or other strong bases. This technique allows one to prepare salt-free solution of pentafluoroethyl lithium; contrary preparation of this organolithium compound using

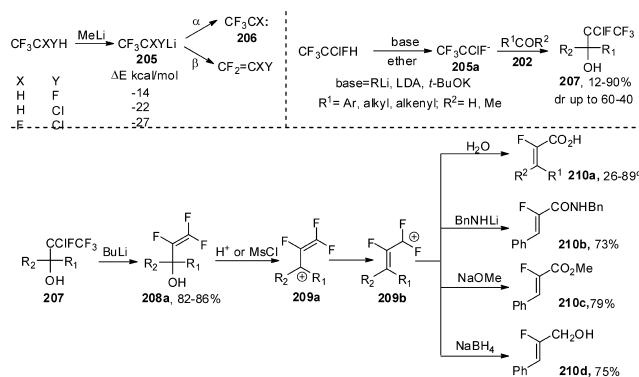
Scheme 112



halogen-lithium exchange gives the complex of this reagent with lithium chloride, bromide, or iodide.²⁶⁹

The deprotonation reaction of fluorinated ethanes CF₃CXYH with MeLi to form the corresponding carbanions (organolithium compounds) **205** was studied theoretically. All of the deprotonation reactions are favored energetically (Δ*E* = 14–27 kcal/mol in ether and THF). Calculations suggest that the formed organolithium compounds **205** have preference for β-elimination over α-elimination. As a result, fluorinated alkenes CF₂ = XY are preferred products in this reaction and carbenes **206** should form quite rarely (Scheme 113).²⁷⁰ These

Scheme 113

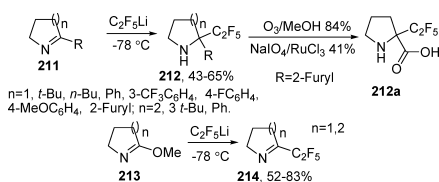


computations showed the possibility to easily deprotonate CF₃CIFH and to use anion **205a** as the fluorinated nucleophile. As a result, 1-chloro-1,2,2,2-tetrafluoroethane can be deprotonated quite easily using various organolithiums, LDA, and even with *t*-BuOK under low temperature (−80 to 0 °C). Fluorinated anion **205a** prepared by this protocol reacts with aldehydes and acetophenone **202** to lead to the corresponding alcohols **207** bearing a CF₃CClF group usually in high yield. The reaction is not selective. Generally two diastereomeric products are formed in 1/1 ratio. CF₃CClF moiety of prepared carbinols **207** can be transformed quite effectively to form another important fluorinated products. For instance, the reaction of **207** with *n*-BuLi gave trifluoroallylic alcohols **208a** by dechlorofluorination reaction (Cl–Li exchange followed by elimination). These fluorinated alcohols were treated with sulfuric acid or transformed to the corresponding mesylates to open a simple route for the preparation of α-fluorinated unsaturated acids **210a** via intermediate formation of allylic cation species **209a**, **209b**. Also, this technique was effectively used for the preparation of other α-fluorocinnamic acid

derivatives **210b**, **210c** and α -fluoroallylic alcohol **210d** (Scheme 113).

Perfluorinated ethyl lithiums have reduced electrophilicity due to electron-withdrawing action of fluorines (or other halogen atoms). Nevertheless, they are sufficiently active to react with other less reactive electrophiles. For example, the reaction of pentafluoroethyl lithium with cyclic ketimines **211** leads to pentafluoroethyl-substituted pyrrolidines, piperidines, and azepanes **212**. This approach was used to develop a simple and convenient pathway to α -pentafluoroethylproline, a prospective candidate for the synthesis and modification of fluorine-containing peptides. A key step of the synthesis of α -pentafluoroethylproline (**212a**) is the addition of C_2F_5Li in the presence of boron trifluoride to 2-furypyrroline followed by oxidative cleavage of furyl group to carboxylic moiety with ozone or using NaO_4 – $RuCl_3$ system.^{269a} The reaction of C_2F_5Li with lactams **213** opens a simple route for the preparation of pentafluoroethylated cyclic imines **214**,^{269a} which were shown to be versatile building blocks for the synthesis of a broad range of interesting fluorinated molecules (Scheme 114).^{269a,271}

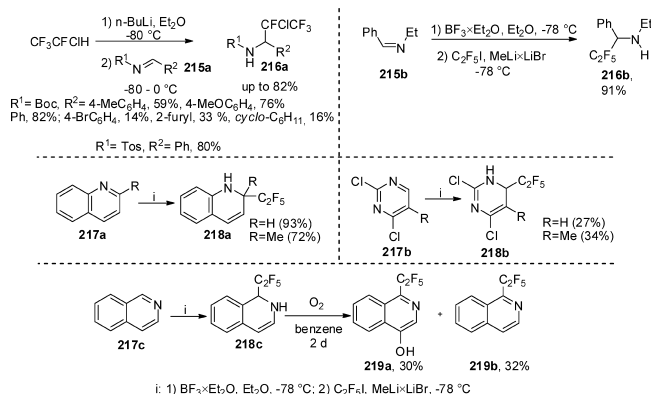
Scheme 114



N-Boc- and *N*-tosyl-substituted imines **215a** are a more reactive type of electrophiles due to the presence of EWGs activating $C=N$ fragment toward an attack of electrophiles. These types of imines were easily involved in the reaction with $CF_3CFClLi$ (prepared by the lithiation of CF_3CFClH with *n*-BuLi) to give the corresponding fluorinated amines **216a**.²⁷² The addition to the less reactive double $C=N$ bonds was performed under $BF_3 \times Et_2O$ activation. Thus, the reaction of pentafluoroethyl lithium with imine **215b** gave fluorinated secondary amine **216b**.²⁷³ Similarly, the addition of pentafluoroethyl lithium to the $C=N$ bond of various azines **217** led to the corresponding derivatives of quinoline **218a**,²⁷⁴ isoquinoline **218b**,²⁷⁵ and pyrimidine **218c**.²⁷⁶ Subsequent oxidation of pentafluoroethylated dihydroisoquinoline **218c** with air gave a 1/1 mixture of 1-pentafluoroethyl-4-hydroxyisoquinoline **219a** and aromatic product **219b** (Scheme 115).

Addition of perfluoroethyl lithium to derivatives of carboxylic acids **220** was investigated intensively. The reaction proceeds through the formation of the intermediate lithiated hemiketals **221**, the thermal stability of which defines the selectivity of the reaction. In the case of relatively unstable ones, extrusion of lithium alkoxide (in the case of esters) or LiX in general case is observed to form fluorinated ketones **222**. Subsequent reaction of **222** with excessive C_2F_5Li resulted in bis(C_2F_5)-substituted carbinols **223**. Alternatively, relatively stable hemiketals **221** gave ketones **222** after workup of the reaction mixture. As a result, in the case of esters, anhydrides, and acids themselves, the reaction pathway is very sensitive for both nature of substrate **220** and the reaction conditions. For example, methyl ester of 5-phenylpentanoic acid gave a mixture of ketone **222a** and carbinol **223a** if the reaction was carried out for 0.5 h, while only alcohol **223a** was isolated in quantitative yield when the

Scheme 115



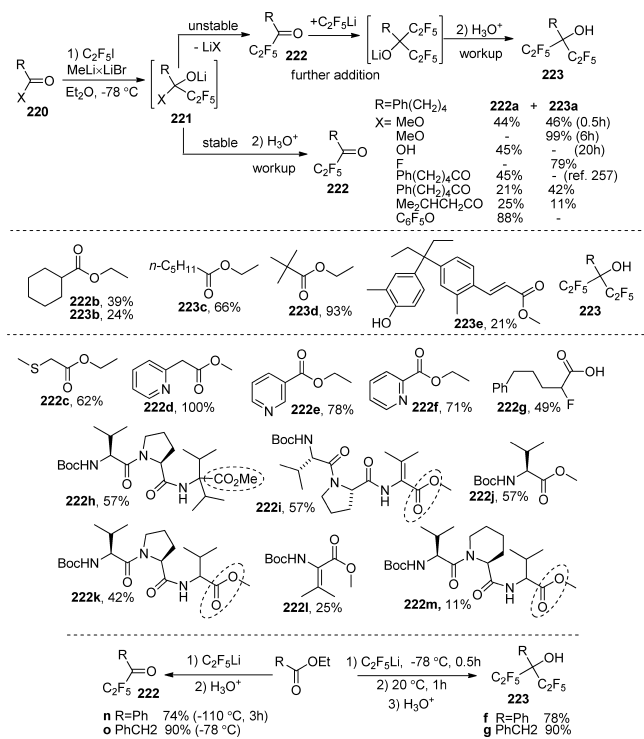
reaction mixture was kept for 6 h before workup.²⁵⁶ Pentafluorophenyl ester and 5-phenylpentanoic acid itself afforded ketone **222a** selectively. In contrast, the fluoroanhydride of this acid gave exclusively carbinol **223a**.²⁵⁶ Anhydride of 5-phenylpentanoic acid and mixed anhydride with 3-methylbutanoic acid reacted with C_2F_5Li low selectively to give a mixture of ketone **222a** and carbinol **223a**.^{256,257} A series of aliphatic esters as well as unsaturated ester led to carbinols **223b–e**.^{254,277} Several substrates, containing electron-withdrawing substituents (pyridyl, fluorine) and methylthio group as well as esters of peptides and amino acids, selectively afforded ketones **222c–m**.^{254,256,278–280} In the case of ethyl benzoate and ethyl phenylacetate, ketone **222** or carbinol **223** can be obtained selectively depending on the reaction conditions used. Thus, if the reaction was carried out and quenched at low temperature, the corresponding ketones **222n** and **222o** were isolated in high yields. If the reaction was allowed to warm to room temperature and stirred for 1 h before workup, carbinols **223f,g** were obtained (Scheme 116).²⁵⁴

Total control of selectivity was achieved by using tertiary amides and Weinreb amides to give ketones **222a,g,p–z,a'** exclusively (Scheme 116, Table 19). No formation of carbinols **223** was observed for the reaction with these carboxylic acid derivatives. Therefore, if the ketone is the desired product, amides should be used as substrates for this reaction. If the desired product is carbinol, the probable solution is to use substrates with rather good leaving groups (for example, esters) and allow the reaction mixture to warm to room temperature during 1 h before workup.

The treatment of isopropylidene protected optically active tartaric acid dichloride **220a** with an excess of C_2F_5Li led to the TADDOL analogue, diol **223h** bearing four C_2F_5 -groups. In contrast, the reaction of isopropylidene protected *meso*-tartaric anhydride **220b** with C_2F_5Li afforded unexpectedly product **223i** having only two pentafluoroethyl groups as a racemic mixture. The reaction proceeded through the formation of *meso*-configured bis-alkoxide diolate **221a**, which transforms into **223i** by transacetalization during acidic workup. The latter process is facilitated by the *cis* orientation of the hydroxyl groups. Stereoselective formation of **223i** with such configuration can be explained by the blocking effect of the acetonide fragment directing the attack of the nucleophile from the opposite face (Scheme 117).²⁸⁷

The addition of pentafluoroethyl lithium to methyl acrylate opens a simple route to the α,β -unsaturated C_2F_5 -ketone **222b'**, which was trapped by the reaction with allyl isothiuronium salt **224** to yield the corresponding derivatives of tautomeric

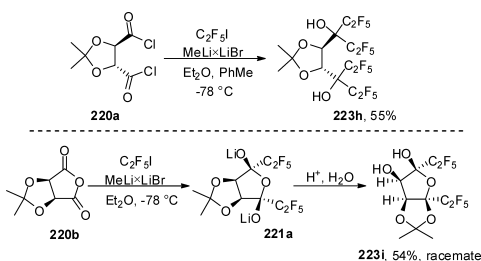
Scheme 116

Table 19. Synthesis of Ketones 222 Using Addition of $\text{C}_2\text{F}_5\text{Li}$ to Amides and Weinreb Amides

R	Yield of 222, %	ref.	R	Yield of 222, %	ref.	R	Yield of 222, %	ref.
$\text{Ph}(\text{CH}_2)_4$	98-99	[25]	$\text{Ph}(\text{CH}_2)_4$	84	[26]	$\text{Ph}(\text{CH}_2)_4$	68	[25]
$\text{Ph}(\text{CH}_2)_4$	84-88	[26]	$\text{Ph}(\text{CH}_2)_4$	68	[26]	$\text{Ph}(\text{CH}_2)_4$	91	[26]
$\text{Ph}(\text{CH}_2)_4$	74	[27]	$\text{BocHN}(\text{CH}_2)_4$	63	[28]	$\text{Ph}(\text{CH}_2)_4$	44	[25]
$\text{BocHN}(\text{CH}_2)_4$	79	[29]	$\text{BocHN}(\text{CH}_2)_4$	71	[30]	$\text{Ph}(\text{CH}_2)_4$	34	[25]
$\text{BocHN}(\text{CH}_2)_4$	38	[31]	$\text{BocHN}(\text{CH}_2)_4$	80	[32]	$\text{Ph}(\text{CH}_2)_4$	90	[25]
$\text{BocHN}(\text{CH}_2)_4$	70-85	[33]	$\text{BocHN}(\text{CH}_2)_4$	given	[34]	$\text{BocHN}(\text{CH}_2)_4$	92	[35]

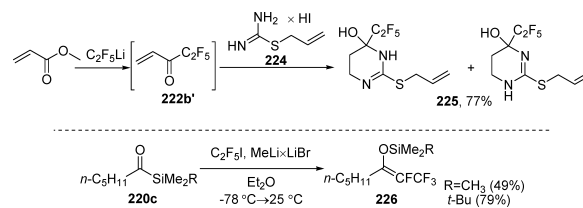
^aYield of alcohol, obtained by the NaBH_4 reduction of ketone.

Scheme 117



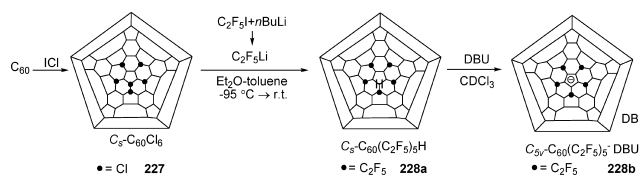
tetrahydropyrimidines 225.²⁸⁸ The reaction with perfluoroethyl lithium afforded perfluorinated silyl enolate 226 via elimination of fluoride in the case of the reaction with acylsilanes 220c (Scheme 118).²⁸⁹

Scheme 118



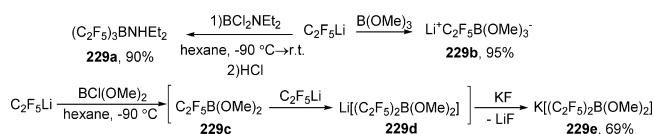
In search of fullerene-based superacids, Strauss and Boltalina performed the synthesis of polyfluoroethylated fullerene 228a. The first step of the synthesis of C_{50}Cl_6 227 was carried out by treatment of fullerene C_{60} with iodine chloride. Next, the reaction of C_{50}Cl_6 227 with $\text{C}_2\text{F}_5\text{Li}$, generated from $\text{C}_2\text{F}_5\text{I}$ and $n\text{-BuLi}$, afforded $\text{C}_{50}\text{Cl}_6(\text{C}_2\text{F}_5)_5\text{H}$ 228a, the structure of which was investigated by X-ray analysis to show C_5 -symmetry of the molecule (Scheme 113). According to the ^1H NMR spectrum, the proton is quite acidic, resonating as a singlet at 5.3 ppm. Treatment of $\text{C}_{50}\text{Cl}_6(\text{C}_2\text{F}_5)_5\text{H}$ 228a with DBU resulted in the formation of $\text{C}_{50}\text{Cl}_6(\text{C}_2\text{F}_5)_5\text{H}^-$ anion 228b. DFT estimated the gas-phase acidity of 228a ($E_{\text{pa}}(\text{A}^-) = 1269 \text{ kJ/mol}$) is higher than that of $\text{CH}_3\text{CO}_2\text{H}$ (1490), $\text{CH}_3\text{SO}_3\text{H}$ (1370), and even $\text{CF}_3\text{SO}_3\text{H}$ (1288), nominating 228a as fullerene superacid. The solvation (DMSO) appeared to be crucial, predicting $\text{pK}_a = 4.5$ to be a considerably weaker acid than $\text{CF}_3\text{SO}_3\text{H}$ (-7.9) and $\text{CH}_3\text{SO}_3\text{H}$ (1.8), but stronger than $\text{CH}_3\text{CO}_2\text{H}$ (17.2). However, experimental measurements were not carried out (Scheme 119).²⁹⁰

Scheme 119



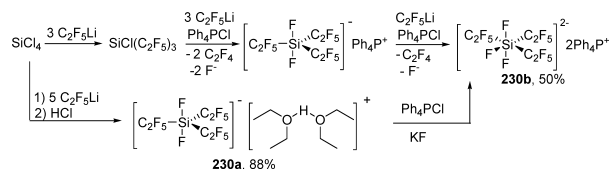
3.2.1.2. Formation of Bond with Heteroatoms. Preparation of Organoelement Compounds with Perfluoroethyl Fragment. Pentafluoroethyl boronate 229b was prepared by the reaction of pentafluoroethyl lithium with trimethylborate. This type of compounds attracts attention as a reagents for perfluoroethylation of different types of organic compounds.²⁶⁹ The reaction of perfluoroethyl lithium with chlorodimethoxyborane led to polyfluoroborates and polyfluoroethyl boranes 229c–e, which are interesting materials for fluoroorganic and elementoorganic chemistry.²⁹¹ In the case of dichloro-(diethylamino)borane, the corresponding diethylamino-derivative 229a was isolated in 90% yield (Scheme 120).²⁹²

Scheme 120



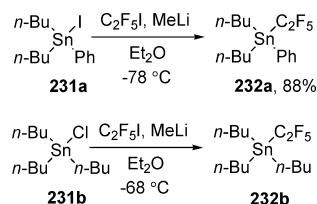
Because of the high fluorophilicity of silicon, perfluoroalkylated silanes are not stable compounds having propensity to the decomposition by difluorocarbene elimination. Recently, Hoge has demonstrated that the reaction of pentafluoroethyl lithium with SiCl_4 leads to the formation of five- or six-coordinate tris(pentafluoroethyl)silicates **230a** and **230b**, which are the first structurally characterized examples of a tris(perfluoroalkyl) silicon compound (Scheme 121).²⁹³

Scheme 121



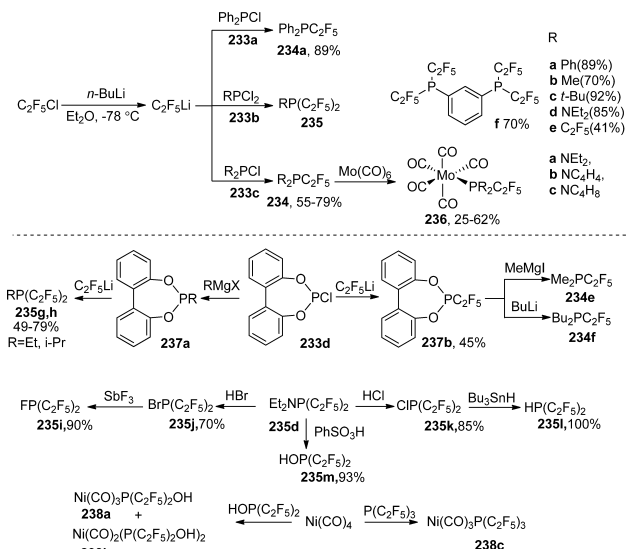
Perfluoroethyl stannane **232a** was prepared by the treatment of the corresponding stannyl iodide **231a** with perfluoroethyl lithium in 88% yield.²⁹⁴ In a similar manner, the reaction of tributyltin chloride **231b** led to the corresponding fluorinated derivative **232b** (Scheme 122).^{254c} The yield of **232b** was not given.

Scheme 122



The reaction of perfluoroethyl lithium with chlorophosphines **233a–c** allowed one to prepare perfluoroethyl **234a–d**²⁹⁵ and bis(perfluoroethyl) **235a–f**²⁹⁶ substituted phosphines in high yields. Pentafluoroethylated phosphines **234b–d** were used for the preparation of complexes with $\text{Mo}(\text{CO})_6$ **236** (structure of **236** has been studied using X-ray). Alternatively, pentafluoroethylated phosphines were prepared using (2,2-biphenylene)-chloridophosphite **233d**, which is readily available by the reaction of 2,2-biphenol with PCl_3 . The treatment of **233d** with $\text{C}_2\text{F}_5\text{Li}$ yields pentafluoroethylphosphonite **237b**, the subsequent treatment of which with Grignard reagents gives monopentafluoroethylated phosphines **234e,f**. Bis-pentafluoroethylated phosphines can also be prepared starting from **233d**. In that case, the first step of the sequence was the reaction of **233d** with Grignard reagents, which afforded phosphines **237a**. Next, phosphines **235g,h** were prepared by the reaction of **237a** with $\text{C}_2\text{F}_5\text{Li}$ (Scheme 123). Bis(pentafluoroethylated) aminophosphane **235d** ($(\text{C}_2\text{F}_5)_2\text{PNEt}_2$) was used as a starting material for the synthesis of other functional bis(pentafluoroethylated) phosphanes **235i–m** having at the phosphorus various groups by substitution of diethylamino fragment (Scheme 123). C_2F_5 -substituted phosphorous derivatives became unique ligands for coordination chemistry due to high steric demand and electron-withdrawing properties of the pentafluoroethyl group. For example, various fluorinated phosphines were studied in substitution reactions with $\text{Ni}(\text{CO})_4$. It was shown that $(\text{C}_2\text{F}_5)_3\text{P}$ makes complex **238c** with substitution of one CO ligand, whereas $(\text{C}_2\text{F}_5)_2\text{POH}$ can form complexes **235b,c** with

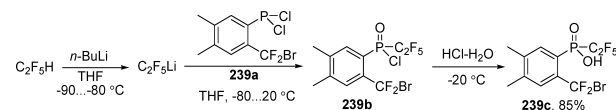
Scheme 123



substitution of one or two carbon monoxide molecules. Some mercury complexes were also prepared. These data are in good agreement with the steric demand of such phosphorus ligands.²⁹⁷

Pentafluoroethylated arylphosphonic acid **239c** was prepared by Rösenthaller and Tverdomed in 85% yield. High selectivity of monosubstitution of chloride should be noted for the reaction of pentafluoroethyl lithium with phosphonic dichloride **239a** to form **239b**. A possible explanation of such selectivity is that the reaction proceeds via addition–elimination mechanism. Because of high steric demand of CF_2Br and C_2F_5 groups, only monosubstitution takes place preventing second addition of organolithiums (Scheme 124).²⁹⁸

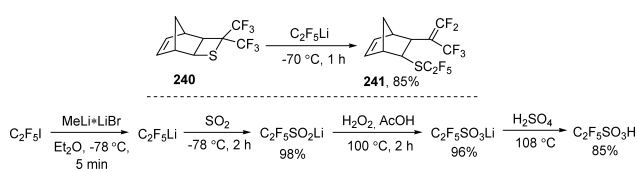
Scheme 124



The reaction of pentafluoroethyl lithium with norbornene derived condensed thietane **240** resulted in opening of the thietan ring to give sulfide **241** in 85% yield.²⁹⁹ The driving force of this reaction is decreasing strain of the polycyclic system. One-pot preparation of pentafluoroethanesulfonic acid was elaborated using reaction of $\text{C}_2\text{F}_5\text{Li}$ with sulfur dioxide, followed by oxidation with H_2O_2 of thus-formed sulfinic acid derivative $\text{C}_2\text{F}_5\text{SO}_2\text{Li}$. Treatment of $\text{C}_2\text{F}_5\text{SO}_3\text{Li}$ with concentrated sulfuric acid gave pentafluoroethanesulfonic (pentfllic) acid in 85% total yield (Scheme 125).³⁰⁰

3.2.2. Fluorinated Vinyl Lithiums. Fluorinated ethanes are very valuable precursors for the preparation of various

Scheme 125



fluorinated vinyl organometallics. Treatment of some freons with bases results in deprotonation followed by fluoride elimination (see preparation of alkenes by elimination). Thus, formed alkene can be deprotonated with excess of a base to give the corresponding organometallic reagent. For example, the reaction of 1,1,1,2-tetrafluoroethane with *n*-BuLi at $-78\text{ }^{\circ}\text{C}$ leads to direct formation of $\text{CF}_2=\text{CFLi}$. This organolithium compound can react with a lot of electrophiles to form various derivatives **208** bearing trifluorovinyl fragment (Scheme 126,

Scheme 126

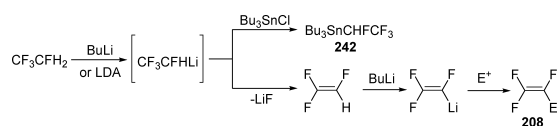


Table 20. Reactions of Trifluorovinylolithium with Some Electrophiles

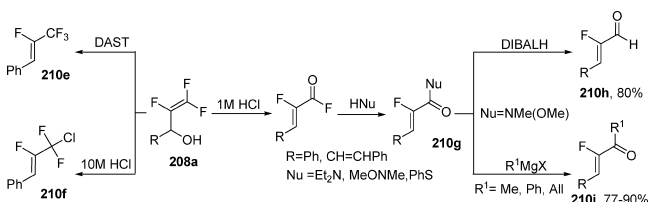
Products of trifluorovinylolithium 208				
$\text{P}(\text{CF}_2\text{CF}_3)_3$, 24%	$\text{P}(\text{CF}_2\text{CF}_3)_2\text{PPh}_2$, 80%	$\text{P}(\text{CF}_2\text{CF}_3)_3$, 67%	$\text{P}(\text{CF}_2\text{CF}_3)_2\text{P}(\text{OMe})_3$, 64%	$\text{P}(\text{CF}_2\text{CF}_3)_3$, 48%
$\text{As}(\text{CF}_2\text{CF}_3)_3$, 90%	$\text{Sb}(\text{CF}_2\text{CF}_3)_3$, 70%	$\text{B}(\text{CF}_3)_3 \times \text{NHMe}_2$, 89%	$\text{B}(\text{OMe})_3$, 80%	SiMe_3 , 80%
BuSiMe_3 , 81%	SnMe_3 , >90%	SnBu_3 , 67%	$\text{Hg}(\text{CF}_2\text{CF}_3)_2$, 93%	$\text{PhHg}(\text{CF}_2\text{CF}_3)$, 79%
HgCl_2 , >90%	$\text{Hg}(\text{CF}_2\text{CF}_3)_2$, >90%	$\text{Mn}(\text{CO})_5$, >90%	$\text{P}(\text{Ph}_3\text{P})_3$, >90%	$\text{P}(\text{Bu}_3\text{P})_3$, >90%

Table 20).³⁰¹ Initially formed CF_3CFHLi is a very unstable organometallic compound; however, the reaction of 1,1,1,2-tetrafluoroethane with LDA followed by addition of Bu_3SnCl gave the corresponding stannane **242** (Scheme 126).

The reaction of trifluorovinyl lithium with carbonyl compounds yields trifluoroallylic alcohols **208a**, which can be converted to different important fluorinated compounds (derivatives of fluorocinnamic acid **210g**, CF_2Cl **210e**, and CF_3 substituted alkenes **210e**, unsaturated aldehydes **210h**, and ketones **210i**, Scheme 127).³⁰²

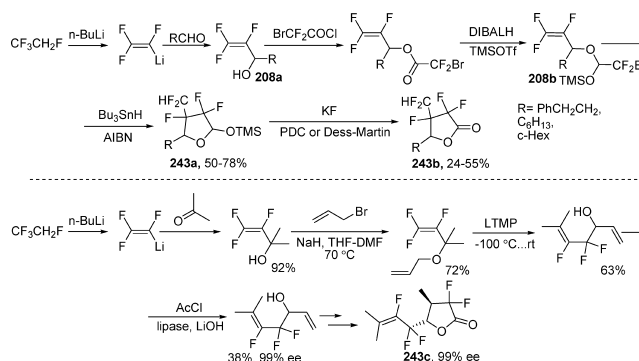
Trifluoroallylic alcohols **208a** prepared by addition of trifluorovinyl lithium were used for preparation of α,α -difluoro- γ -lactones **243b**. A key step of sequence is radical cyclization of trifluoroallyl *O*-(trimethylsilyl)- α -bromo- α,α -difluoroacetals **208b** to give γ -lactols **243a** in good yield.

Scheme 127



Lactols **243a** are then converted into the corresponding α,α -difluoro- γ -lactones **243b** by the oxidation with PDC or Dess-Martin periodinane (Scheme 128). Electrophysiological tests

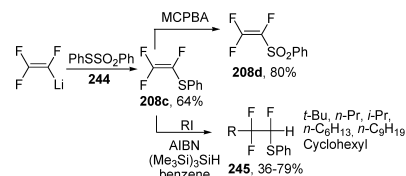
Scheme 128



revealed that the difluorinated analogues were as active as the natural eldanolide on the olfactory receptors. Using a similar approach, the same authors synthesized fluorinated analogues **243c** of the sex pheromone of the male African sugar cane borer in optically active forms and tested their pheromone activities. The study of activity of these fluorinated pheromone analogues **243c** indicates that the point of fluorination has a critical influence on the pheromone activity.³⁰³

Phenyl trifluorovinyl sulfide **208c** was prepared by the reaction of trifluorovinyl lithium with *S*-phenyl benzenethiosulfonate (**244**). Subsequent oxidation with MCPBA resulted in the corresponding sulfone **208d** in 80% yield. This sulfide is quite reactive toward radical species generated from alkyl halides. As a result, this approach affords an effective synthesis of various fluorinated sulfur derivatives **245** (Scheme 129).³⁰⁴

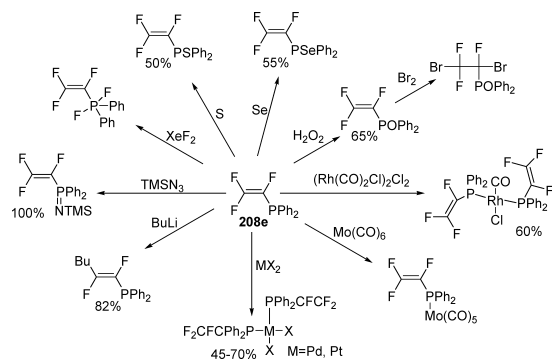
Scheme 129



Trifluorovinyl lithium has been studied very thoroughly for the synthesis of various organoelement compounds and especially in phosphorus chemistry. Such derivatives were synthesized directly by reactions with the corresponding phosphorus electrophiles (Table 20). Also, the modification of diphenyl(trifluorovinyl)phosphine **208e** was studied intensively in terms of the phosphorus fragment modification as well as the preparation of metal complexes (Scheme 130).^{302,305}

Similarly, the reaction of $\text{CF}_3\text{CH}_2\text{Cl}$ with butyl lithium opens a direct approach to 2,2-difluoro-1-chlorovinyl lithium $\text{CF}_2=\text{CClLi}$ (Scheme 131). This organometallic reagent was found also very useful to prepare the corresponding derivatives **246** bearing 2,2-difluoro-1-chlorovinyl moiety.³⁰⁶ For example, reaction with carbonyl compounds leads to the corresponding allylic alcohols, 2,2-difluoro-1-chlorovinyl-substituted silanes, sulfides, phosphines,³⁰⁷ stibines,³⁰⁸ which were prepared using this approach (Table 21). It should be noted that 2,2-difluoro-1-chlorovinyl lithium synthesized by the reaction of $\text{CF}_3\text{CH}_2\text{Cl}$

Scheme 130



Scheme 131

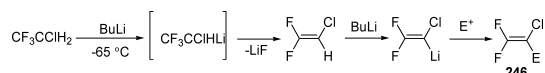
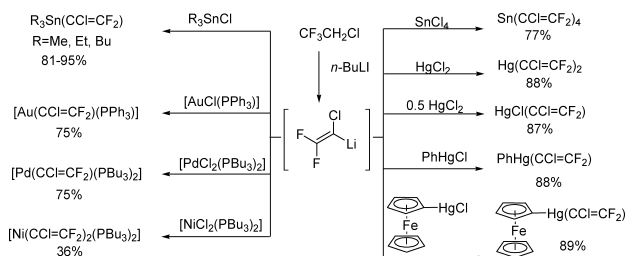


Table 21. Reactions of 2,2-Difluoro-1-chlorovinyl Lithium with Electrophiles

with butyl lithium gives much better results than direct metalation of 1-chloro-2,2-difluoroethene.

2,2-Difluoro-1-chlorovinyl lithium appeared to be a very useful precursor for the synthesis of a broad spectra of fluorinated organometallic complexes. Thus, fluorinated derivatives of tin, mercury, nickel, palladium, and gold were effectively prepared using this high-yielding, one-pot synthetic approach (Scheme 132).^{301g,308}

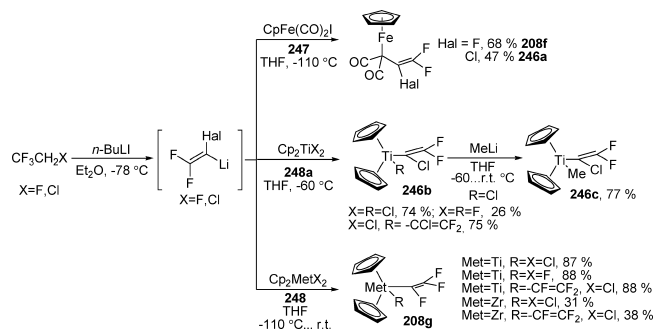
Scheme 132



A series of cyclopentadienyl complexes with polyfluorovinyl substituents were prepared using the reaction of fluorovinyl lithiums with halo-precursors **247**, **248**. Thus, polyfluorovinyl iron complexes **208f**, **246a** were synthesized by transmetalation reaction from $\text{CpFe}(\text{CO})_2\text{I}$ **247**.³⁰⁷ Similarly, titano- and zirconocenes **246b,c** and **208g** were obtained in good to high yields starting from metallocenes **248** (Scheme 133).³⁰⁹ Detailed structural analysis of prepared complexes was conducted using X-ray and other spectroscopic methods.

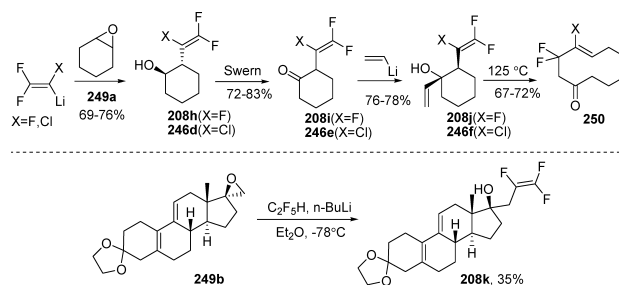
Ring-opening of epoxide **249a** with trifluorovinyl lithium and 1-chloro-2,2-difluorovinyl lithium was used by Percy as a key

Scheme 133



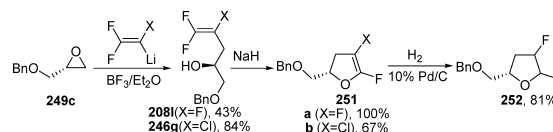
step for the preparation of the homoallylic alcohols **208j**, **246f** studied in oxy-Cope rearrangement. Homoallylic alcohols **208h**, **246d** gave γ,γ -difluorinated β,γ -enones **208i**, **246e** by Swern oxidation. Subsequent reaction with vinyl lithium afforded substituted *trans*-1,2-divinylcyclohexanols **208j**, **246f** used for the synthesis of fluorine-containing cyclodecenones **250** by oxy-Cope rearrangement.³¹⁰ The addition of trifluorovinyl lithium to steroidal epoxide **249b** gave fluorinated derivative **208k** selectively in 35% yield (Scheme 134).²⁶⁷

Scheme 134



A simple stereoselective three-step synthesis of 2-fluoro- and 2-chloro-2,3-dideoxy arabinose derivatives **251** from the readily available (*R*)-glycidol and either 1,1,1,2-tetrafluoroethane or 1-chloro-2,2,2-trifluoroethane was described. A key step of the method is the opening of epoxide **249c** with $\text{CF}_2=\text{CFLi}$ and $\text{CF}_2=\text{CClLi}$ generated from the above-mentioned freons. Subsequent cyclization of **208i**, **246g** induced by sodium hydride as a base gave a short route to valuable fluorinated sugar derivatives **251a,b** in stereoselective manner. Further transformation of **251a** was performed by addition of hydrogen to the double bond to give fluorinated tetrahydrofuran **252** (Scheme 135).³¹¹

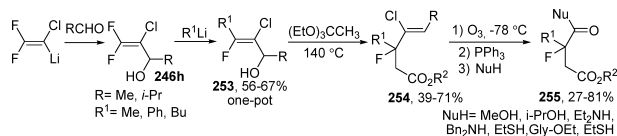
Scheme 135



A very effective sequence for the synthesis of fluorinated succinic acid derivatives was elaborated using 1-chloro-2,2-difluorovinyl lithium.³¹² The first step of the transformation was the addition of $\text{CF}_2=\text{CClLi}$ to carbonyl group of aldehydes to form fluorinated allylic alcohols **246h**. Next, the substitution of the fluorine with organolithiums gave product

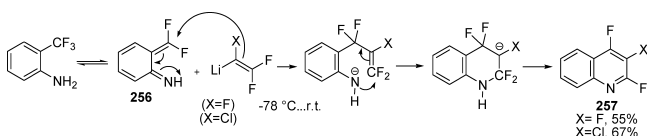
253 having *E*-configuration of the double bond. Fluorinated allylic alcohols **253** are transformed into 3-fluoro-4-halo-4-alkenoates **254** through [3,3]-sigmatropic rearrangement, which is performed at elevated temperatures in orthoether. Ozonolysis of the haloalkenes **254** affords the corresponding acyl halides, which can be trapped by nucleophiles to afford fluorinated succinic acid derivatives **255**, having two differentiated carboxylic groups (Scheme 136).

Scheme 136



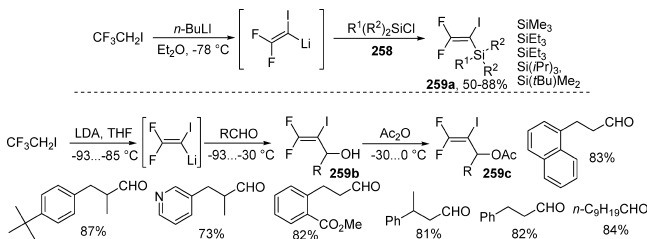
A very effective synthesis of fluorinated quinolines is based on application of trifluorovinyl lithium and 1-chloro-2,2-difluorovinyl lithium. The reaction of this organolithiums with 2-trifluoromethyl aniline leads to the formation of quinoneimine intermediate **256** by dehydrofluorination. Subsequent addition of fluorinated vinyl lithiums gave 2,3,4-trifluoroquinoline and 3-chloro-2,4-difluoroquinoline **257** in good yields (Scheme 137).³¹³

Scheme 137



Application of CF₃CH₂I has expanded the synthetic scope of the method on the synthesis of 2,2-difluoro-1-iodovinyl derivatives. Difluoroiodovinyl lithium CF₂=CILI was generated by the treatment of CF₃CH₂I with LDA in THF at low temperature. Thus, a number of iodovinylsilanes **259a** were prepared by the treatment of CF₂=CILI with chlorotrialkylsilanes **258**.³¹⁴ Effective one-pot pathway to fluorinated allyl acetates **259c** was elaborated using the reaction of CF₂=CILI with aldehydes followed by the acylation of intermediately formed allylic alcohols **259b** with acetic anhydride. The corresponding allyl acetates **259c** were isolated in high total yield (Scheme 138).³¹⁵

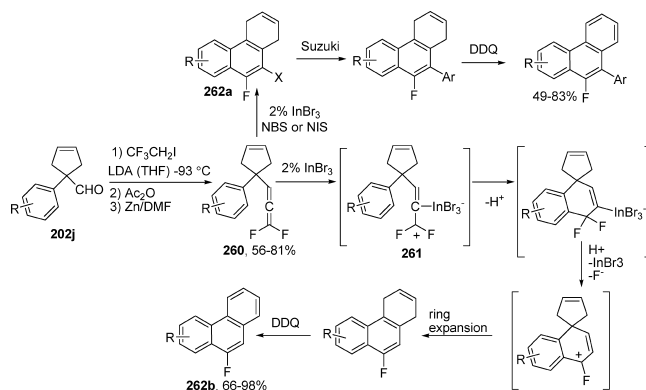
Scheme 138



Recently, Ichikawa proposed a simple three-step synthesis of 1,1-difluoroallenes **260** starting from aldehydes **202j**. Addition of CF₂=CILI to **202j** led to the corresponding allylic alcohols, which gave allylic acetates by the reaction with acetic anhydride. Subsequent treatment of allylic acetates with Zn provided target 1,1-difluoroallenes **260** in high total yields (56–81% starting

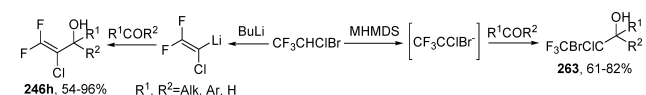
from CF₃CH₂I). Cyclopentenyl derived allenes **260** were used for the synthesis of fluorine-containing polyaromatics (up to five condensed benzene rings) using indium(III)-catalyzed domino reaction. The InBr₃ generated 1,1-difluoroallylic cations **261** readily underwent Friedel–Crafts-type cyclization/ring expansion and subsequent aromatization with DDQ to afford F-PAHs **262b**. Alternatively, the reaction of **260** with InBr₃ in the presence of *N*-bromosuccinimide (NBS) or *N*-iodosuccinimide (NIS) allowed the preparation of bromine- and iodine-substituted fluorophenanthrenes **262a**, which were used for cross coupling with boronic acids (Scheme 139).³¹⁶

Scheme 139



Metalation of Halothane with lithium and sodium hexamethyldisilazide followed by the reaction with carbonyls gave the corresponding alcohols **263** bearing CF₃CClBr fragment. Contrary, the reaction with butyl lithium gave the products of 1,1-difluoro-2-chlorovinylation **246h** through the formation of 1,1-difluoro-2-chlorovinyl lithium and subsequent addition to the carbonyl group (Scheme 140).³¹⁷

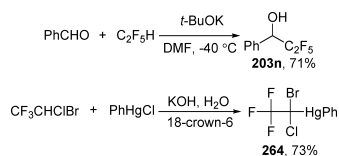
Scheme 140



3.2.3. Polyfluorinated Ethyl Sodium and Potassium Compounds. In this section, reactions of anions, generated from PFEs by treatment with strong bases with sodium and potassium counter-anions, are collected. This is very rare type of perfluorinated organometallic compounds, and only a few examples of such derivatives are known to date. As it was mentioned in the previous section, PFEs can be deprotonated using strong bases. Thus, pentafluoroethane treated with *t*-BuOK in DMF at low temperature gave perfluoroethyl anion, which then reacted with benzaldehyde to form fluorinated alcohol **203n**.³¹⁸ Using KOH as a base, CF₃CBrCl-mercury derivative **264** was prepared by reaction of Halothane with PhHgCl in phase-transfer conditions (Scheme 141).³¹⁹

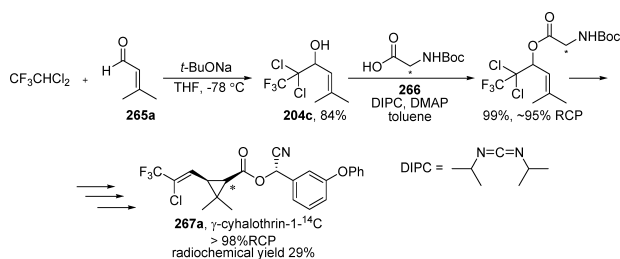
Cyhalothrin is a synthetic pyrethroid insecticide. γ -Cyhalothrin **267a** is the most active form of this insecticide out of 16 possible isomers.³²⁰ Cyclopropyl carbon-14 labeled γ -cyhalothrin **267a** was prepared and examined in a variety of metabolism and environmental fate studies to get a deeper understanding of the biological action of this single isomer. The first step of the synthesis was deprotonation of CF₃CCl₂H by treatment with sodium *t*-BuOK followed by the reaction with

Scheme 141



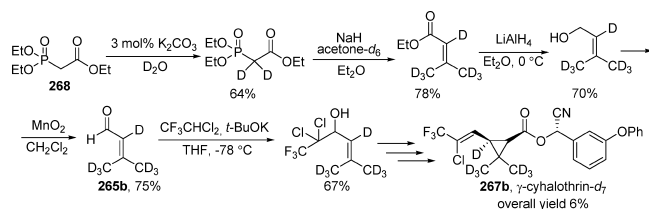
aldehyde **265a**, which affords the allyl alcohol **204c**. Carbon-14 was then incorporated by the esterification reaction with commercially available *N*-*t*-Boc-glycine-2-¹⁴C **266** in the presence of *N,N*-diisopropylcarbodiimide and catalytic amounts of DMAP. The desired molecule **267a** was synthesized in seven chemical steps in >98% radiochemical purity and an overall radiochemical yield of 29% (Scheme 142).³²¹

Scheme 142



γ -Cyhalothrin-*d*₇ **267b** was prepared using a very similar approach. Addition of in situ generated CF₃CCl₂K to the deuterated aldehyde **265b** was performed to introduce deuterium label into the molecule. The deuterated aldehyde **265b** was synthesized from phosphonate **268** in four steps as depicted on Scheme 143. The overall yield of the synthetic sequence is 6% starting from acetone-*d*₆ (Scheme 143).³²²

Scheme 143

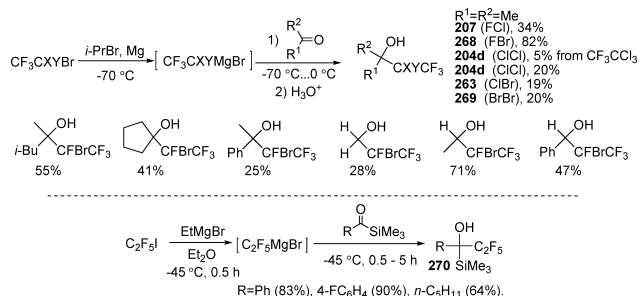


3.2.4. Polyfluorinated Ethylmagnesium Compounds.

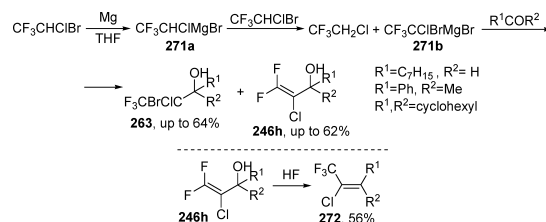
As compared to lithium analogues, perfluoroethyl magnesium halides are more stable, but less reactive at the same time. The addition of various polyfluorinated ethyl magnesium halides to carbonyl compounds led to fluorinated alcohols in moderate to good yields (Scheme 144).^{37b} Alkyl (aryl) perfluoroethyl trimethylsilyl carbinols **270** were synthesized using the corresponding acyltrimethylsilanes as electrophiles (Scheme 144).³²³

The reaction of Halothane with magnesium was studied intensively. The Grignard reagent **271a** formed at the first step of the reaction acts as a base to form metalated Halothane **272b** and 1-chloro-2,2,2-trifluoroethane. As a result, a mixture of two products **263**, **246h** is formed in variable yields. Product **246h** having 1,1-difluoro-2-chlorovinyl moiety can be easily converted into trifluoromethylated alkene **272** by treatment with HF (Scheme 145).³²⁴

Scheme 144

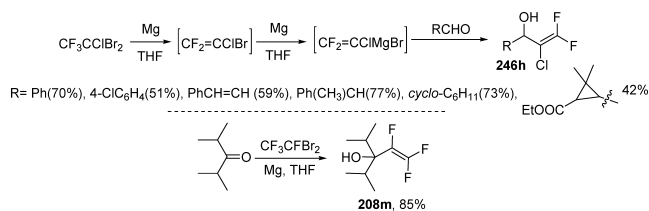


Scheme 145



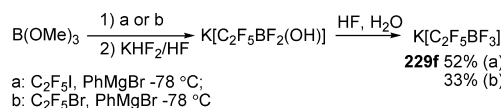
An interesting approach to difluoroethylenes was elaborated starting from CF₃CClBr₂. Treatment of this reagent with Mg in THF gave 1-bromo-1-chloro-2,2-difluoroethene, which formed fluorinated vinyl magnesium bromide CF₂=CClMgBr by the reaction with a second equivalent of magnesium. The addition of this organomagnesium reagent to aldehydes led to a series of fluorinated allyl alcohols **246h** in good yields.³²⁵ A similar alcohol **208m** having trifluorovinyl fragment was obtained using di(iso-butyl)ketone and CF₃CFBr₂ (Scheme 146).³²⁶

Scheme 146



Perfluoroalkyltrifluoroborate **229f** was prepared for creation of novel electrolyte salts starting from trimethylborate and pentafluoroethyl magnesium iodide.³²⁷ Using pentafluoroethyl bromide to generate the Grignard reagent resulted in lower yield (33%) of **229f** (Scheme 147).³²⁸

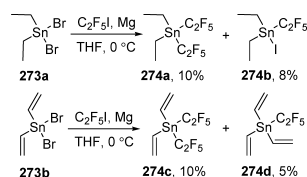
Scheme 147



Perfluoroethyltin compounds **274a,c** were synthesized via Barbier-type reaction from the corresponding ethyl- and vinyl tin bromides **273a,b**. The iodide transhalogenation product **274b** and the vinyl group redistribution product **274d** as well as the expected products were also isolated (Scheme 148).³²⁹

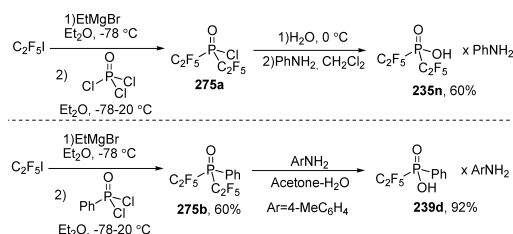
An efficient pathway toward perfluoroalkylphosphinic acids was elaborated by Caffyn.³³⁰ Pentafluoroethyl magnesium bromide was prepared from perfluoroethyl iodide by trans-

Scheme 148



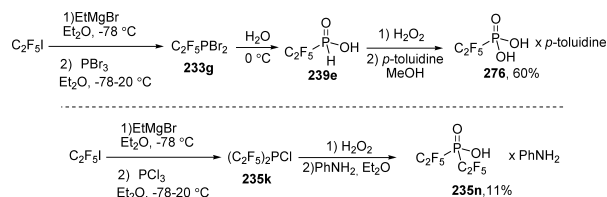
metalation with EtMgBr to form less basic organometallics. The reaction of this perfluoroethyl Grignard reagent with phosphoryl chloride gave selectively bis(perfluoroalkyl)-phosphonyl chloride **275a**, which afforded bis(perfluoroethyl)-phosphinic acid **235n** after aqueous workup. Similarly, perfluoroethyl(phenyl)phosphinic acid **239d** can be obtained from phenylphosphonic dichloride in two steps. Both perfluoroalkylphosphinic acids **235n** and **239d** were isolated as salts with anilines. Free acids can be prepared by passing through an Amberlite IR120A cation exchange column (H⁺ form), eluting with water (Scheme 149).

Scheme 149



Caffyn has also reported the convenient one-pot synthesis of perfluoroalkyl phosphonic acids.³³¹ The reaction of C₂F₅MgBr with PBr₃ led to the fluorinated phosphorus dibromide **233g**, which was hydrolyzed into perfluoroalkylphosphinic acid **239e**. Next, the oxidation of **239e** with hydrogen peroxide led to crude C₂F₅PO(OH)₂ converted quantitatively into the corresponding salt **276** by treatment with *p*-toluidine. The reaction of PCl₃ with 6-fold excess of C₂F₅MgBr led to bis(perfluoroalkyl)chlorophosphine **235k**. Oxidative hydrolysis of **235k** afforded bis(perfluoroethyl)phosphinic acid **235n** in very low yield (Scheme 150).

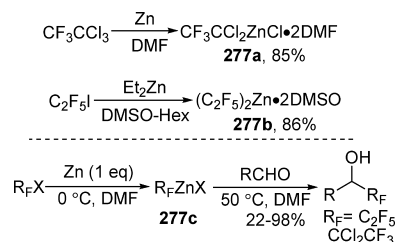
Scheme 150



3.2.5. Organozinc Compounds. Fluorinated ethanes react easily with Zn in DMF to give the corresponding organozinc reagents. Alternatively, these organozinc derivatives were prepared from diethyl zinc; for example, reaction of Et₂Zn with pentafluoroethyl iodide in DMSO gave pentafluoroethyl zinc as a complex with the solvent. This type of fluorinated organometallics is considerably more stable in comparison with lithium and magnesium analogues. Some polyfluorinated ethyl zinc compounds form complexes with DMF **277a**³³² and DMSO **277b**,³³³ which are stable enough to be isolated

(Scheme 151). Polyfluorinated ethyl zinc reagents **277c** were used successfully in reactions with various electrophiles. Thus,

Scheme 151



treatment of carbonyl compounds with polyfluoroethanes and 1 equiv of zinc afforded various fluorinated alcohols (Scheme 151, Table 22).

Table 22. Reaction of Polyfluorinated Ethyl Zinc Reagents with Aldehydes

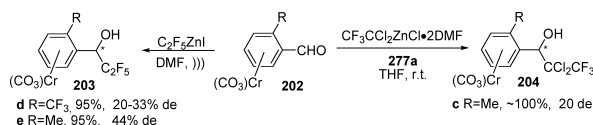
Product Yield, % X	ref	Product Yield, % X	ref	Product Yield, % X	ref
 87, Cl	334	 80, Cl	63, 335	 96, Cl	63, 335
 86-95, Cl	63, 334, 335	 61 ^a , Cl	63, 335	 87, Cl	63, 335
 23, Br	63, 335	 69-82, Cl	63, 334, 335	 21 ^a , I	341
 27, Br	63, 335	 60, Cl	63, 335	 50, I	336
 60, Cl	334	 98, Cl	334	 70, Cl	337
 70-80, Cl	200b, 334	 54 ^b , I	340	 71-75, Cl	335, 338, 339
 41 ^a -67, I	340, 341, 342	 63-72 ^a , Cl	63, 334, 335	 35, Br	335
 42 ^a -48 ^a , I	340, 341	 40 ^a , I	340	 15-31, Cl	344
 33, Cl	340	 R = PhCH2, 4-ClC6H4, 4-MeOC6H4	340		

^aIn the presence of NiCl₂(PPh₃)₂ (1 mol %). ^bUnder ultrasound irradiation. ^cIn the presence of CuCl (5 mol %). ^dIn the presence of PdCl₂(PPh₃)₂.

Efforts were made to perform asymmetric perfluoroalkylation using polyfluoroethyl zinc derivatives. For example, the addition of C₂F₅ZnI and DMF complex **277a** to chromium tricarbonyl derivative **202** led to a mixture of diastereomers **203d**, **203e**, or **204c** in nearly quantitative yield, but low diastereomeric excess (up to 44%) was observed (Scheme 152).^{258,268b}

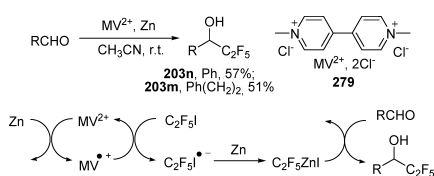
The addition of such organozinc species to aldehydes in a Barbier-type reaction was also performed. Using an electron-transfer system containing the reducing agent (Zn) and methyl viologen MV²⁺ **279** as an electron-transfer catalyst, compounds **203n** and **203m** were isolated. Under these conditions, formation of organozinc compounds was detected at room

Scheme 152



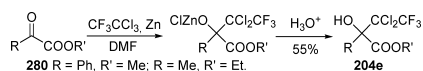
temperature, while without methyl viologen the reaction proceeded only at heating (Scheme 153).³⁴⁵

Scheme 153



The addition of polyfluorinated ethyl zinc derivatives to esters of ketoacids **280** followed by the acidic workup afforded esters of fluorinated hydroxy acids **204e** in good yields (Scheme 154).⁶³

Scheme 154



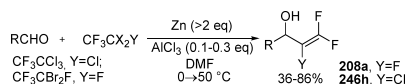
Using more than 2 equiv of Zn in the presence of AlCl_3 , fluorinated allyl alcohols **208a**, **246h** were prepared in good to high yields from the corresponding aldehydes and generated in situ organozinc derivatives (Table 23, Scheme 155).

Table 23. Reaction of Polyfluorinated Ethyl Zinc Reagents with Aldehydes in the Presence of AlCl_3

Product, yield, %	ref	Product, yield, %	ref	Product, yield, %	ref
	63,346		63,346		63,346
	63,346		347		344
	63,346		347		

^a1:1 diastereomeric mixture.

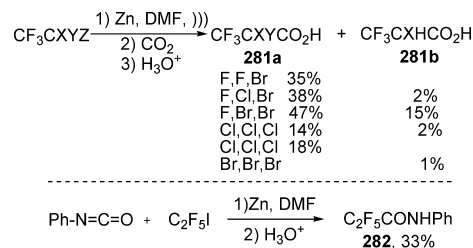
Scheme 155



The reaction of fluoroethyl zinc with carbon dioxide led to fluorinated propionic acids **281a** with minor admixture of partially reduced form **281b**.^{63,348} Phenylamide of perfluoropropionic acid **282** was synthesized by the addition of perfluoroethyl zinc species to phenyl isocyanate in moderate yield (Scheme 156).³⁴⁹

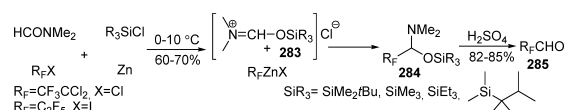
Treated with trialkylsilyl chlorides R_3SiCl , DMF formed iminium salt **283**. This electrophile gave trialkylsilyloxy

Scheme 156



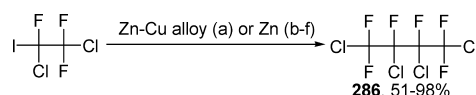
dimethylamines **284** in high yields by the reaction with some polyfluorinated ethyl zinc derivatives, generated in situ from CF_3CCl_3 and $\text{C}_2\text{F}_5\text{I}$. The reaction was performed by simple mixing of all reagents under slight cooling condition to prevent overheating.^{63,350} Acidic workup of trialkylsilyloxy dimethylamines **284** afforded fluorinated aldehydes **285** in high yields (Scheme 157).³⁵¹

Scheme 157



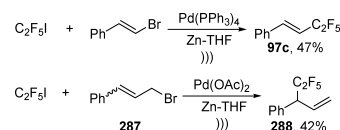
Homocoupling of ClCF_2CFCl was performed under treatment with zinc–copper alloy or metallic zinc to give partially chlorinated polyfluorobutane **286** in 51–98% yields depending on conditions used. It should be noted that only C–I bond participated in the transformation (Scheme 158).³⁵²

Scheme 158



Cross-coupling reactions of perfluoroethyl zinc were also reported. *E*-Bromostyrene gave stereoselectively *E*-isomer of pentafluoroethylated styrene **97c** under $\text{Pd}(\text{PPh}_3)_4$ catalysis. The reaction of allyl type bromide **287** was accompanied by rearrangement to afford alkene **288** in moderate yield (Scheme 159).³⁴⁰

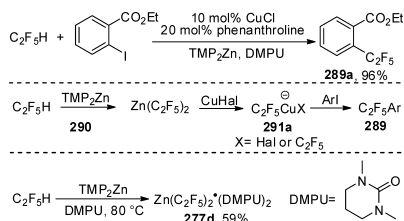
Scheme 159



Recently, a new protocol for copper-catalyzed coupling of iodoarenes and fluorinated organometallics was proposed by Daugulis.³⁵³ Pentafluoroethane and other perfluorinated alkanes having C–H bond can be used for the coupling with some iodoarenes in the presence of zinc bis-2,2,6,6-tetramethylpiperidide **290** as a base through the formation of the perfluorinated organocuprate **291a** (see also next section of this Review). Thus, the fluorinated ethyl benzoate **289a** was prepared in almost quantitative yield using this approach. Perfluoroethyl zinc intermediate **277d** can be isolated as an

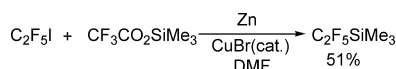
adduct with 1,3-dimethyl-tetrahydropyrimidin-2(1H)-one (DMPU), which is the solvent in this reaction (Scheme 160).

Scheme 160



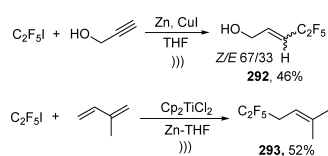
A useful approach to C_2F_5TMS and other fluorine-containing silanes was elaborated by Igumov.³⁵⁴ The reaction of C_2F_5I with trimethylsilyl trifluoroacetate $CF_3CO_2SiMe_3$ in a polar solvent, preferably DMF or dimethylacetamide in the presence of Zn activated by CuBr or CF_3CO_2H , afforded C_2F_5TMS in 51% yield (Scheme 161).

Scheme 161



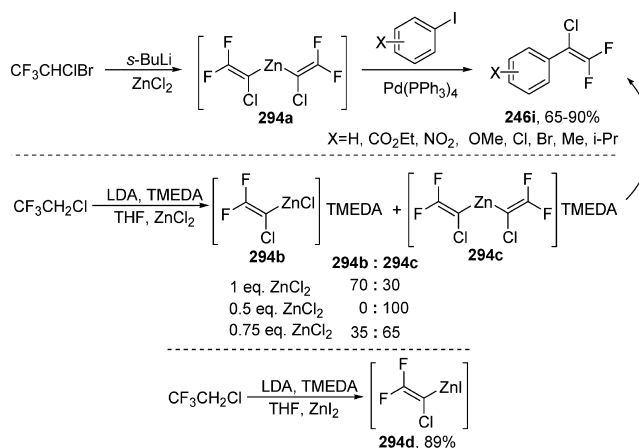
Ultrasound-promoted hydro(perfluoroalkyl)ation of alkynes with perfluoroalkylcuprate, which was formed in situ from C_2F_5I and zinc in the presence of CuI in THF, led regioselectively to fluorinated allyl alcohol **292** as a mixture of diastereomers. Similarly, ultrasound-promoted (Cp_2TiCl_2 -catalyzed) hydro(perfluoroalkyl)ation of isoprene afforded the product of its pentafluoroethylation **293** in 52% yield (Scheme 162).³⁴⁰

Scheme 162



3.2.6. Perfluorinated Vinyl Zinc Derivatives. Perfluorinated vinyl organozinc species prepared from C2 freons were found to be extremely useful for the synthesis of fluorinated styrenes via a palladium-catalyzed cross-coupling. For example, 1-chloro-2,2-difluorostyrenes **246i** were prepared by coupling of bis(1-chloro-2,2-difluorovinyl) zinc **294a** with various aryl iodides. The source of bis(1-chloro-2,2-difluorovinyl) zinc **294a** is the reaction of Halothane with *sec*-BuLi followed by zinc chloride addition (Scheme 163). The reaction proceeds at room temperature to give target alkenes **246i** in up to quantitative yields using $Pd(PPh_3)_4$ (1.5–6 mol %). The only exception is 2-nitrophenyl iodide giving low yield of the corresponding styrene (13%).³⁵⁵ 2-Thiophenyl-substituted alkene was also prepared by this approach in 65% yield. Alternatively, the same transformation can be done with 1-chloro-2,2-difluorovinyl zinc chloride **294b** prepared by the reaction of CF_3CH_2Cl with LDA and excess of $ZnCl_2$. The structure of these organozinc reagents was studied by ^{19}F NMR. Depending on the ratio of the reagents, all types of zinc species can be obtained, and usually the equilibrium between mono **294b** and divinyl zinc **294c** species is observed.³⁵⁶ When

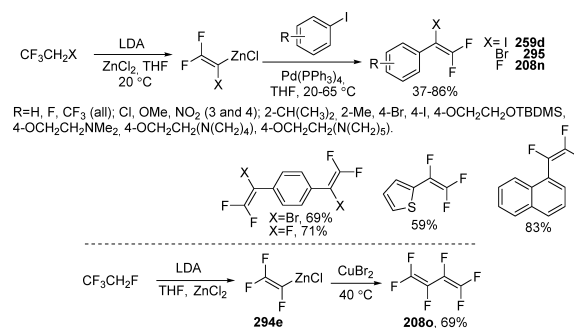
Scheme 163



$ZnCl_2$ was replaced with ZnI_2 in this metalation reaction, the resultant $CF_2=CClZnI$ **294d** was formed in 89% yield (Scheme 163).

Trifluorosubstituted styrenes **208n** as well as 1-bromo **295** and 1-iodo-2,2-difluorostyrenes **259d** were obtained in similar way proposed by Burton.³⁵⁷ Polyfluorovinyl lithiums were generated from CF_3CH_2X ($X = Br, F, \text{ or } I$) by metalation with LDA–TMEDA system. The subsequent reaction with zinc halides gave organozinc derivatives again existing as a mixture of mono- and divinylsubstituted zinc organometallics. The coupling with aromatic iodides takes place at room temperature or at 65 °C giving target trifluorostyrenes in the presence of $Pd(PPh_3)_4$. Conversions are almost quantitative (NMR), and preparative yields are in the range of 37–86%. Bromobenzene can also be involved in this type of coupling, however, to give lower yield (48%). Oxidative coupling of trifluorovinyl zinc chloride **294e** was performed under action with copper(II) bromide to give perfluorobutadiene (**208o**) in 69% yield (Scheme 164).³⁵⁸

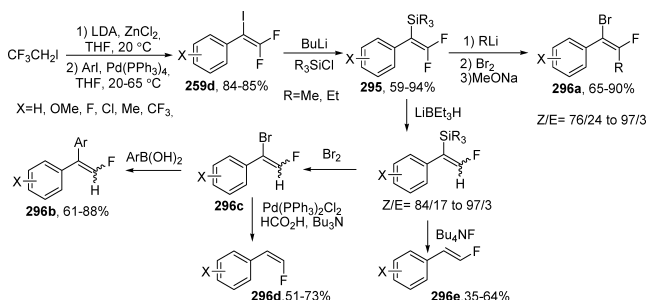
Scheme 164



A manifold of very useful approaches to both diastereoisomers of monofluorostyrenes as well as some difluorostyrenes was elaborated by Paquin starting from 1-iodo-2,2-difluorostyrenes **259d**, which were prepared from CF_3CH_2I . These compounds can be transformed into the corresponding difluorovinyl silanes **295**, which are key compounds for these syntheses. Reduction of **295** with $LiBEt_3H$ under appropriate conditions followed by protodesilylation induced with Bu_4NF results in stereoselective synthesis of *trans*-monofluorostyrenes **296e** in good total yield. Alternatively, intermediate monofluorovinyl silanes were converted into bromides **296c**. The

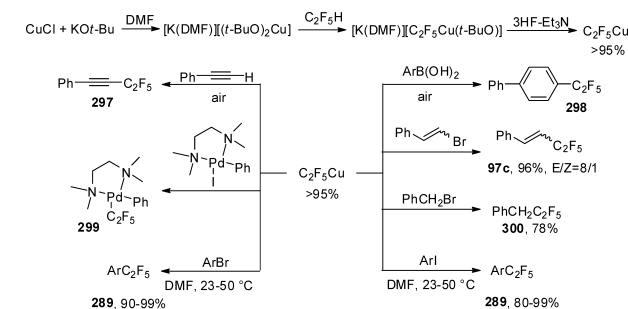
reduction of these bromides with $\text{Pd}(\text{PPh}_3)_2\text{Cl}_2\text{--HCO}_2\text{H--Bu}_3\text{N}$ system opens the stereoselective route to *cis*-mono-fluorostyrenes **296d**. The Suzuki coupling of bromides **296c** with a variety of boronic acids gives access to the desired 1,1-diaryl-2-fluoroethenes **296b**.³⁵⁹ The treatment of difluorovinyl silanes **295** with alkyl or aryllithiums followed by the reactions with bromine and MeONa afforded fluorostyrenes **296a** in high yields and stereoselectivity (Scheme 165).³⁶⁰

Scheme 165



3.2.7. Organocopper Derivatives. Recently, Grushin demonstrated the very straightforward synthesis of pentafluoroethyl copper species by cupration of pentafluoroethane with $[\text{K}(\text{DMF})][(\text{t-BuO})_2\text{Cu}]$.³⁶¹ The base was prepared in situ from CuCl and *t*-BuOK in DMF at room temperature. It was found that smooth cupration of $\text{C}_2\text{F}_5\text{H}$ took place to form compound $[\text{K}(\text{DMF})_2][\text{Cu}(\text{Ot-Bu})\text{C}_2\text{F}_5]$ isolated in pure form and characterized by X-ray diffraction. This reagent was converted to $\text{C}_2\text{F}_5\text{Cu}$ by the reaction with $\text{Et}_3\text{N--3HF}$. This cuprate has demonstrated high synthetic potential to form various pentafluoroethylated derivatives **97c**, **289**, **297–300** (reaction with benzyl bromide, preparation of C_2F_5 -substituted alkenes, reactions with boronic acid and acetylenes in oxidative conditions). The authors also prepared and isolated pentafluoroethylated palladium complex **299**, which is difficult to synthesize (see also Scheme 166). It should be noted that the

Scheme 166



reagent is tolerated to many functional groups. Very effective coupling with aryl iodides and bromides provided a simple route to pentafluoroethylated arenes prepared generally in very high yields (Scheme 166, Table 24).

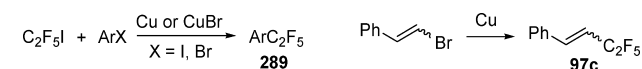
The copper-mediated Ullmann-type cross-coupling of $\text{C}_2\text{F}_5\text{I}$ with aryl and vinyl bromides or iodides was reported in several works. As is usual for Ullmann reaction, couplings were carried out in polar aprotic solvents at elevated temperatures to give the desired fluorinated aromatic compounds in moderate to high yields. It was postulated that the reaction includes formation of fluorinated organocopper species as intermediates;

Table 24. Preparation of ArC_2F_5 by the Reaction of $\text{C}_2\text{F}_5\text{Cu}$ with ArI or ArBr

ArC_2F_5	Yield, % (Hal)	ArC_2F_5	Yield, % (Hal)	ArC_2F_5	Yield, % (Hal)
	95 (I) 95 (Br)		96 (I)		97 (I) 94 (Br)
	98 (I)		99 (I)		94 (I) 96 (Br)
	90 (I)		95 (I)		97 (I) 99 (Br)
	97 (I)		96 (I) 98 (Br)		94 (I)
	92 (I)		97 (I)		95 (I)
	99 (Br)		80 (Br)		90 (Br)
	97 (Br)		96 (Br)		93 (Br)
	97 (Br)		95 (Br)		

however, the mechanism was not studied (Scheme 167, Table 25).

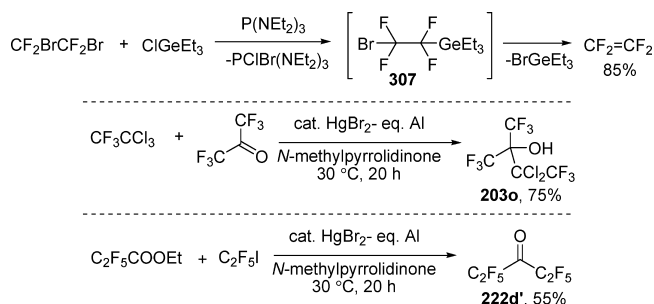
Scheme 167

Table 25. Coupling of Aryl- and Vinylbromides or Iodides with $\text{C}_2\text{F}_5\text{I}$ under Copper Assistance

ArX	Conditions	Product	Yield %	ref
	Cu, HMPA, 110 °C, 16 h		37	362
	Cu, HMPA, 110-120 °C, 3 h		17	363
	Cu, DMF, 140 °C, 9 h		22	364
	Cu, DMF, 100 °C, 16 h		54	365
	Cu(bromide), DMF, 110-120 °C, 16 h		70-95 ^a	366
	Cu(bromide), DMF, 120 °C, 3.5 h		66	367
	CuBr, DMF, 60 °C sealed tube		80	368
	Cu, DMSO, 80 °C, 16 h		41	369
	Cu, DMSO, 130 °C, 28 h		27	370
	Cu, DMSO, 110-120 °C, 4 h		88	371

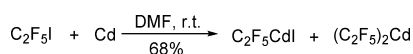
^aThe exact value was not given.

Scheme 173



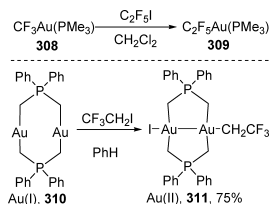
powder gave a mixture of $\text{C}_2\text{F}_5\text{CdI}$ and $(\text{C}_2\text{F}_5)_2\text{Cd}$ at room temperature in DMF (Scheme 174).³⁸³

Scheme 174



Perfluoroethyl gold derivative **309** was detected in the reaction mixture obtained by the treatment of gold complex **308** with $\text{C}_2\text{F}_5\text{I}$. Compound **309** was not isolated, and its existence was confirmed by means of ^{19}F NMR only.³⁸⁴ In contrast, gold(I) ylide dimer **310** reacts with $\text{CF}_3\text{CH}_2\text{I}$ to give gold(II)–alkyl halide complex $[\text{Au}(\text{CH}_2)_2\text{PPh}_2](\text{CF}_3\text{CH}_2)\text{I}$ **311** in high yield. The structure of **311** was established by X-ray analysis, showing that the Au–Au bond has a length of 2.681 Å (Scheme 175).

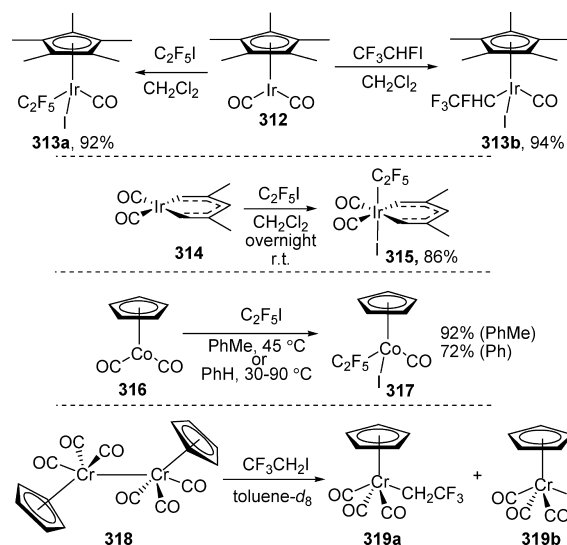
Scheme 175



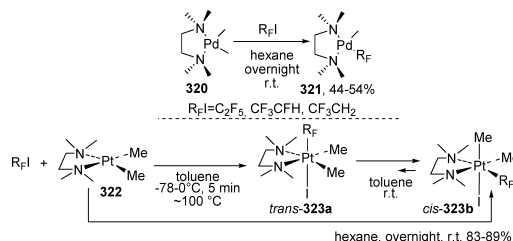
Oxidative addition of $\text{C}_2\text{F}_5\text{I}$ or $\text{CF}_3\text{CH}_2\text{I}$ to $[\text{Ir}(\eta\text{-C}_5\text{Me}_5)(\text{CO})_2]$ **312** gave the corresponding iridium complexes **313a**, **313b** in 92% and 94% yield, respectively.³⁸⁵ Similar reaction of $\text{C}_2\text{F}_5\text{I}$ with Ir(I) **314** or Co **316** precursors proceeded smoothly to afford octahedral Ir(III) product **315**³⁸⁶ and cobalt complex **317**.³⁸⁷ Sterically strained dimeric chromium cyclopentadienyl complex **318** reacted with $\text{CF}_3\text{CH}_2\text{I}$ to form after Cr–Cr bond cleavage CF_3CH_2 -substituted chromium complex **319a** as well as iodo-derivative **319b** (Scheme 176).³⁸⁸

A series of square planar Pd(II) complexes **321** were prepared by the reaction of $\text{Pd}(\text{TMEDA})(\text{CH}_3)_2$ (**320**) with fluoroalkyl iodides R_fI . A possible mechanism of the transformation involves oxidative addition followed by reductive elimination of CH_3I . Contrary to palladium complexes, polyfluoroalkyl iodides give octahedral Pt(IV) **323a** complexes by the reaction with $\text{Pt}(\text{TMEDA})(\text{CH}_3)_2$ (**322**). Performed at low temperatures, the reaction affords *trans*-Pt(TMEDA)- $(\text{CH}_3)_2(\text{R}_f)\text{I}$ **323a** as the kinetic products, while thermodynamically favored *cis*-complexes **323a** are formed at room temperature. In addition, isomerization of *trans*-isomers **323a** into the corresponding *cis*-isomers **323b** is observed in solution at room temperature (Scheme 177).³⁸⁹

Scheme 176

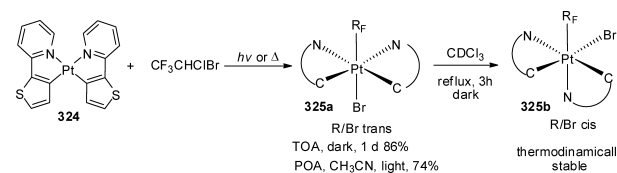


Scheme 177



Both thermal oxidative addition (TOA) and photochemical oxidative addition (POA) of Halothane to *cis*-bis[2-(2'-thienyl)pyridine] platinum(II) (**324**) led stereoselectively to the fluorinated platinum complex **325a**, which has *trans*-configuration. Isomerization of *trans*-**325a** into thermodynamically stable *cis*-**325b** was performed by refluxing in CDCl_3 in the dark (Scheme 178).³⁹⁰

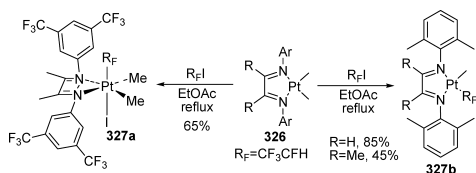
Scheme 178



A result of the reaction of $\text{Pt}(\text{diimine})\text{Me}_2$ complex **326** with CF_3CFHI depends on sterical demand of these diimine complexes. Pt(II) complexes **327b** were isolated as a result of reductive elimination of MeI for sterically hindered complexes with diimine ligands containing methyl groups in the 2,6-positions of the aryl group. In contrast, a less bulky complex, having two CF_3 -groups in the 3,5-positions, affords Pt(IV) complex **327a** exclusively (Scheme 179).³⁹¹

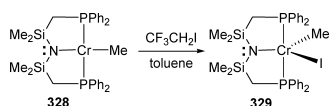
Unusual five-coordinated chromium(III) complex **329** was obtained by Fryzuk. It was found that coordinatively and electronically unsaturated chromium(II) complex **328** reacts with $\text{CF}_3\text{CH}_2\text{I}$ to undergo one-electron oxidation with formation of five-coordinated chromium(III) complex **329**. This chromium derivative **329** was tested as an ethylene

Scheme 179



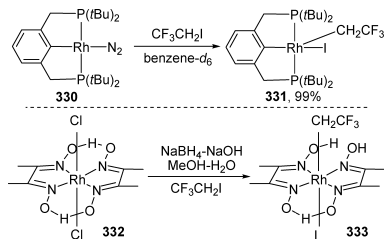
polymerization catalyst; however, low efficiency of **329** was shown despite the presence of an open coordination site. Complex **329** catalyzed polymerization but was quickly deactivated (Scheme 180).³⁹²

Scheme 180



Five-coordinated rhodium complex **331** was synthesized in quantitative yield by oxidative addition of $\text{CF}_3\text{CH}_2\text{I}$ to planar nitrogen–rhodium complex **330**.³⁹³ Octahedral rhodium complex **332** was converted into fluorinated derivative **333** by the reduction with NaBH_4 followed by oxidative addition of $\text{CF}_3\text{CH}_2\text{I}$ (Scheme 181).³⁹⁴

Scheme 181

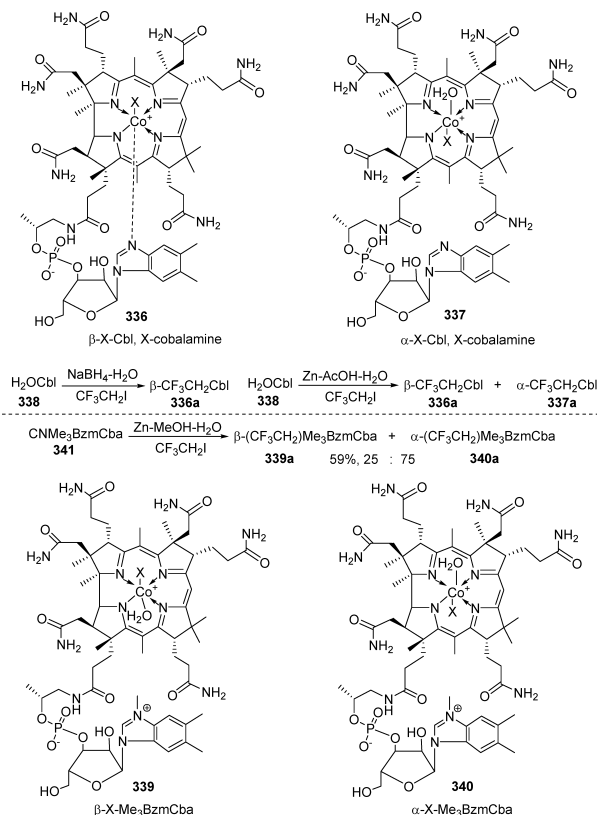


Reactions of PFEs with analogues of naturally occurred or natural like metal complexes were also reported. For instance, iron(II)-tetraphenylporphyrin complex **334** reacts with Halothane (CF_3CHClBr) in the presence of sodium dithionite to give σ -alkyl-Fe(III)-(TPP)-(CHClCF₃) complex **335a**. In contrast, the reaction with CF_3CCl_3 led to Fe(II) complex with chloro(trifluoromethyl)carbene **335b** (Scheme 182), with yields of more than 90%.³⁹⁵

A lot of investigations were carried out to understand the mechanism of action and properties of cobalamins. As a part of this study, several reactions of PFEs with cobalamins and analogues were maintained. Thus, the treatment of aquacobalamin H_2OCbl **338** with NaBH_4 followed by the reaction with 5–10-fold excess of $\text{CF}_3\text{CH}_2\text{I}$ led to β -isomer of $\text{CF}_3\text{CH}_2\text{Cbl}$ cobalamin **336a**.³⁹⁶ In contrast, using Zn as a

reductant afforded a mixture of β - and α -isomers **336a**, **337a**.³⁹⁷ Similarly, a 25:75 mixture of β -**339a** and α -isomers **340a** was obtained in the reaction of $\text{CF}_3\text{CH}_2\text{I}$ with cobalamide $\text{CNMe}_3\text{BzmCba}$ **341**, containing Me-group in the 3-position of benzimidazole ring.³⁹⁸ To clarify the terms β - and α -isomers, the definition is as follows. β -Isomer (**336**, **339**) has substituent and benzimidazole moiety on the opposite sides of the plane, containing corrin ring. In α -isomer (**337**, **340**), a substituent is situated “below” the corrin ring blocking the possibility of coordination of benzimidazole with cobalt atom (Scheme 183).

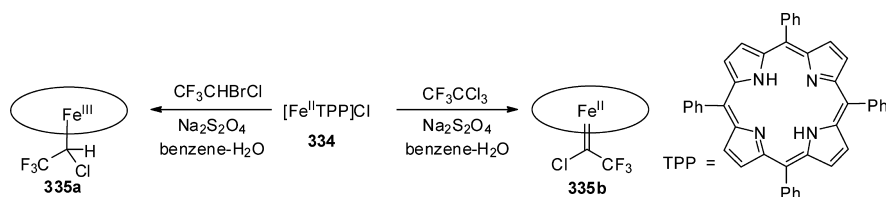
Scheme 183



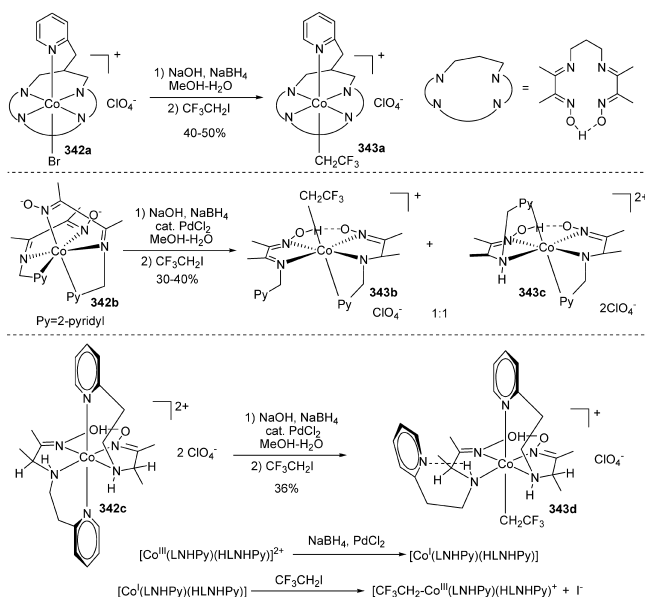
It was also shown that cobaloximes can be used as simple but effective models of cobalamins. A series of polyfluoroalkylated cobaloximes were prepared and used as such models. For that purpose, precursors **342** were reduced in situ with NaBH_4 in aqueous MeOH and treated with excessive $\text{CF}_3\text{CH}_2\text{I}$ to form trifluoroethylated complexes **343** in moderate yields (Scheme 184).³⁹⁹

Another good model of cobalamins is cobalt complexes with so-called “Salen” type ligands derived from salicylaldehyde and ethylenediamine. Standard two-step one-pot procedure (reduction with NaBH_4 followed by oxidative addition of $\text{CF}_3\text{CH}_2\text{I}$)

Scheme 182

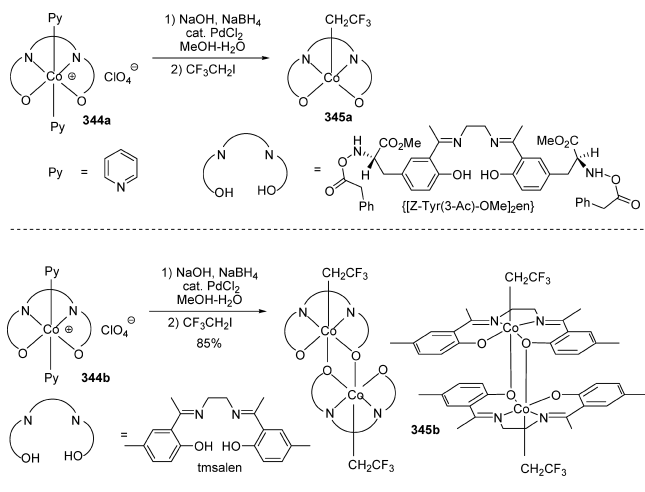


Scheme 184



was used to transform tyrosine derivative [Co(III){[Z-Tyr(3-Ac)-OMe]₂en}(Py)₂}]ClO₄ **344a**⁴⁰⁰ and [Co(tmsalen)(Py)₂]-ClO₄ **344v**⁴⁰¹ into the corresponding fluorinated cobalt complexes **345a**, **345b**. In the case of less sterically hindered tmsalen ligand, dimeric organometallic complex **345b** has been formed in high yield (Scheme 185).

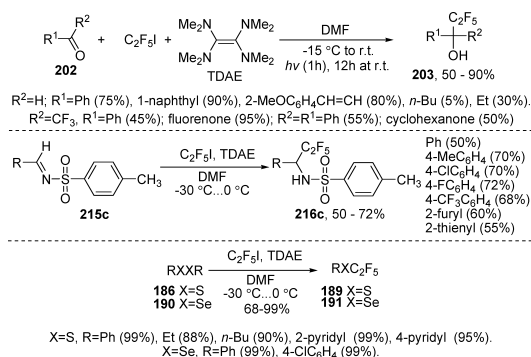
Scheme 185



3.2.9. Generation of Perfluorinated Anions Using Tetrakis(dimethylamino)ethylene (TDAE). An effective method for generation of C₂F₅-anion under mild conditions was elaborated by Pooput. It was found that TDAE can easily reduce C₂F₅I into C₂F₅-anion, forming dication TDAE²⁺. Thus, obtained C₂F₅-anion reacted with carbonyls **202**, imines **215c**, disulfides **186**, and diselenides **190** to afford the corresponding fluorinated alcohols **203**, amines **216c**, sulfides **189**, and selenides **191** in high yields (Scheme 186).⁴⁰²

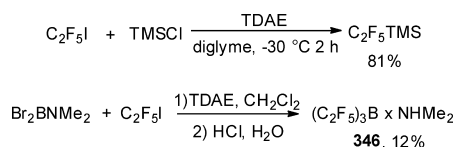
Pentafluoroethyltrimethyl silane (C₂F₅TMS) is a useful reagent for anionic pentafluoroethylation. It was synthesized by silylation of C₂F₅-anion generated under treatment of C₂F₅I with TDAE in diglyme.^{402b} Tris-(perfluoroethyl)borane **346**

Scheme 186



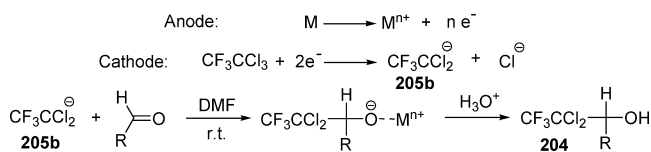
was prepared in low yield by the reaction of Br₂BNMe₂ with C₂F₅I in the presence of TDAE (Scheme 187).⁴⁰³

Scheme 187



3.2.10. Generation of Perfluoroethyl Anions by Electrolysis. Polyfluoroethyl anions can be generated also using electrolysis. Thus, electroreduction of CF₃CCl₃ in the presence of carbonyl compounds gave the corresponding fluorinated alcohols **204** by the addition of polyfluoroethyl anion **205b** to carbonyls. Electrolysis was carried out at room temperature in DMF at constant electric current in an undivided cell fitted with a magnesium or aluminum sacrificial anode and a nickel grid cathode freshly coated with an electrolytic deposit of cadmium (Scheme 188).⁴⁰⁴

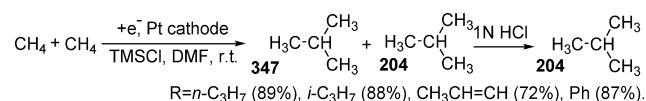
Scheme 188



Another variation of this approach includes the electroreductive coupling of CF₃CCl₃ with aldehydes. This technique used Pt cathode and a carbon anode in dry DMF containing Et₄NOTs. The cathodic and anodic chambers were separated by a ceramic diaphragm, and TMSI was added to the catholyte. After electrolysis and aqueous workup, a crude mixture of alcohol **204** and its TMS-ether **347** was treated with HCl to give pure final alcohol **204** in high yield (Scheme 189).⁴⁰⁵

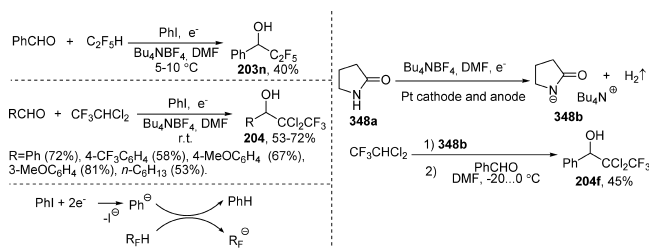
In the case of hydrogen-containing PFEs, the formation of anions was performed via the reaction with electrogenerated bases. Thus, electroreduction of phenyl iodide gives phenyl

Scheme 189



anion, which reacts with $\text{C}_2\text{F}_5\text{H}^{406}$ or $\text{CF}_3\text{CHCl}_2^{404}$ to form polyfluoroethyl anions. The addition of these species to aldehydes affords fluorinated alcohols **203n**, **204** in moderate to good yield. Alternatively, this transformation was performed using 2-pyrrolidone anion **348b** generated from 2-pyrrolidone (**348a**). The corresponding fluorinated benzyl alcohol **204f** was isolated in 45% yield. This method has been also used for trifluoromethylation of aldehydes with fluoroform (Scheme 190).⁴⁰⁷

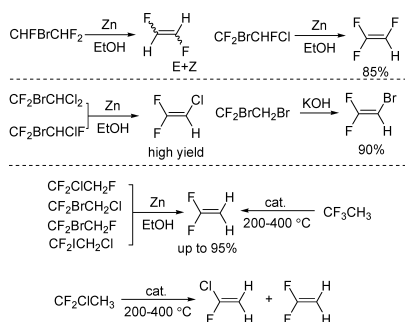
Scheme 190



3.2.11. Synthesis of Fluorinated Alkenes by β -Elimination Reaction. Fluorinated ethanes are a valuable laboratory source of fluorinated alkenes. A base- or zinc-induced elimination opens an excellent route for the preparation of variety of fluorinated alkenes by this way.^{29,31,32} For example, 1,1-difluoroethylene, 1,2-difluoroethylene, and 2-chloro-1,1-difluoroethylene can be prepared in high yield from the corresponding PFEs using zinc-induced elimination. It should be noted that this reaction requires the presence in the structure of starting freon halogens heavier than fluorine to form intermediate organozinc derivatives. However, there is no significant difference in yields and the reaction direction for chlorine-, bromine-, or iodine-substituted fluorinated ethanes. Alternatively, target fluorinated alkenes (for example, $\text{CF}_2=\text{CBrH}$) can be prepared by heating of C–H bond containing freons with solid KOH opening an excellent possibility for interconversion of alkenes through the halogenation–dehydrohalogenation procedure (Scheme 5, section 2).³¹ 1,1-Difluoroethylene and 1-chloro-1-fluoroethylene were prepared also by catalyzed dehydrohalogenation. These alkenes were prepared in up to 94% yield under a range of temperatures (200 – 400°C) using NiO, Fe_2O_3 , ZnO, Ag, charcoal catalysts (Scheme 191).⁴⁰⁸

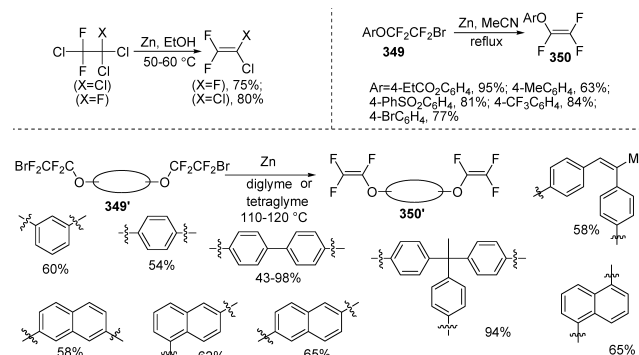
Dehalogenation of exhaustively halogenated PFEs can be also achieved using zinc to form the corresponding fluorinated alkenes. These reactions were carried out in EtOH,⁴⁰⁹ $\text{CH}_3\text{CN}^{410}$ under reflux or in diglyme and tetraglyme at

Scheme 191



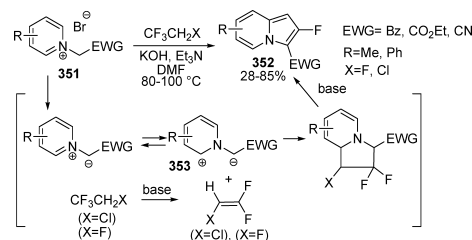
110 – 120°C (Scheme 192). The corresponding polyalkenes were also obtained in the case of derivatives of bis- and tris(hydroxy)arenes, synthesized by reaction of PFEs with phenols (Scheme 228, Table 31).⁴¹¹

Scheme 192



A series of fluoroindolizines **352** were synthesized using 1,3-dipolar [3 + 2] cycloaddition of *N*-ylides **353** to fluorinated ethenes $\text{CF}_2=\text{CXH}$. Pyridinium and isoquinolinium *N*-ylides **353** as well as fluorinated ethenes $\text{CF}_2=\text{CXH}$ were generated in situ from pyridinium or isoquinolinium bromides **351** and 1-chloro-2,2,2-trifluoroethane or 1,1,1,2-tetrafluoroethane correspondingly (Scheme 193).⁴¹²

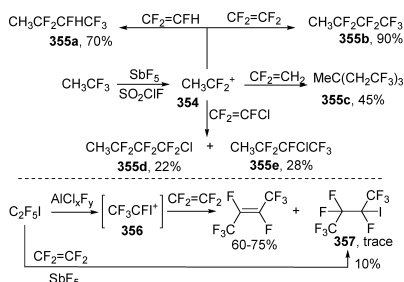
Scheme 193



3.3. Reaction of PFE Involving Carbocations

Fluorinated carbocations are very rare type of intermediates.^{1d,413} However, in some cases, such intermediates play an important role in the synthesis and chemical transformations of fluorinated ethanes. For example, halogen exchange and isomerization of freons with Lewis acids can proceed via formation of carbocation intermediates. Nevertheless, rather stable 1,1-difluoroethyl carbocation **354** can be prepared by reaction of SbF_5 with 1,1,1-trifluoroethane in SO_2ClF . This intermediate is stable enough at -80°C to be characterized using NMR spectroscopy. The ion shows a triplet at 4.50 ppm in ^1H NMR spectrum ($J_{\text{H-F}} = 17$ Hz) and a quadruplet at -96.4 ppm ($J_{\text{H-F}} = 17$ Hz) in ^{19}F NMR.⁴¹⁴ Cation **354** generated from SbF_5 with 1,1,1-trifluoroethane reacted with several fluorinated alkenes to form fluorinated adducts **355a–e** in good to high yields.⁴¹⁵ Also, the corresponding cationic intermediate **356** was postulated in the reaction of $\text{C}_2\text{F}_5\text{I}$ with such Lewis acids as SbF_5 and fluorinated aluminum chloride.⁴¹⁶ However, the reaction of this cation with tetrafluoroethylene gave only a trace amount of the desired adduct **357**. The main reaction product was perfluorinated but-2-ene (Scheme 194).

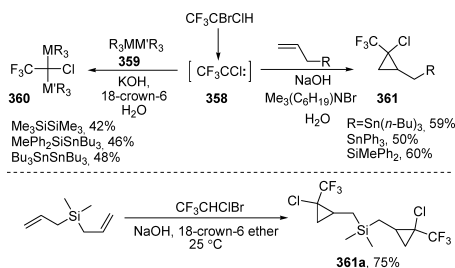
Scheme 194



3.4. Reactions of PFE Involving Carbenes

Because of the high electronegativity of the fluorine and other halogens, the generation of carbenes from PFEs can be performed easily using standard basic conditions for carbene reactions by α -elimination. Thus, using different phase transfer catalysts, trifluoromethylchloro carbene **358** (generated in situ from Halothane) reacts easily with organometallic compounds **359** to give insertion products **360**.⁴¹⁷ The addition of trifluoromethylchloro carbene **358** to alkenes affords trifluoromethyl-substituted cyclopropanes **361**, **361a** in good yields (Scheme 195).^{417,418}

Scheme 195



3.5. Reactions of PFE with Nucleophiles

Nucleophilic substitution of a halogen at a saturated (sp^3 hybridized) carbon atom is a classical reaction in organic chemistry. The mechanism of this transformation has been investigated very thoroughly and has become an essential part of any organic chemistry textbook for students. One could expect that the reaction of nucleophilic substitution in the case of C-2 freons is an obvious process as well. However, this is not true because a number of various mechanisms take place for this transformation in the case of fluorinated ethanes. For these types of compounds, at least four possible mechanisms of nucleophilic substitution are known. Therefore, all of these possibilities should be taken into account (Scheme 196).

Thus, in the case of CF_3CH_2Hal (Hal is mostly I and Br), the reaction of nucleophilic substitution can proceed as the classical S_N2 process (Scheme 196).

The elimination–addition sequence (E-Ad) can take place for PFEs containing one or more hydrogens. For example, a base-induced (as a rule the used nucleophile plays also the role of the base in this case) elimination of hydrogen halide from **22a** followed by an addition of the nucleophile to the intermediately formed alkene **362a** gives anion **363a**, which can form final product **364a** by protonation (the formal result of the nucleophilic substitution) or alkene **365a** by a halogen β -elimination. The regioselectivity of a nucleophilic addition is

controlled by fluorines, which direct the attack of a nucleophile on the fluorine adjacent carbon.

The $S_{RN}1$ mechanism is often realized in the reactions with S-nucleophiles, which one can explain by reductive properties of S-nucleophiles. Single electron transfer (SET) from sulfur nucleophile (behaving as a reducing agent) to the molecule of PFE initiates the reaction to give radical **366a** and radical-anion **23**, which transforms into fluorinated radical **24** by the elimination of the most heavy halide (weakest C–Hal bond of PFE is broken). Next, radical-anion **366b**, formed by the reaction of fluorinated radical **24** with a nucleophile, reacts with another molecule of PFE to form substitution product **364** and radical-anion **23**, which continues the process.

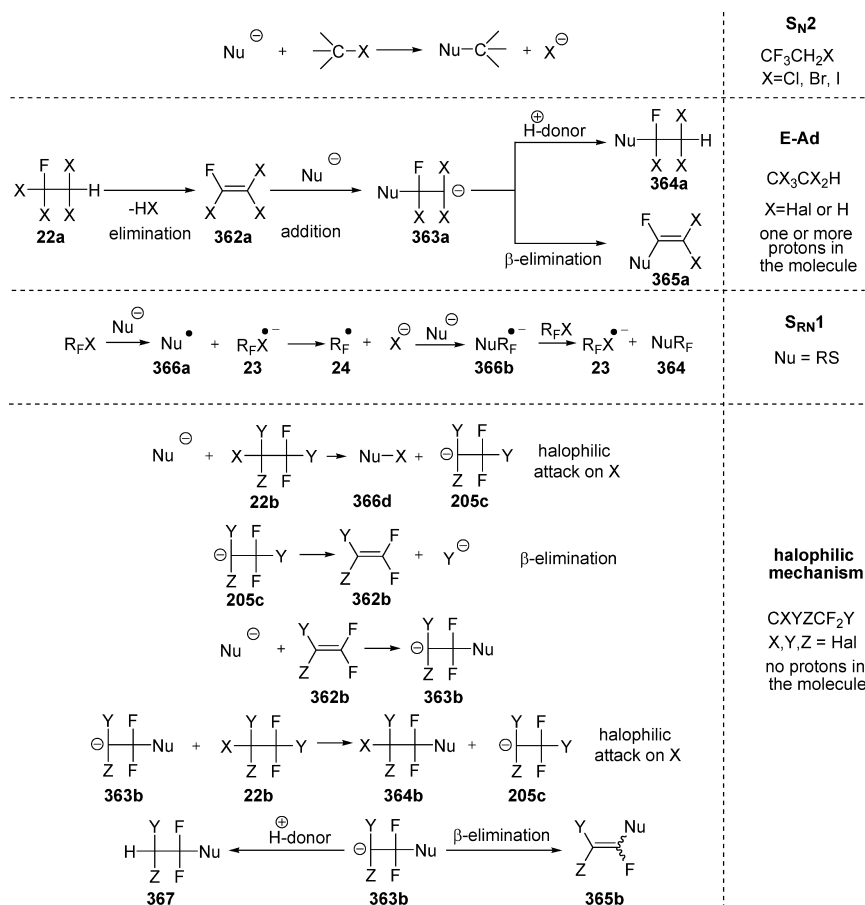
In the case of PFE **22b**, containing only halogens, so-called halophilic substitution is mostly observed due to that the sp^3 carbon atom is shielded by bulky groups, for example, by CF_3 and heavy halogens.^{203b} The first step of halophilic mechanism is the attack of a nucleophile on PFE, which gives anion **205c** and initiates the cascade transformation. β -Elimination in **205c** affords alkene **362b** and yields anion **363b** by the addition of nucleophile. Next, halophilic attack on **363b** results in final product **364b** and anion **205c** to continue the processes. Side products **367** and **365b** are formed by protonation and β -elimination correspondingly. It should be noted that the formation of side products **367** and **365b** is the sharp distinction of nucleophilic substitution in PFE from a general case (Scheme 196).

To illustrate the variety and the complexity of nucleophilic substitutions in case of PFE, several typical examples will be given. After that the main part of this section organized by types of nucleophiles will be presented. Thus, 2,2,2-trifluoroethyl bromide looks as a very simple substrate to be used in reactions with nucleophiles. Despite the presence of electron-withdrawing CF_3 group, this alkylation agent is much less reactive in S_N2 reaction than expected. The kinetic study of the reaction of some alkyl bromides with potassium iodide in acetone reveals a decrease of rate of the reaction of 2,2,2-trifluoroethyl bromide as compared to propyl bromide by a factor of 6350.⁴¹⁹ The rate of the reaction of CF_3CH_2I with thiophenolates in methanol is lower by a factor 1.7×10^4 as compared to ethyl iodide.⁴²⁰ That means that reaction is retarded significantly but the steric factor could not explain such a significant change of the reaction rate. The CF_3 group has a conformation energy (A-value) almost insignificantly higher than ethyl and methyl groups, 2.1, 1.75, and 1.7 kcal/mol, respectively. Therefore, not only steric but electronic influence of the CF_3 group (probably destabilizing the S_N2 transition state by a repulsion of a nucleophile with fluorine lone pair of CF_3 group takes place) should be taken into account, and modern theoretical study of reactions of fluorinated ethanes with nucleophiles is very desirable.

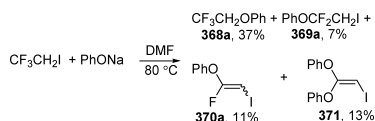
When nucleophile is a strong base itself or the reaction is performed in the presence of strong base, the mechanism is changed totally. For example, the reaction of sodium phenolate with CF_3CH_2I in DMF gave not only the expected S_N2 product **368a** but also ether **369a** as well as alkenes **370a**, **371**, which are the result of E1cb elimination to form 1,1-difluoro-2-iodoethene followed by the addition of phenolate and protonation or F^- elimination (E-Ad mechanism). Product **371** is a result of the double substitution with the participation of phenolate (Scheme 197).⁴²¹

The influence of leaving group in the case of nucleophilic substitution with freons is very important. Fluorine and

Scheme 196



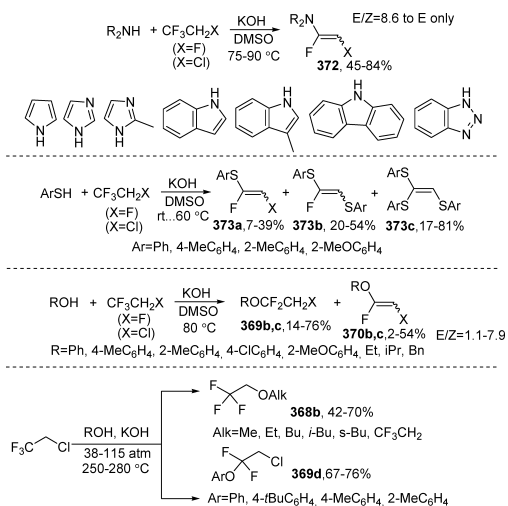
Scheme 197



chlorine substituents being poor nucleofuges shifted the mechanism of nucleophilic substitution reaction from $\text{S}_{\text{N}}2$ to the elimination–addition mechanism. For example, freons $\text{CF}_3\text{CH}_2\text{Cl}$ and $\text{CF}_3\text{CH}_2\text{F}$ reacted with various *O*-, *S*-, and *N*-nucleophiles in DMSO using potassium hydroxide as a base to give only the products resulted from elimination of HF –addition of nucleophile sequence. Only products of fluorovinylations 372,373 were formed in the reaction with thiols and amines. In the case of thiols, total halogen substitution can be performed to give 373c. Contrary, a mixture of products 369b,c, 370b,c is formed in the case of reaction with phenols and alcohols.⁴²² Changing the reaction conditions from heating at 80°C in DMSO to heating in neat at $250\text{--}280^\circ\text{C}$ led to selective formation of saturated $\text{S}_{\text{N}}2$ products 368b in the case of alcohols or E-Ad products 369d in the case of less nucleophilic phenols (Scheme 198).⁴²³

Generally, the reaction of $\text{CF}_3\text{CH}_2\text{X}$ with thiols proceeds at higher rate and better yields, as compared to the reactions with *O*-, *N*-nucleophiles. Such acceleration is a result of $\text{S}_{\text{RN}}1$ process, and the transformation is accelerated by illumination, UV-irradiation, or heating and suppressed by a radical scavengers such as 1,3-dinitrobenzene or hydroquinone. Moreover, the formation of radical intermediates was confirmed

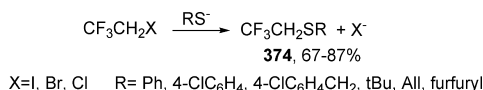
Scheme 198



by ESR spectra. It should be noted that $\text{S}_{\text{RN}}1$ mechanism was only confirmed for iodo, bromo, and chloro derivatives but not for $\text{CF}_3\text{CH}_2\text{F}$. Corresponding substitution products 374 were generally obtained in high yields (Scheme 199).⁴²⁴

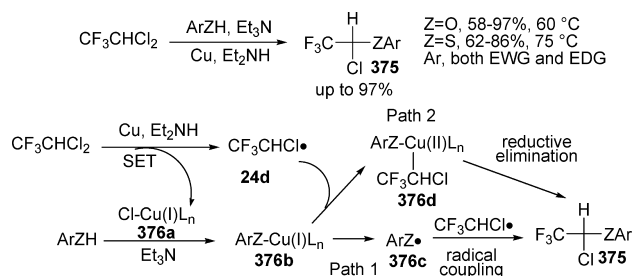
In addition to the above-mentioned mechanisms, there is an alternative pathway of the nucleophilic substitution, which is connected with application of metallic copper as a catalyst. It was shown that the copper-mediated cross-coupling reaction of CF_3CHCl_2 with phenols and thiophenols leads to a series of fluorinated ethers and thioethers 375 in high yields (Scheme

Scheme 199

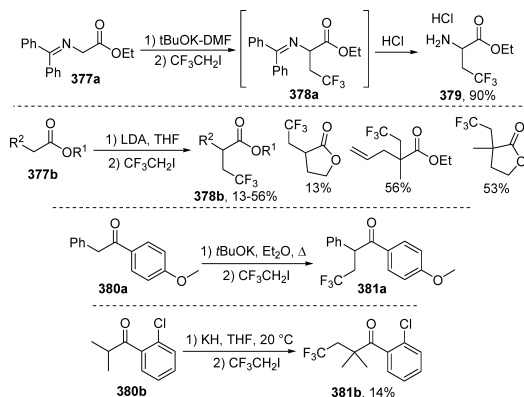


200). The proposed mechanism for the transformation can be realized as follows (Scheme 201). First, single electron transfer

Scheme 200



Scheme 201



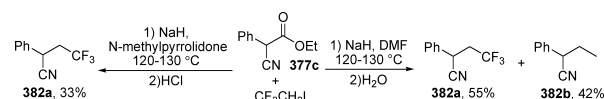
(SET) from Cu to CF₃CHCl₂ affords radical **24d** and copper chloride complex **376a**, which reacts with nucleophile to form intermediate **376b**. Next, **376b** eliminates copper chloride to form radical **376c**, which couples with fluorinated radical **24d** to give final product **375** (path 1). Alternatively, radical **24d** reacts with **376b** to give copper(II) intermediate **376d**, which transforms into **375** by reductive elimination (path 2).⁹⁰

3.5.1. Reactions with C-Nucleophiles. Enolates, derived from ketones and esters, can be easily alkylated with CF₃CH₂I to provide a versatile approach to fluorinated ketones and esters. Trifluomethylated alanine ester **379** can be prepared quite easily by alkylation of ethyl *N*-(diphenylmethylene) glycinate **377a** with 2,2,2-trifluoro-1-iodoethane to afford the desired product **378a** almost quantitatively. Hydrolysis of **378a** with HCl afforded amino ester **379** in 90% overall yield after crystallization as its hydrochloride salt.⁴²⁵ Similarly, a series of trifluoromethylated esters **378b** and ketones **381** was prepared successfully starting from **377b** and **380** (Scheme 201).⁴²⁶

Alkylation of phenyl cyanoacetate **376c** with CF₃CH₂I in DMF afforded a mixture of fluorinated and nonfluorinated butanenitriles **382a** and **382b**. These two products are formed due to that the iodide anion formed in situ attacks the ethoxy moiety of the ester group to induce nucleophilic substitution and decarboxylation. The iodoethane formed as a byproduct behaves as alkylating agent to form **382b**. In the case of

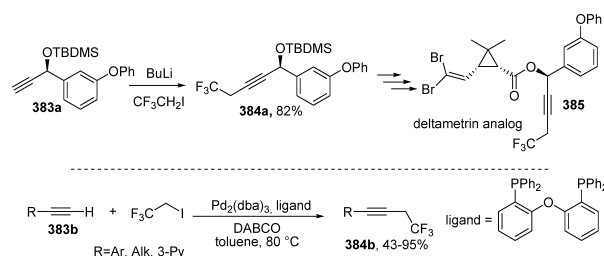
application of *N*-methylpyrrolidone as a solvent, only fluorinated butanenitrile **382a** was isolated in moderate yield after acidic workup (Scheme 202).⁴²⁷

Scheme 202



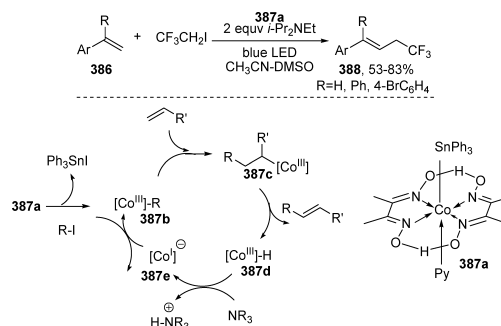
The reaction of CF₃CH₂I with lithium acetylide of **383a** allowed one to prepare trifluoroethylated acetylene **384a**, which was used as a key building block in the synthesis of the pyrethroid deltamethrin analogue **385** having ethynyl group instead of cyano moiety in the structure.⁴²⁸ Pyrethroid insecticides, which affect sodium channels in neurons, are widely used to control insect pests in agriculture and in the home. The biological study of prepared derivatives of deltamethrin showed that not only the cyano but also the ethynyl group can stimulate neurons, although the efficiencies on the inductive level were different. The trifluoroethylation of acetylenes can be also performed in conditions of the Sonogashira reaction. Thus, the interaction of terminal acetylenes **383b** with CF₃CH₂I was carried out in toluene using a Pd₂(dba)₃–phosphine–DABCO catalytic system to give trifluoroethylated acetylenes **384b** in up to 95% yields (Scheme 203).⁴²⁹

Scheme 203



An interesting photochemical Heck-type coupling reaction catalyzed by simple cobalt complexes was reported recently by Carreira.⁴³⁰ The reaction of styrenes **386** with CF₃CH₂I under catalysis with cobaloxime-complex **387a** and irradiation with blue LED affords alkenes **388** in good to high yields, providing a useful bond construction strategy for the preparation of allylic trifluoromethanes (Scheme 204). The process may be also carried out in photochemical flow reactor, giving a noticeable productivity increase. A mechanism of the transformation was

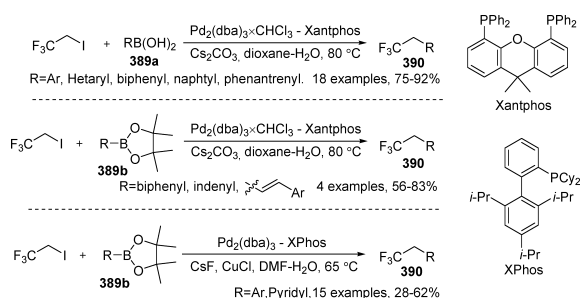
Scheme 204



not discussed in the report. However, in previous works, the authors proposed a hypothesis for cobalt-catalyzed alkyl Heck-type couplings. Alkylation of **387a** with alkyl iodide affords Co(III)–R complex **387b**, to initiate the process. Next, the addition of **387b** to the alkene leads to cobalt(III) intermediate **387c**, which transforms into the 1,2-disubstituted alkene by elimination of hydrido-cobalt intermediate **387d**. Proton transfer from **387d** to amine base leads to ammonium salt and the regenerated form of cobalt catalyst **387e** (Scheme 204).

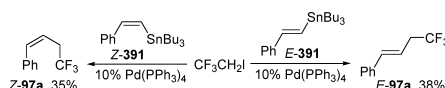
$\text{CF}_3\text{CH}_2\text{I}$ was successfully involved in the Suzuki coupling with boronic acids **389a**. The reaction was carried out in a dioxane–water mixture using $\text{Pd}_2(\text{dba})_3$ -Xantphos catalysis to give **390** in high yields.⁴³¹ In the case of dioxaborolanes **389b**, the corresponding compounds **390** were obtained in lower yields. In addition to $\text{Pd}_2(\text{dba})_3$ -Xantphos, $\text{Pd}_2(\text{dba})_3$ -Xphos was also used as a catalytic system in this case (Scheme 205).⁴³²

Scheme 205



1-Iodo-2,2,2-trifluoroethane can be also used in the palladium-catalyzed cross-coupling of the Stille type. Thus, both *Z*- and *E*-isomers of styrene **97a** were prepared in moderate yields starting from *Z*- and *E*-alkenyl stannanes **391** (Scheme 206).⁴³³

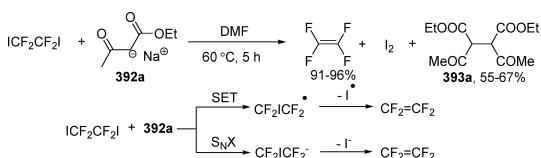
Scheme 206



In contrast, exhaustively halogenated PFEs rarely reacted with C-nucleophiles to give substitution products with creation of a new C–C bond. Usually the formation of fluorinated ethylenes as products of dehalogenation (see the corresponding part) or dimerization of carbanions was observed. For example, sodium acetoacetate **392a** reacted with $\text{ICF}_2\text{CF}_2\text{I}$ to give dimer **393a** and tetrafluoroethylene in high yields.⁴³⁴ The formation of tetrafluoroethylene can be explained in terms of SET reduction or the halophilic mechanism (Scheme 207).

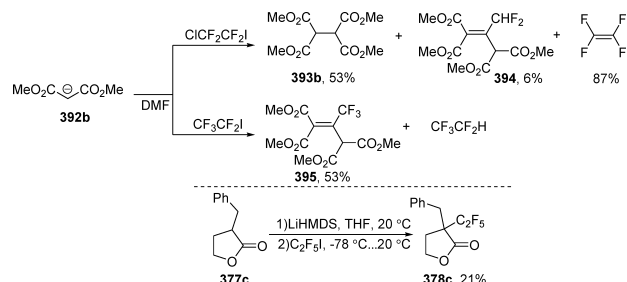
Similarly, the reaction of dimethyl malonate **392b** with $\text{ClCF}_2\text{CF}_2\text{I}$ afforded dimer **393b** and tetrafluoroethylene with a minor admixture of “double substitution” product **394**. In contrast, the reaction with $\text{C}_2\text{F}_5\text{I}$ afforded the substitution

Scheme 207



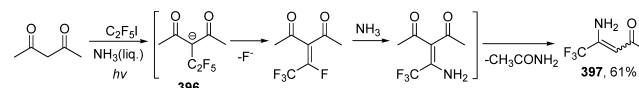
product in higher yield. The corresponding ethylene **395** was isolated in 53% yield.⁴³⁵ Perfluoroalkylation of γ -lactone **377c** with $\text{C}_2\text{F}_5\text{I}$ led to fluorinated derivative **378c** in 21% yield (Scheme 208).⁴³⁶

Scheme 208



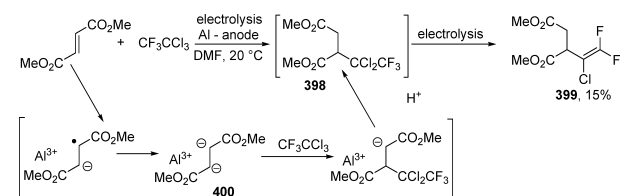
When perfluoroalkylation of acetylacetone was performed in liquid ammonia under UV-irradiation, CF_3 -substituted enaminoketone **397** was isolated. The transformation starts from the formation of α -pentafluoroethylated anion **396**, and subsequent elimination of fluoride resulted in intermediate α,β -unsaturated diketone having a double bond highly activated for Michael addition. As a result, the reaction with ammonia to give enaminoenone followed by extrusion of acetamide affords the final product **397** (Scheme 209).⁴³⁷

Scheme 209



Dianion **400** was generated electrolytically from dimethylfumarate using aluminum anode. This nucleophile reacted with CF_3CCl_3 to yield alkene **399** in low yield. The possible mechanism postulates formation of the nucleophilic substitution product **398**, which gives the final compound by dehalogenation (Scheme 210).⁴³⁸

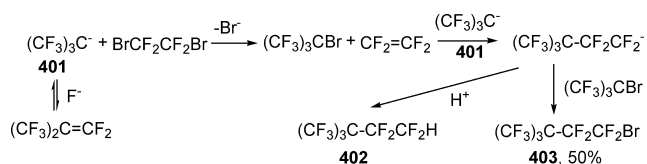
Scheme 210



Perfluorinated *tert*-butyl anion **401** can be obtained very easily by the addition of fluoride to perfluoroisobutylene in diglyme. The reaction of the formed anion with $\text{BrCF}_2\text{CF}_2\text{Br}$ gave simultaneously the corresponding bromide **403** and reduced product **402**, the formation of which can be explained by halophilic attack (Scheme 211).⁴³⁹

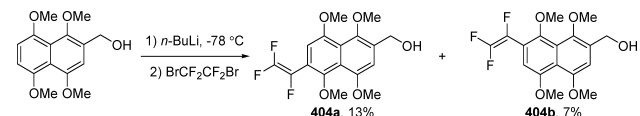
In rare instances, the reaction of PFE with organolithium compounds leads to formal products of nucleophilic substitution in low yields. The major reaction product is trifluorovinyl-substituted derivative formed from the starting carbanion. Thus, formation of **404a**, **404b** is the result of nucleophilic addition of organolithium compound to tetra-

Scheme 211



fluoroethylene, followed by elimination of LiF (Scheme 212).⁴⁴⁰

Scheme 212



3.5.2. Reactions with *N*-Nucleophiles. Reaction of $\text{CF}_3\text{CH}_2\text{I}$ and $\text{CF}_3\text{CH}_2\text{Br}$ with *N*-nucleophiles (amides, imides, sulfamides, nitrogen heterocycles) was performed usually in DMF to give $\text{S}_{\text{N}}2$ -type products in yields strongly depending on nucleophile structure. Generally, better yields were achieved in the case of amides (Table 27).

The *N*7- and *N*9-regioisomers of trifluoroethyl-substituted adenine **405a**, **405b** were prepared regioselectively by the direct alkylation of *N*6-[(dimethylamino)methylene]adenine **406a** (protection of NH_2 group can be performed by the reaction with DMF acetal **407**) or adenine **406b**. However, yields of target derivatives are rather moderate (Scheme 213).⁴⁵⁴

Ionic liquids became more and more popular in application as solvents for organic materials and green chemistry. This new type or media for various reactions is arguably the key to clean processes and green chemistry. It is known that imidazolium salts having a fluorinated anion have a number of benefits such as low melting point, viscosity, and lipophilicity. Trifluoroethylated 1-methylimidazolium salt **408a** was prepared using the reaction of *N*-methylimidazole with 1,1,1-trifluoroethyl iodide. Quite long heating in a sealed vessel gave target product **408a** only in 20% yield.⁴⁵⁵ Better results were obtained in the case of pyridine ring. 1,2-Dibromotetrafluoroethane reacted easily with DMAP under copper assistance to yield pyridinium salt **409**. When *p*-dinitrobenzene was added to the reaction mixture, no product was observed. This was considered by the authors as a possibility of $\text{S}_{\text{RN}}1$ sequence to occur initiated by a single-electron transfer (Scheme 214).⁴⁵⁶

Double substitution takes place in the reactions of CF_3CHCl_2 with a number of different secondary amines to form 1,1-bis(dialkylamino)-2,2,2-trifluoroethanes **410** generally in good yield. The subsequent reaction of these fluorinated aminsals **410** with BuLi opens a route to 1,1-bis(dialkylamino)-2,2-difluoroethene **411**, which is a very useful building block and can participate in reactions with nucleophiles, electrophiles, [2 + 4], and even [2 + 2] cycloadditions to form valuable fluorinated building blocks **412a–i**.⁴⁵⁷ Inverse electron demand Diels–Alder reaction takes place for acrolein and methyl vinyl ketone, but in the case of unsaturated esters and nitriles **413** the formation of [2 + 2] cycloadducts **412h** is observed (Scheme 215).

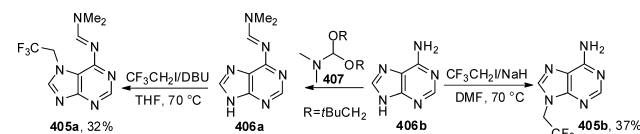
Secondary amines were successfully involved into the reaction with some PFE. Thus, $\text{BrCF}_2\text{CF}_2^-$ and CFBrClCF_2^- derivatives **414a** were obtained with an admixture of reduced products **414b** by treatment of $\text{BrCF}_2\text{CF}_2\text{Br}$ and $\text{CFBrClCF}_2\text{Br}$

Table 27. Reactions of $\text{CF}_3\text{CH}_2\text{I}$ and $\text{CF}_3\text{CH}_2\text{Br}$ with *N*-Nucleophiles

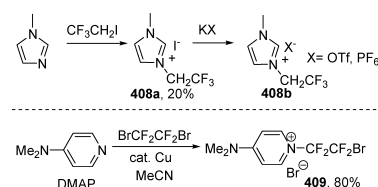
PFE reaction conditions	Ref	Product	PFE reaction conditions	Ref	Product
$\text{CF}_3\text{CH}_2\text{I}$, K_2CO_3 , DMF, 20–85 °C, 138 h	441		$\text{CF}_3\text{CH}_2\text{I}$	442	
$\text{CF}_3\text{CH}_2\text{I}$, K_2CO_3 , DMF, 70 °C	443		$\text{CF}_3\text{CH}_2\text{I}$, NaH	444	
$\text{CF}_3\text{CH}_2\text{I}$, Cs_2CO_3 , DMF, 80 °C	445		$\text{CF}_3\text{CH}_2\text{I}$, NaH, DMF, 60 °C, 24 h	446	
$\text{CF}_3\text{CH}_2\text{I}$, K_2CO_3 , DMF, 80 °C, 12 h	447		$\text{CF}_3\text{CH}_2\text{I}$, K_2CO_3 , DMF, 40 °C, 20 h	448	
$\text{CF}_3\text{CH}_2\text{I}$, K_2CO_3 , DMF, 20 °C	449		$\text{CF}_3\text{CH}_2\text{I}$, NaH, DMF, 50–60 °C, 3 h	450	
$\text{CF}_3\text{CH}_2\text{I}$, NaH, DMF, 150 °C, 4 h	451		$\text{CF}_3\text{CH}_2\text{I}$, NaH, DMF, 50–60 °C, 3 h	452	
$\text{CF}_3\text{CH}_2\text{I}$, Cs_2CO_3 , DMF	452		$\text{CF}_3\text{CH}_2\text{Br}$, NaH, DMF, 20 °C	453	

^aThe yield was not given.

Scheme 213



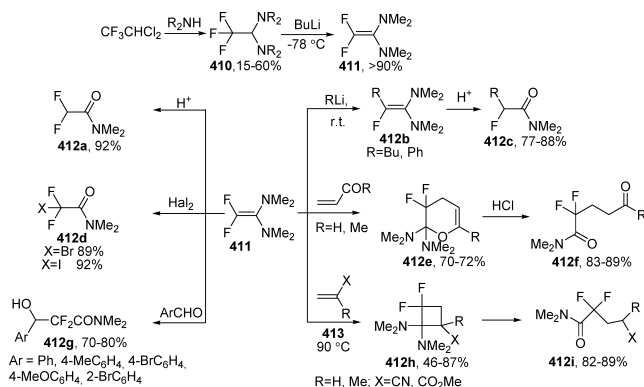
Scheme 214



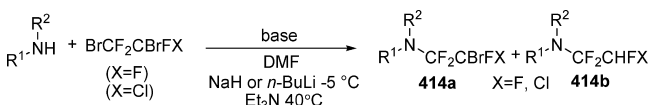
with amines in DMF. Higher yields of target derivatives **414a** were observed in the presence of strong bases (Scheme 216, Table 28).⁴⁵⁸

Generated by treatment of heterocycles **415a** with NaH, azolium salts **415b** reacted easily with $\text{BrCF}_2\text{CF}_2\text{Br}$, $\text{ClCF}_2\text{CFCl}_2$, $\text{ClCF}_2\text{CF}_2\text{I}$, and ClCF_2CFCl under Bu_4NI catalysis. The corresponding fluorinated derivatives **416a–d**

Scheme 215



Scheme 216

Table 28. Reactions of BrCF₂CF₂Br and CBrCF₂CF₂Br with Secondary Amines

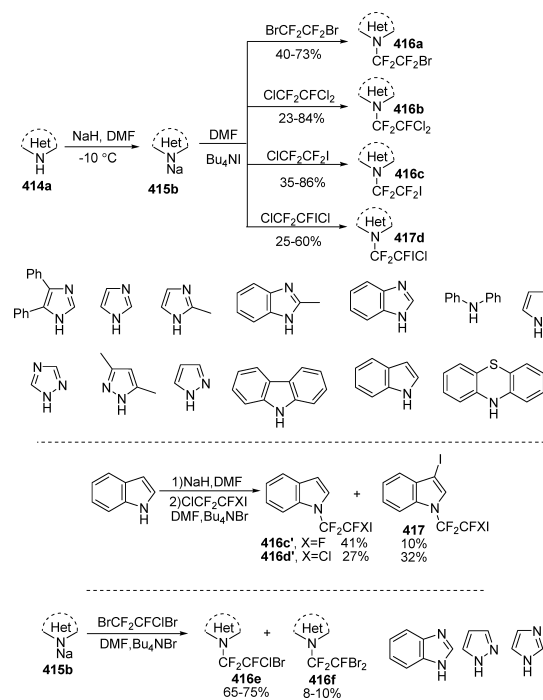
R ¹ R ² NH	X	conditions	yield of 414a, %	yield of 414b, %
Et ₂ NH	F	Et ₃ N, 40 °C	29	57
		<i>n</i> -BuLi, Et ₃ N, -5 °C	42	6
(CH ₂) ₅ NH		Et ₃ N, 40 °C	12	35
		NaH, Et ₃ N, -5 °C	45	1
(CH ₂) ₄ NH		NaH, Et ₃ N, -5 °C	31	4
Et ₂ NH	Cl	without base	21	4
		Et ₃ N, 40 °C	57	2
		<i>n</i> -BuLi, -5 °C	32	0
		<i>n</i> -BuLi, Et ₃ N, -5 °C	53	0
		NaH, Et ₃ N, -5 °C	77	5
(CH ₂) ₃ NH		Et ₃ N, 40 °C	45	8
		NaH, Et ₃ N, -5 °C	87	3
(CH ₂) ₄ NH		Et ₃ N, 40 °C	25	0
		NaH, Et ₃ N, -5 °C	68	1
<i>i</i> -Pr ₂ NH		<i>n</i> -BuLi, Et ₃ N, -5 °C	5	0

were isolated in good to high yields. It should be noted that substitution of chlorine took place instead of iodine atom in the case of iodo-perfluoroethanes ClCF₂CF₂I and ClCF₂CF₂Cl, which is consistent with the halophilic reaction mechanism. In the case of iodo-perfluoroethanes ClCF₂CF₂I and ClCF₂CF₂Cl, perfluoroalkylation of indole afforded ring-iodination products **417** in addition to the products of *N*-alkylation **416c'**, **416d'**. The reaction with BrCF₂CF₂Br gave desired CF₂CF₂Br-azoles **416e** and minor amounts of CF₂CF₂Br₂-azoles **416f** as byproducts (Scheme 217).⁴⁵⁹

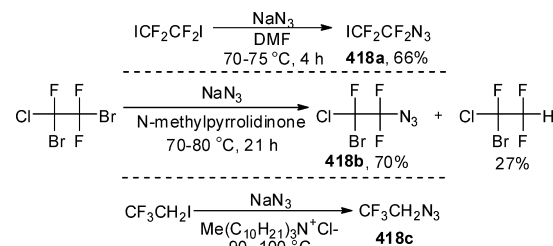
Treatment of ICF₂CF₂I and ClCFBrCF₂Br with NaN₃ at 70–80 °C afforded the corresponding fluorinated azides **418a**, **418b** in high yields.⁴⁶⁰ The reaction of CF₃CH₂I with NaN₃ was also performed at 90–100 °C in conditions of phase transfer catalysis to give azide **418c** (Scheme 218).⁴⁶¹

3.5.3. Reactions with P-, As-, and Sb-Reagents. The reaction of ethyl phosphonites **420** with 2,2,2-trifluoroethyl iodide gave ethyl (2,2,2-trifluoroethyl)phosphinates **419** in moderate yields by the Michaelis–Arbuzov reaction.⁴⁶² Similarly, the reaction of ethyl diphenylphosphinite **421** with 2,2,2-trifluoroethyl iodide gave 2,2,2-trifluoroethyldiphenyl-

Scheme 217

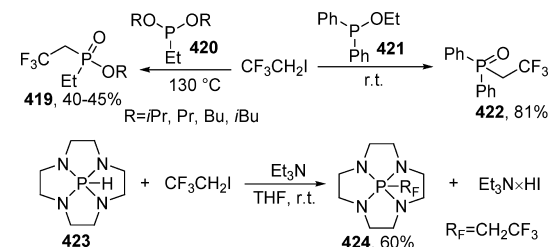


Scheme 218



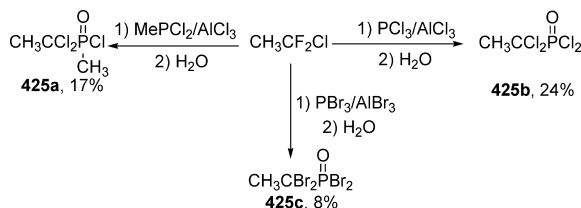
phosphine oxide **422**. Cyclenphosphorane **423** reacts with 2-iodo-1,1,1-trifluoroethane in the presence of triethylamine at room temperature in tetrahydrofuran to form 1,1,1-trifluoroethylcyclenphosphorane **424** in 60% yield (Scheme 219).⁴⁶³

Scheme 219



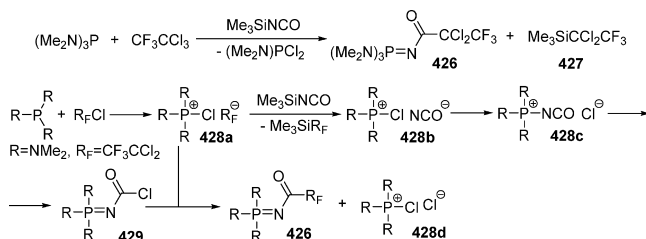
The reaction of CH₃CF₂Cl with phosphorus trichloride, phosphorus tribromide, and CH₃PCl₂ in the presence of aluminum chloride or bromide (Kinnear–Perren reaction) was investigated. It was found that the result of this transformation is the formation of chlorinated or brominated phosphonic or phosphinic halides **425a–c**. Complete defluorination takes place during the reaction to form chlorinated or brominated products, however, in low yields (Scheme 220).⁴⁶⁴

Scheme 220



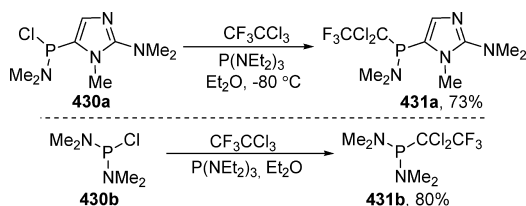
The three-component reaction, involving CF_3CCl_2 , $\text{P}(\text{NMe}_2)_3$, and trimethylsilyl isocyanate, led to iminophosphite **426** and perfluoroethylated organosilicon derivative **427**. The first step of the reaction is halophilic attack on CF_3CCl_2 , forming chlorophosphonium salt **428a**, which reacts with TMSNCO to give cyanate **428b**. The rearrangement of **428b** affords derivative **428c**, which is attacked by chloride to yield compound **429**. The last step of the reaction is the substitution of chloride in compound **429**, leading to fluorinated iminophosphite **426** and dichlorophosphine **428d** (Scheme 221).⁴⁶⁵

Scheme 221



The substitution of chloride at phosphorus atom in compound **430a** by the CF_3CCl_2 moiety was achieved by the treatment of phosphorus electrophile **430a** with CF_3CCl_2 in the presence of $\text{P}(\text{NEt}_2)_3$. The corresponding derivative **431a** was isolated in 73% yield.⁴⁶⁶ Similarly, polyfluoroethylated phosphine **431b** was synthesized starting from diamidochlorophosphite **431b** (Scheme 222).⁴⁶⁷

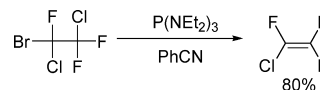
Scheme 222



It should be noted that the reaction of PFE with $\text{P}(\text{NEt}_2)_3$ carried out without any electrophiles leads to dehalogenation. This transformation can be explained by halide extrusion from anion formed in the reaction mixture by halophilic attack. For example, reaction of $\text{CF}_2\text{ClCFBrCl}$ with $\text{P}(\text{NEt}_2)_3$ led exclusively to 1-chloro-1,2,2-trifluoroethene in 80% yield (Scheme 223).⁴⁶⁸

Bis(diethylamido) phosphites **432** are very valuable reagents for the conversion of alcohols to a variety of important products. These reagents can be prepared very easily by the phosphitylation of alcohols with $(\text{Et}_2\text{N})_2\text{PCl}$. Bis(diethylamido) phosphites reacted smoothly with CF_3CCl_2 to yield poly-

Scheme 223



fluoroalkylated derivatives **433**. Intermediate phosphonium salt behaves as alkylating agent transferring an alkyl group to the CF_3CCl_2 anion. On the basis of that reaction, a very convenient and generally applicable method for a two-step substitution of hydroxy groups in primary alcohols for a wide range of different fluorinated species was elaborated (Scheme 224).^{469,470}

Perfluoroalkylation of trimethylphosphine, arsine, stibine was carried out under treatment with $\text{C}_2\text{F}_5\text{I}$ to yield $\text{R}_2\text{XC}_2\text{F}_5$ **435**⁴⁷¹ and the corresponding salts **434**. In the case of trimethylsilylphosphines **436**, the reaction with $\text{C}_2\text{F}_5\text{I}$ afforded pentafluoroethylphosphines **234** and trimethylsilyl iodide (Scheme 225).^{301c,472}

Trimethylsilylpentafluoroethane was prepared by three-component reaction of $\text{C}_2\text{F}_5\text{I}$, TMSCl , and $\text{P}(\text{NEt}_2)_3$ in benzonitrile or butyronitrile (Scheme 226).⁴⁷³

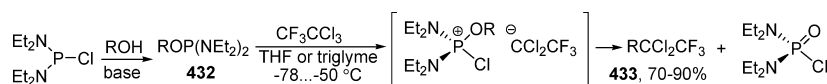
It was reported that 1,2-dibromotetrafluoroethane was used in the Atherton–Todd reaction.⁴⁷⁴ Dialkyl phosphites **437** were transformed into phosphoramidates **438** by treatment with 1,2-dibromotetrafluoroethane in the presence of diethylamine.⁴⁷⁵ Heating of $\text{C}_2\text{F}_5\text{I}$ with a mixture of copper powder and finely ground red phosphorus, previously reduced by heating in a stream of hydrogen, afforded a mixture of $(\text{C}_2\text{F}_5)_2\text{PI}$ and $(\text{C}_2\text{F}_5)\text{PI}_2$ in low yield.⁴⁷⁶ The reaction of 1-bromo-1,1-difluoroethane with sodium dimethylphosphide gave the corresponding nucleophilic substitution product **439** in 32% yield under heating at 120 °C during 48 h (Scheme 227).

3.5.4. Reactions with O-Nucleophiles. A series of ethers **368** bearing a CF_3CH_2 group was prepared via the Williamson reaction. The corresponding reaction of $\text{CF}_3\text{CH}_2\text{I}$ and $\text{CF}_3\text{CH}_2\text{Br}$ with O-nucleophiles was performed in polar solvents (DMF, DMSO, and acetone) using K_2CO_3 , Cs_2CO_3 , or NaH as bases. Phenols, allylic, and heterocyclic alcohols were mostly used as nucleophiles to give $\text{S}_\text{N}2$ -type products in variable yields (Scheme 228, Table 29).

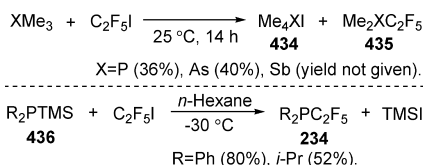
Halothane is widely used in medicine as an anesthetic. A number of fluorinated ethers have been synthesized for evaluation as volatile general anesthetics. Those ethers having one H with at least two halogens other than fluorine or two or more hydrogens with at least one bromine or one chlorine atom were found the best anesthetics. For example, the reflux of sodium methoxide with Halothane in methanol resulted in the formation of a mixture of the corresponding ethers **440a** and **441**, the formation of which can be rationalized as an elimination–addition tandem to form the products of formal fluorine substitution.⁴⁸⁸ An addition of a phase transfer catalyst allowed one to perform the reaction selectively to give only one product. The reaction of Halothane with phenols and alkoxides prepared from the corresponding alcohols resulted in perfluoroethylated ethers **440a–e** (formal products of one fluorine substitution) in moderate yields.⁴⁸⁸ Very similarly, propargyl derived ethers **440f–h** were synthesized in high yields from various propargyl alcohols.⁴⁸⁹ Tyrosine derivative **440i**⁴⁹⁰ and several heterocyclic compounds **440j–l**⁴⁹¹ having $\text{OCF}_2\text{CHClBr}$ fragment were also prepared in moderate yields using crown ether **442** (Scheme 229).

An attempt to prepare the corresponding ether from $\text{CF}_2\text{ClCH}_2\text{I}$ led to isolation of methyl iodoacetate **444a** in

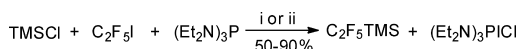
Scheme 224



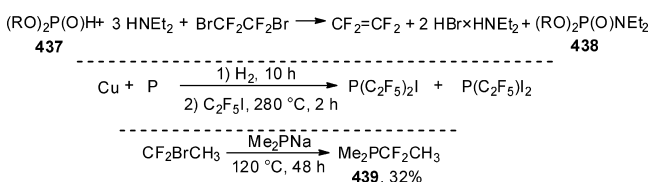
Scheme 225



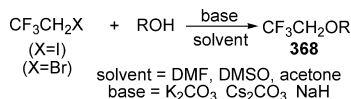
Scheme 226

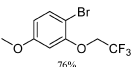
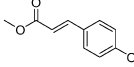
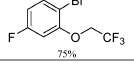
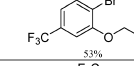
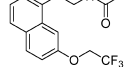
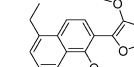
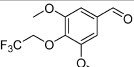
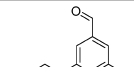
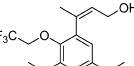
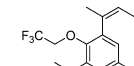
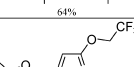
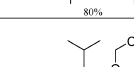
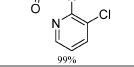
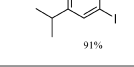


Scheme 227



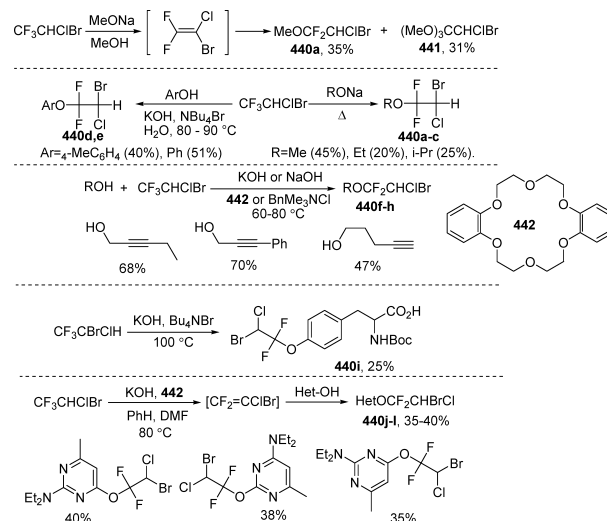
Scheme 228

Table 29. Reactions of CF₃CH₂I and CF₃CH₂Br with O-Nucleophiles

PFE Reaction conditions ref	Product	PFE Reaction conditions ref	Product
CF ₃ CH ₂ I K ₂ CO ₃ , DMF 70 °C, 144 h	 76%	CF ₃ CH ₂ I 1) NaH, DMSO 0 °C; 2) DMSO 80 °C, 24 h	 52%
CF ₃ CH ₂ I K ₂ CO ₃ , DMF 70 °C, 144 h	 75%	CF ₃ CH ₂ I K ₂ CO ₃ , DMF 70 °C, 144 h	 53%
CF ₃ CH ₂ I Bu ₄ NBr, K ₂ CO ₃ , acetone, reflux	 10%	CF ₃ CH ₂ I NaH, DMF 0 °C	 10%
CF ₃ CH ₂ I K ₂ CO ₃ , DMF 80-110 °C	 50%	CF ₃ CH ₂ I K ₂ CO ₃ , DMF 50 °C, 168 h	 50%
CF ₃ CH ₂ I Cs ₂ CO ₃ , DMF	 64%	CF ₃ CH ₂ I Cs ₂ CO ₃ , DMF	 80%
CF ₃ CH ₂ I K ₂ CO ₃ , DMF 100 °C	 99%	CF ₃ CH ₂ Br Cs ₂ CO ₃ , DMF 50 °C	 91%
CF ₃ CH ₂ I K ₂ CO ₃ , NaH, DMF 80-120 °C	 42%	CF ₃ CH ₂ Br not given	 not given

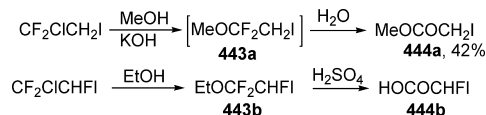
42% yield as a result of ether **443a** formation followed by hydrolysis of two C–F bonds to the carbonyl moiety. Quite

Scheme 229



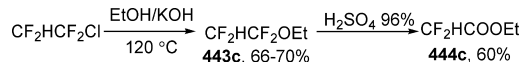
interesting is the chemoselectivity of the substitution. No substitution at the iodomethylene fragment but the reaction on the CF₂Cl moiety takes place.⁴⁹² Similarly, the reaction of CF₂ClCHFI with ethanol gives substitution product at CF₂Cl group **443b**, subsequent hydrolysis of which permits the preparation of iodoethanoic acid **444b** (Scheme 230).^{31c}

Scheme 230



The reaction of 1-chloro-1,1,2,2-tetrafluoroethane (freon 124A) with ethanol in the presence of potassium hydroxide gave nucleophilic substitution product **443c** in good yield (most probably via formation of tetrafluoroethylene). Subsequent acidic hydrolysis in concentrated sulfuric acid opens a simple route to ethyl difluoroacetate **444c** (Scheme 231).⁴⁹³

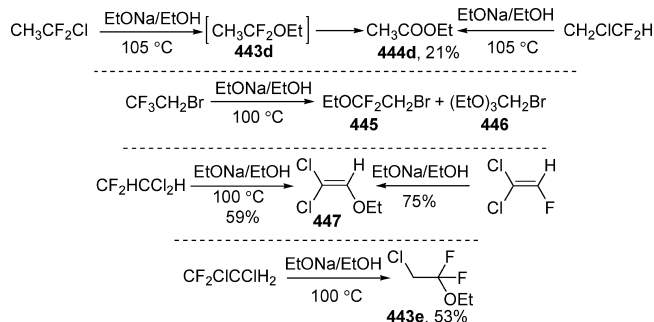
Scheme 231



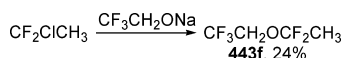
The reaction of other C-2 freons with sodium ethoxide gave an additional confirmation that this is not a simple displacement reaction of the Williamson type in most instances. The products formed in a number of reactions can be accounted for only by assuming that these saturated compounds are converted to fluoroolefins, which then reacted with the alcohol to give the final fluoroether (Scheme 232).⁴⁹⁴

Reaction of sodium trifluoroethoxide, prepared from trifluoroethanol and sodium, with 1-chloro-1,1-difluoroethane gave the corresponding ether **443f** in 24% yield (Scheme 233).⁴⁹⁵

Scheme 232

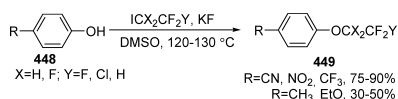


Scheme 233



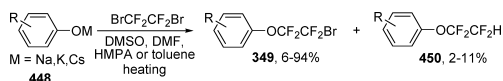
The alkylation of substituted phenols **448** with fluoroethyl iodides, affording the corresponding fluoroalkyl phenol ethers **449**, can be performed using KF as a base. These conditions permit one to avoid undesirable multiple reaction paths to give only the products of the iodide substitution. Nearly neutral reaction conditions, high yields (generally), short reaction times, and a simple workup procedure are advantages of the method that work well with various fluorinated ethyl iodides. It was postulated that KF forms a complex with phenols to activate the nucleophilic substitution (Scheme 234).⁴⁹⁶

Scheme 234



Reaction of BrCF₂CF₂Br with *O*-nucleophiles mostly resulted in substitution of bromine with phenolates, generated from phenols **448** by means of KOH, Cs₂CO₃, or NaH. In the general case, target fluorinated ethers **349** were obtained in high yields by heating in DMF, HMPA, or DMSO, and only reaction in the KOH–DMF system afforded desired ethers **349** in low yields. As in the case of *N*-nucleophiles, formation of minor reduced products **450** was also documented. The mechanism of the transformation is the same as that mentioned above for the general case (Scheme 235, Table 30).

Scheme 235



Heteroalkoxides were also used for substitution. Thus, sodium salt of 3-hydroxyindazole **451** gave the corresponding product **349a** after treatment with BrCF₂CF₂Br in DMF (Scheme 236).⁵⁰⁶

Fluorinated chlorosulfonyl chloride **349b** was synthesized using BrCF₂CF₂Br. Nucleophilic substitution with 4-bromophenol was performed as the first step. Next, the second bromide was substituted with sodium dithionite. As the last step, the treatment with chlorine afforded **349b** in 61% total yield (Scheme 237).⁵⁰⁷

Diphenoxides, obtained from biphenols **448'**, reacted with BrCF₂CF₂Br to give bis(CF₂F)₂ derivatives **349'** in high yields. In most cases, compounds **349'** were converted into polyfluorodienes **350'** by the treatment with Zn. These dienes are very useful monomers for the preparation of fluorine-containing polymers (Scheme 238, Table 31).

Reactions of other exhaustively halogenated PFEs with *O*-nucleophiles can be found rarely in the literature. BrCF₂CFBrCl,⁵⁰⁰ ClCF₂CFCl₂, ClCF₂CCl₃, and CF₃CCl₃ afforded the corresponding fluorinated ethers **453**, **455** in good to high yields by the reaction with alkoxides. It should be noted that in all cases products with RO-substituent attached to CF₂-group were obtained. Byproducts of the transformation, compounds **454**, **456**, were obtained only in minor amounts (Scheme 239, Table 32).

The substitution of both iodine atoms in ICF₂CF₂I was achieved with methoxy group by a two-step procedure. The treatment of ICF₂CF₂I by methoxide in DMSO gave intermediate fluorinated alkoxide **457a**, which was treated with methyl *p*-toluenesulfonate to afford fluorinated dimethoxyethane **457b** in 75% yield (Scheme 240).⁵¹¹

3.5.5. Reactions with S-Nucleophiles and Other Chalcogen Nucleophiles. Similarly to *N*- and *O*-nucleophiles, *S*-nucleophiles can react with CF₃CH₂I and CF₃CH₂Br to form S_N2-type products. Using this approach, a series of thioethers **367**, containing alkyl-, aryl, hetaryl moieties, and CF₃CH₂ group, was prepared (Scheme 199, Table 33, Scheme 241).

The reaction of 1-bromo-1-chloro-2,2,2-trifluoroethane with various aliphatic and aromatic thiols in the presence of sodium dithionite and sodium bicarbonate leads to the corresponding sulfides **458**. Selective substitution of bromine indicates that most probably the reaction proceeds via a radical mechanism initiated by the sulfoxylate radical anion generated from the dithionite anion. The reaction proceeds at 40–45 °C giving 1-chloro-2,2,2-trifluoroethyl sulfides **458** with hydroxy, amino, amide, and ester functional groups in 32–80% yield. However, only disulfide was formed in the case of *t*-butyl thiol under the reaction conditions.⁵²¹ This reaction has been also studied in pure nucleophilic conditions (4-MeC₆H₄SH–NaH) resulting in up to 73% yield of sulfide in NMP.⁵²² High selectivity of bromine substitution should be noted for these conditions (Scheme 242, Table 34).

Similarly to *N*- and *O*-nucleophiles, the reaction of exhaustively halogenated PFEs with *S*-nucleophiles gave a mixture of target substitution product with an admixture of reduced product. Thus, the reaction of BrCF₂CF₂Br with thioalkoxides in benzene afforded the corresponding products **459a**, **460** in low yields.^{497a} In contrast, using DMF as a solvent allowed one to increase the yields dramatically.^{502,523} C₂F₅I was also involved in the reaction with thiolates to provide compounds **459b** in good to high yields starting from aminothiols **461** (Scheme 243).⁵²⁴

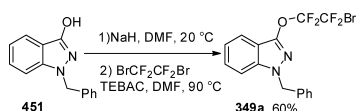
In the case of CF₃CCl₃, the reaction with thiophenoxide is not selective and results in the formation of a complex mixture of fluorinated products **459c**, **459d**, **462a**, **462b**, and disulfide **186b**. The reaction is very sensitive to the nature of solvent to give complete inversion of chemoselectivity of the substitution. Such results can be explained by processes shown in Scheme 244.⁵²⁵ The key step of this interaction is the halophilic reaction with thiophenoxide.

As it was mentioned above, *S*-nucleophiles have a tendency to react with PFEs by means of S_{RN}1 mechanism involving

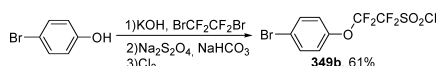
Table 30. Reactions of BrCF₂CF₂Br with Phenoxides

R	yield of 349, %	ref	R	yield of 349, %	ref
H	26	497	4-CH ₃ O	85	498,504
4-Cl	20	497a	4-CN	94	504
4-NO ₂	6	497a	4-ClC ₆ H ₄ CO	86	497a
2-allyl	7.5	497a,499	4-CO ₂ Et	95	410a,b
H	83	500	4-CO ₂ Me	63	410a,b
4-Br	92	501,502	H	65	503
3,5-di(CO ₂ Me) ₂	62	501d	4-Me	49	13c,411a
4-Me	70	410b	3-CF ₃	85	13c,411a
3-F, 5-Br	95	504	4-PhSO ₂	81	410c
H	84	504	2,5-diCl	not given	505
4-Cl	89	504	2,5-diBr	not given	505
2-CH ₃	80	504	2,3,5,6-tetraF	not given	505

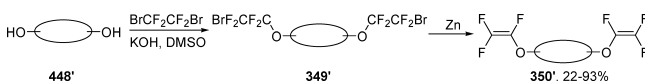
Scheme 236



Scheme 237



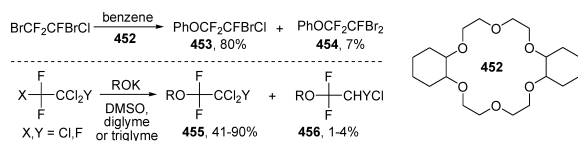
Scheme 238

Table 31. Reactions of BrCF₂CF₂Br with Bisphenoxides

Ar	Yield of 350', % ^a	ref	Ar	Yields of 350', % ^a	ref
	50	411a		56	411d
	53	411a		43	411d
	65 ^a	411f		60 ^a	411f
	70	411f		22	411a
	93	411a		78	411a
				41 ^b	508

^aTotal yield after treatment with Zn. ^bYield of bis(ether) 349'.

Scheme 239

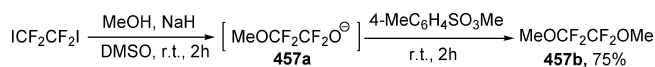


radicals. In the case of exhaustively halogenated PFEs, this mechanism is observed most frequently. Substitution reaction is assisted by irradiation, SO₂, or Na₂SO₃, which are good mediators of electron transfer through the formation of radical anions SO₂^{•-} and SO₃^{•-} (see Scheme 40). The addition of the

Table 32. Reactions of Exhaustively Halogenated PFEs with O-Nucleophiles

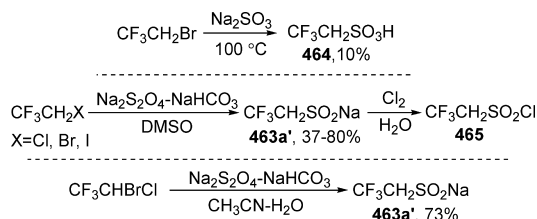
R	X	Y	yield of 455, %	yield of 456, %	ref
Ph	Cl	F	37–79	1–9	609,610
4-MeC ₆ H ₄	Cl	F	35	9	509
4-MeOC ₆ H ₄	Cl	F	31	10	509
4-ClC ₆ H ₄	Cl	F	8	3	509
PhCH ₂	Cl	F	53	<1	510
C ₂ H ₅ C(CH ₃) ₂	Cl	F	75	<1	510
Ph	F	Cl	78	<1	510
4-MeOC ₆ H ₄	F	Cl	76	1	510
PhCH ₂	F	Cl	76	2	510
C ₂ H ₅ C(CH ₃) ₂	F	Cl	41	<1	510
Ph	Cl	Cl	88	2	510
4-MeC ₆ H ₄	Cl	Cl	91	4	510
4-NH ₂ C ₆ H ₄	Cl	Cl	74	2.4	510
4-ClC ₆ H ₄	Cl	Cl	70	3.7	510
PhCH ₂	Cl	Cl	61	2.4	510
C ₂ H ₅ C(CH ₃) ₂	Cl	Cl	40	<1	510

Scheme 240

Table 33. Reactions of CF₃CH₂I and CF₃CH₂Br with S-Nucleophiles

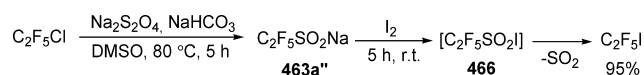
PFE Reaction conditions	Ref	Product	PFE Reaction conditions	Ref	Product
CF ₃ CH ₂ I 1) KF, MeOH 60 °C, 0.34 h 2) DMSO 120 °C, 16 h	512		CF ₃ CH ₂ I KOH, MeOH heating, 1 h	513	
CF ₃ CH ₂ I DBU DMF, benzene Ambient temperature, 4 h	514		CF ₃ CH ₂ I K ₂ CO ₃ , DMF	515	
CF ₃ CH ₂ I K ₂ CO ₃ , acetone Ambient temperature, 1 h	516		CF ₃ CH ₂ I DBU, DMF 45–55 °C, 14 h	517	
CF ₃ CH ₂ Br K ₂ CO ₃ , acetone heating, 5 h	518		CF ₃ CH ₂ I MeONa, MeOH	519	
			CF ₃ CH ₂ Br DMF	520	

Scheme 247



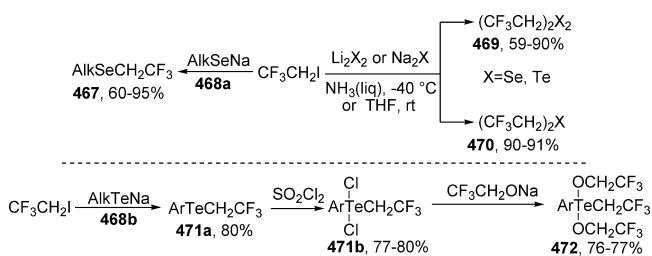
An efficient two-step method for the preparation of pentafluoroiodoethane from chloropentafluoroethane was elaborated by Zhang.⁵³⁴ The treatment of $\text{C}_2\text{F}_5\text{Cl}$ with $\text{Na}_2\text{S}_2\text{O}_4\text{--NaHCO}_3$ in DMSO afforded the corresponding fluorinated sulfinate **463a''**, which was transformed into $\text{C}_2\text{F}_5\text{I}$ by treatment with iodine in 95% yield (Scheme 248). The reaction proceeds through the formation of unstable sulfonyl iodide **466**, which extrudes easily SO_2 to give the final product.

Scheme 248



The reaction of 2,2,2-trifluoro-1-iodoethane with lithium and sodium diselenide or ditelluride performed in THF at room temperature gave the corresponding diselenides and ditellurides **469** in good yields. This method was found to be very general to synthesize various diselenides and ditellurides from alkylating agents.⁵⁴¹ Symmetrical selenides and tellurides **470** were prepared by alkylation of sodium selenide and telluride.⁵⁴² The reaction of sodium alkylselenolates **468a** with 2,2,2-trifluoro-1-iodoethane has been used to prepare unsymmetrical selenides **467** with 1,1,1-trifluoroethyl fragment. Similarly, the corresponding tellurides **471a** have been synthesized from **468b**. Chlorination of these tellurides with sulfuryl chloride yielded the corresponding tellurium dichlorides **471b**, which on further treatment with $\text{CF}_3\text{CH}_2\text{ONa}$ in trifluoroethanol gave the corresponding bis(2,2,2-trifluoroethoxides) **472** (Scheme 249).⁵⁴³

Scheme 249

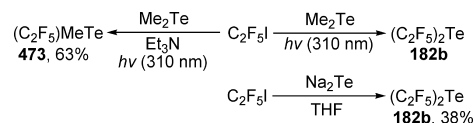


Perfluorinated organotellurium compounds were prepared via perfluoroalkylation of sodium telluride or dimethyltellurium by $\text{C}_2\text{F}_5\text{I}$. Reactions were photochemically or thermally induced to afford the corresponding fluorinated organotellurium compounds **182b**, **473** in good yields (Scheme 250).⁵⁴⁴

3.6. Olefination Using PFE

3.6.1. Wittig, Horner, and Fujita–Hiyama Olefination. Synthesis of alkenes starting from carbonyl compounds, so-called carbonyl olefination, is one of the most powerful tools for the creation of a new $\text{C}=\text{C}$ double bond. So, it is not

Scheme 250



surprising to see PFEs involved in various olefination techniques to construct various fluorinated alkenes. CF_3CCl_3 was used in the synthesis of alkenes via modification of the Wittig reaction. These syntheses were carried in DMF,⁵⁴⁵ CH_3CN ,⁵⁴⁶ and even in solventless conditions⁵⁴⁷ to give the corresponding β -chloro- β -trifluoromethylstyrenes **272** in mostly good to high yields. Stereoselectivity of the reaction depends strongly on the carbonyl compound nature to afford 1/1 mixtures of isomers or pure Z-isomer in the case of cinnamic aldehyde, for instance (Scheme 251, Table 37).

Scheme 251

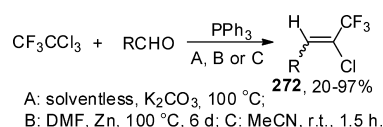


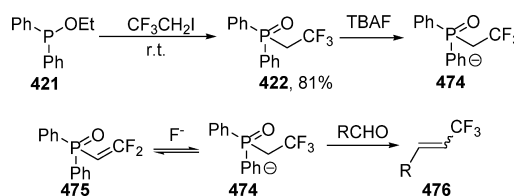
Table 37. Wittig Olefination Using PFEs

R	Yield of 272 , % (Z/E), conditions, ref. ^a	R	Yield of 272 , % (Z/E), conditions, ref. ^a
4-ClC ₆ H ₄	86 (75:25)	3-ClC ₆ H ₄	84 (75:25)
4-MeC ₆ H ₄	77 (75:25)	2-ClC ₆ H ₄	76 (85:15)
4-BrC ₆ H ₄	97 (100:0)	3-BrC ₆ H ₄	94 (98:2)
4-NO ₂ C ₆ H ₄	93 (54:46)	Ph	77 (68:32)
4-MeOC ₆ H ₄	73 (100:0)	3,4-(OCH ₂ OC ₆ H ₄)	65 (86:14)
4-CN ₂ C ₆ H ₄	68 (100:0)	PhCH ₂ CH ₂	48 (65:35)
4-FC ₆ H ₄	84 (97:3)	PhCH(Me)	55 (57:43)
2-NO ₂ C ₆ H ₄	75 (100:0)	PhCH=CH	49 (100:0)
1-Naphthyl	32 (100:0)		43 (92:8) ^b
Ph	56 (90:10) ^c		20 ^c

^aConditions A, ref 547, if otherwise noted. ^bConditions B.⁵⁴⁵
^cConditions C.⁵⁴⁸

Treatment of trifluoromethylated phosphine oxide **422**, obtained by the alkylation of **421** with $\text{CF}_3\text{CH}_2\text{I}$, with aromatic aldehydes in the presence of TBAF at room temperature afforded the corresponding trifluoromethylated alkenes **476** via Horner reaction in good yields. The reaction proceeds via intermediate anion **474**, which exists in the equilibrium with difluoroalkene **475**. As a rule, a mixture of E- and Z-isomers is formed, and the stereoselectivity of the reaction is difficult to predict (Scheme 252, Table 38).⁵⁴⁹

Scheme 252



Another well-known olefination method, which employs PFEs, is the reaction elaborated by Fujita and Hiyama. The treatment of carbonyl compounds with a PFE and Zn (2 equiv and more) in DMF gave fluorinated alcohols **204**, which were converted into acetates **477** by the reaction with Ac_2O . Subsequent treatment with Zn afforded final alkenes **272**.^{338,339}

Table 38. Reactions of PFE with Thiolate 76

R	yield of 476, E/Z	R	yield of 476, E/Z
Ph	63%, 4/1	4-MeOC ₆ H ₄	75%, 19/1
4-NCC ₆ H ₄	64%, 100/0	4-MeOCOC ₆ H ₄	65%, 19/1
2-indolyl	73%, 100/0	3-Py	66%, 4/1
PhCHCH	83%, 3/2	PhCH ₂ CH ₂	60%, 1/5
4-ClC ₆ H ₄	74%, 4/1	4-Py	58%, 100/0
4-MeCOC ₆ H ₄	24%, 2/3	2-naphthyl	98%, 6/1

One-pot variant of the synthesis was also elaborated. For example, using both mentioned methods, a wide variety of alkenes 272 was prepared in high yields^{63,346} (Scheme 253, Table 39).

Scheme 253

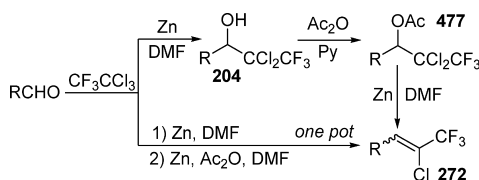


Table 39. Fujita–Hiyama Carbonyl Olefination

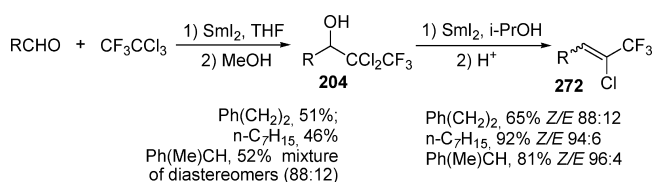
Three-step procedure				
R	yield of 204, %	yield of 477, %	yield of 272 (E/Z), %	ref.
	Et	93	86 (86:14)	338, 339
	3-PhOC ₆ H ₄ CH ₂	74	98	74 (86:14)
	2-Me-3-PhOC ₆ H ₄ CH ₂	86	100	95 (89:11)
	Ac	70	98	88
Me2CH-CH=	-a	-a	82 (85:15)	338
One-pot procedure				
R	yield of 272 (E/Z), %	R	yield of 272 (E/Z), %	ref.
Ph	75-78 (87:13)	Me2C=CH-	53 (88:12)	63, 346
3,4-(OCH2OC ₆ H ₃)	81 (87:13)	4-ClC ₆ H ₄	73-76 (88:12)	63, 346
E-Ph-CH=CH-	72 (89:11)	2-BrC ₆ H ₄	67 (82:18)	549
cyclo-C ₆ H ₁₁	50 (85:15)	2-ClC ₆ H ₄	72 (83:17)	549
1-Naphthyl	62 (88:12)	1-furyl	54 (86:14)	550
	54 (84:16)		46 (81:19)b	551
R = 2-Me-3-PhC ₆ H ₃ CH ₂			47 (95:5)b	551
	50 (79:21)b		40 (89:11)b	551
R = PhCH ₂	36		44 (93:7)b	551
4-ClC ₆ H ₄	26			
4-MeOC ₆ H ₄	28			

^aThe yield is not given. ^bAcetylation of phenolic OH is also observed.

A similar approach to chlorotrifluoromethylstyrenes 272 was elaborated using samarium(II) iodide. Being a strong reducing agent, samarium(II) iodide triggers the sequence of transformations to result in the formation of polyfluoroethyl anion followed by its addition to the aldehyde. Formed fluorinated alcohols 204 were converted into alkenes 272 by subsequent treatment with an additional amount of samarium(II) iodide (Scheme 254).⁵⁵⁵

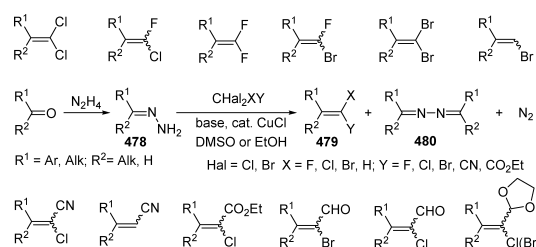
3.6.2. Catalytic Olefination Reaction (Nenajdenko–Shastin Olefination). In 1999 a novel catalytic olefination

Scheme 254



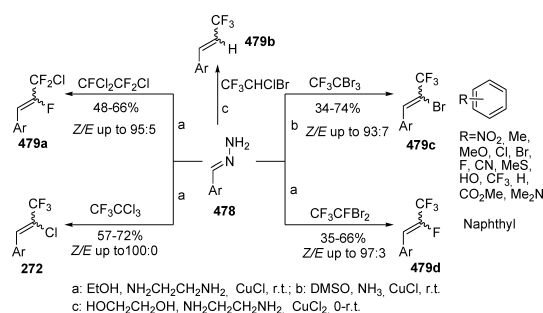
reaction was elaborated by Nenajdenko and Shastin.⁵⁵⁶ It was found that *N*-unsubstituted hydrazones 478 can be converted into alkenes 479 by the treatment with polyhalogenated alkanes in the presence of a base (usually ammonia or ethylenediamine) and catalytic amounts of copper salts. The only side products of the transformation are symmetrical azines 480, which are usually formed in minor amounts and can be separated very easily. The reaction has a wide synthetic scope allowing one to prepare both alkyl and aryl halogenoalkenes, including fluorinated ones and derivatives with functional groups.⁵⁵⁷ A simple experimental procedure, which does not require use of organometallic or toxic organophosphorous compounds, has affordable price, availability of starting materials, high yields, and stereoselectivity as distinct advantages of the reaction (Scheme 255).

Scheme 255



In the case of PFEs, catalytic olefination reaction opened a novel stereoselective pathway to the β -halo- β -trifluoromethylstyrenes. Using CF₃CFBr₂, CF₃CCl₃, and CF₃CBr₃, β -fluoro-479d,⁵⁵⁸ β -chloro-272,⁵⁵⁹ and β -bromo- β -trifluoromethylstyrenes 479c⁵⁶⁰ were synthesized stereoselectively in good to high yields. When CF₂ClCF₂Cl was used in catalytic olefination reaction, the corresponding β -fluoro- β -chlorodifluoromethylstyrenes 479a was prepared in good yields.⁵⁵⁹ Z-Isomers dominate in the isomeric mixtures of alkenes, and in some cases the reaction led to Z-isomeric alkenes only (Scheme 256, Table

Scheme 256



40). Because of the presence of labile β -halogen atom, β -halogenostyrenes, obtained easily by Nenajdenko–Shastin olefination, appeared to be very valuable building blocks for the synthesis of various fluorinated compounds by reaction with nucleophiles and heterocyclizations.⁵⁶¹ Recently, Hanamoto demonstrated⁵⁶² a practical and convenient synthesis of (*E*)- β -(trifluoromethyl)styrenes by the reaction of Halothane (HCFC-123B1) with aromatic hydrazones in the presence of 1,2-ethylenediamine and a catalytic amount of CuCl₂·2H₂O at room temperature. Catalytic olefination provided acceptable to high yields and high to excellent stereoselectivity of target trifluoromethylated styrenes 479b in contrast to other methods

Table 40. Nenajdenko–Shastin Catalytic Olefination Using Freones

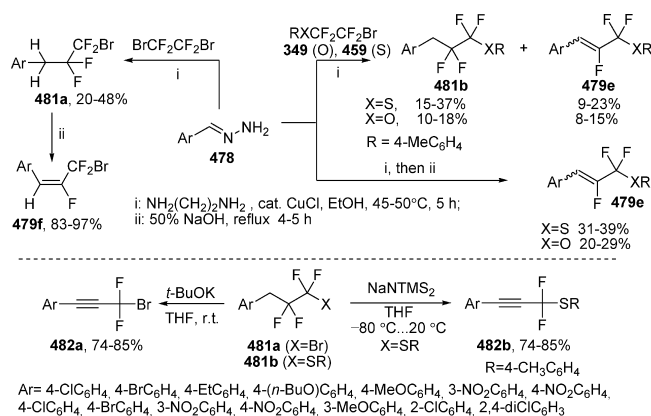
PFE: Ar:	yield, % (Z:E)				
	CF ₃ CCl ₃ 272	CF ₃ CBr ₃ 479c	CF ₂ ClCFCl ₂ 479a	CF ₃ CFBr ₂ 479d	CF ₃ CHClBr 479b
4-NO ₂ C ₆ H ₄	67 (88:12)		63 (86:14)	66 (90:10)	95 (98:2)
3-NO ₂ C ₆ H ₄	57 (75:25)		63 (80:20)	61 (89:11)	
4-ClC ₆ H ₄	73 (75:25)	78 (88:12)	49 (91:9)	52 (93:7)	79 (94:6)
3-ClC ₆ H ₄	57 (80:20)		48 (92:8)		
4-MeOC ₆ H ₄	56 (91:9)	74 (89:11)	49 (95:5)	47 (95:5)	46 (93:7)
4-BrC ₆ H ₄	63 (86:14)		55 (86:14)	53 (92:8)	
Ph	27 (86:14)	61 (93:7)	39 (93:7)		
4-CNC ₆ H ₄	61 (75:25)		35 (86:14)		67 (97:3)
4-MeC ₆ H ₄	72 (86:14)	74 (90:10)	66 (94:6)	61 (93:7)	
4-FC ₆ H ₄	34 (80:20)		36 (93:7)		
2-FC ₆ H ₄	32 (75:25)		33 (80:20)		
2-ClC ₆ H ₄	40 (80:20)		24 (87:13)		
terephthalic aldehyde	24 (66:34)	17 (78:22)	19 (87:13)	69 (86:14)	
	(Z,Z:Z,E)	(Z,Z:Z,E)	(Z,Z:Z,E)	(Z,Z:Z,E)	
isophthalic aldehyde	50 (65:35)	48 (83:17)		33 (86:14)	
	(Z,Z:Z,E)	(Z,Z:Z,E)		(Z,Z:Z,E)	
1-Naphtyl	47 (100:0)		48 (100:0)		80 (98:2)
4-Me ₂ NC ₆ H ₄	31 (100:0)	44 (82:18)			
2-MeOC ₆ H ₄	22 (86:14)				
2,4-diClC ₆ H ₄				53 (94:6)	
4-EtC ₆ H ₄				51 (94:6)	
4-(<i>n</i> -C ₄ H ₉)C ₆ H ₄				49 (94:6)	
4-CO ₂ MeC ₆ H ₄	55 (80:20)	60 (83:17)		62 (97:3)	
4-CF ₃ C ₆ H ₄		31 (84:16)		33 (95:5)	
2-NO ₂ C ₆ H ₄	68 (83:17)			48 (91:9)	
4-MeSC ₆ H ₄		60 (89:11)			
4- <i>t</i> -BuC ₆ H ₄		69 (84:16)			
4-HOC ₆ H ₄		45 (86:14)			
2-MeC ₆ H ₄		47 (89:11)			
4-PhC ₆ H ₄					95 (98:2)
3-HOC ₆ H ₄					58 (95:5)
3-HOC ₆ H ₄					45 (98:2)
3,4-diMeOC ₆ H ₃					54 (97:3)
3,5-diMeOC ₆ H ₃					74 (97:3)
2-naphthyl					72 (99:1)

for the preparation of trifluoromethylated alkenes (Wittig, Horner, or Julia–Kocienski reactions), which demonstrated low stereoselectivity.^{545–549,627,563}

Unexpectedly, this reaction with BrCF₂CF₂Br led to saturated products **481a** instead of alkenes. Compounds **481a** were easily converted into alkenes **479f** by treatment with NaOH in EtOH.⁵⁶⁴ Similarly, the reaction of fluorinated derivatives **349**, **459** afforded alkanes **481b** as a mixture with alkenes **479e**. The treatment of alkanes **481b** and the reaction mixtures of **481b** and **479e** with 50% NaOH gave pure alkenes **479e** in moderate yields.^{502,565} The corresponding fluorinated alkynes **482a**, **482b** were prepared from alkanes **481a**, **481b** by elimination of two HF molecules under treatment of *t*-BuOK⁵⁶⁴ or NaNTMS₂.^{502,565} As a result, on the basis of the catalytic olefination reaction, a whole family of CF₃-, CF₂-, and CF₂CF₂-bridged compounds was synthesized (Scheme 257).

Two mechanistic schemes were proposed for catalytic olefination reaction. Accordingly to the first one,⁵⁶⁶ the reaction is initiated by Cu^{II} as a catalyst formed by oxidation of CuCl with polyhalogenated alkane (PHA). Cu^{II} oxidizes hydrazone **478** to give diazoalkane **483**, which gives copper-carbene complex **484** after copper-catalyzed extrusion of

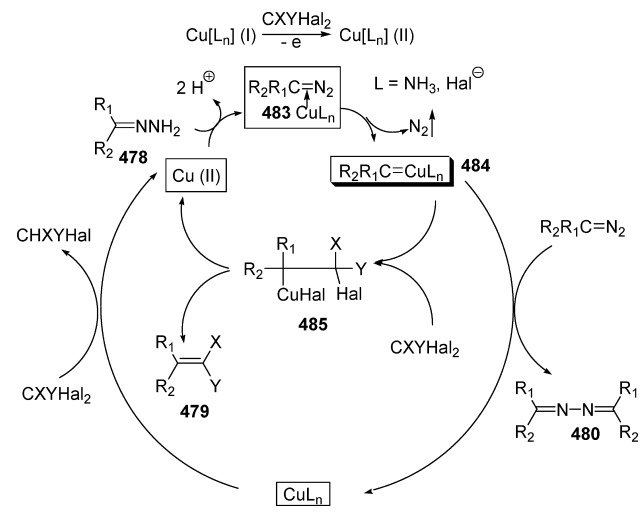
Scheme 257



nitrogen. Next, complex **484** reacts with PHA to form organocopper intermediate **485**, which forms alkene **479** by elimination of Cu^{II} halide and regeneration of catalyst-Cu^{II} (inner cycle). Formation of azine **480** is a result of the dimerization of the diazoalkane or the reaction of copper-

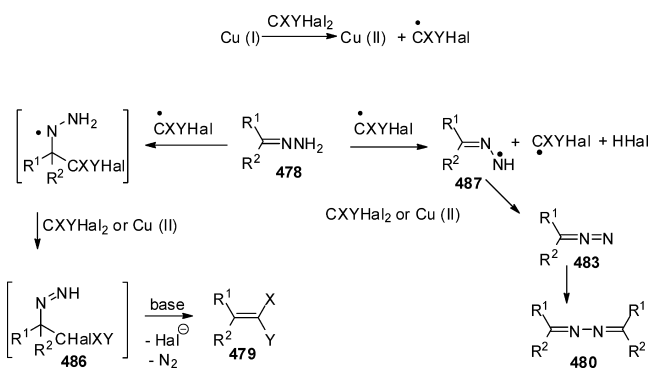
carbene complex **484** with diazoalkane **483** (outer cycle). A base plays the role of a ligand for copper and an acceptor of hydrogen halide formed in the course of the reaction (Scheme 258).

Scheme 258



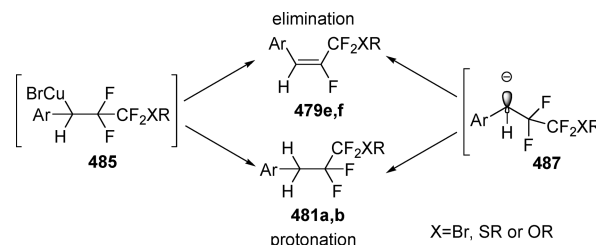
An alternative mechanistic scheme includes formation of radical particles by reaction of PHA with Cu^{I} to initiate the process. Next, the addition of radical formed to the $\text{C}=\text{N}$ bond of hydrazone leads to diazene **486** by means of sequential transformations. Decay of diazene **486** led to elimination of molecular nitrogen HHal to form target alkenes **479**. The formation of azine **480** is a result of hydrazone **478** oxidation followed by dimerization of diazoalkane **483** formed (Scheme 259).

Scheme 259



Despite the principal difference, both of the reaction schemes postulate cleavage of $\text{C}-\text{Hal}$ bond at the last step of an alkene formation. In the case of intermediates **485** and **487**, cleavage of $\text{C}-\text{F}$ bond is more difficult than in the common case, because of the higher energy of $\text{C}-\text{F}$ bond, as compared to other $\text{C}-\text{Hal}$ bonds (115.7 ($\text{C}-\text{F}$), 77.2 ($\text{C}-\text{Cl}$), 64.3 ($\text{C}-\text{Br}$), and 50.7 kcal/mol ($\text{C}-\text{I}$)).⁵⁶⁷ As a result, the cleavage of $\text{C}-\text{F}$ bond proceeds considerably slowly, allowing concurrent protonation of intermediates **485** and **487** to play a greater role to give alkanes **481a,b** (Scheme 260).

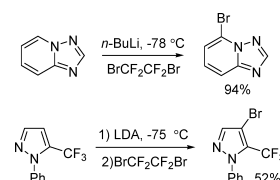
Scheme 260



3.7. Halogenation of Carbanions Using PFE

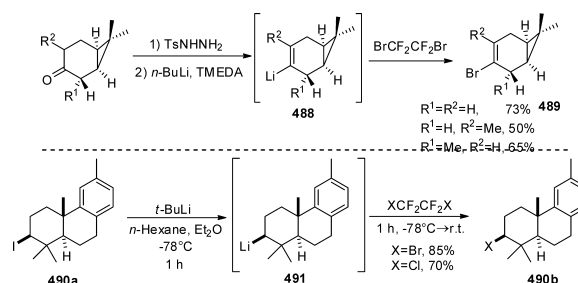
Another very important application of PFEs is halogenation of various carbanions. Although this reaction does not involve the fluorinated core of PFE, giving compounds without additional fluorines, this application of PFEs should be surveyed due to their very high synthetic value. Mild conditions, high tolerance against functional groups, selectivity, and universality are their advantages, making the halogenation using PFE a very useful method. To confirm that, a number of examples can be found in the literature. Only several typical examples are included in the Review. For example, using 1,2-dibromotetrafluoroethane bromination of aryl- and heteroaryl lithiums can be achieved (Scheme 261).⁵⁶⁸

Scheme 261



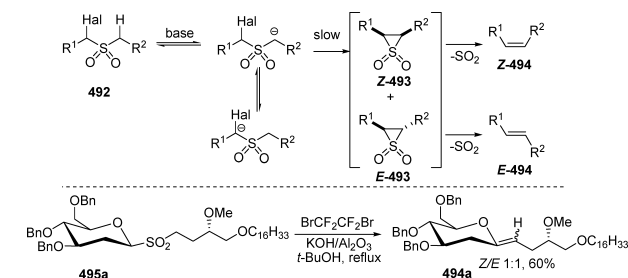
Aliphatic and vinyl lithiums can be also involved in this reaction. Thus, vinyl lithium intermediates **488** obtained by the Shapiro reaction were converted into bromoalkenes **489** in good yields.⁵⁶⁹ Applied to transformation of iodide **490a** into bromide **490b** through a vinyl lithium intermediate **491**, this method gave stereoselectively bromine-substituted polycyclic compound **490b** in 85% yield (Scheme 262).⁵⁷⁰

Scheme 262



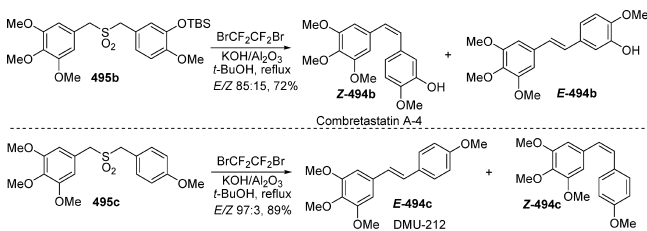
$\text{BrCF}_2\text{CF}_2\text{Br}$ was used as a very effective brominating agent in the Ramberg–Backlund route to alkenes. The Ramberg–Backlund rearrangement is the base-induced rearrangement of α -halogenated sulfones **492** via episulfone intermediates **493** to produce alkenes **494**. For example, *exo* glycol **494a** was prepared in 60% yield from sulfone **495a** in mild conditions. The treatment of sulfone **495** with $\text{KOH}/\text{Al}_2\text{O}_3$ in $\text{BrCF}_2\text{CF}_2\text{Br}$ -*t*-BuOH mixture afforded alkene **494a** as a 1:1 mixture of *Z/E* isomers (Scheme 263).⁵⁷¹

Scheme 263



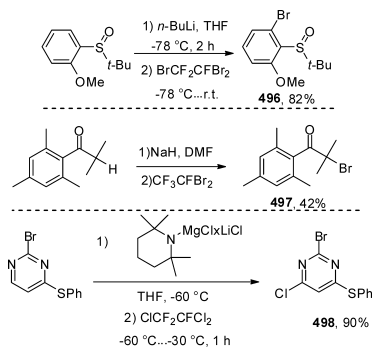
The Ramberg–Bäcklund approach has been also used for the synthesis of stilbenoid anticancer agents combretastatin A-4 **494b** and *E*-stilbene DMU-212 **494c**. Compounds **494b**, **494c** were prepared stereoselectively in high yields, treating sulphones **495b** and **495c** with KOH/Al₂O₃ in the presence of BrCF₂CF₂Br (Scheme 264).⁵⁷²

Scheme 264



A few examples of bromination by another PFE were reported. Thus, BrCF₂CFBr₂⁵⁷³ and CF₃CFBr₂⁵⁷⁴ were used for the preparation of bromides **496** and **497**. The chlorination of organometallics was performed by ClCF₂CF₂Cl⁵⁷⁰ and ClCF₂CFCl₂ to give compounds **490b**, **498** in high yield (Schemes 262, 265).⁵⁷⁵

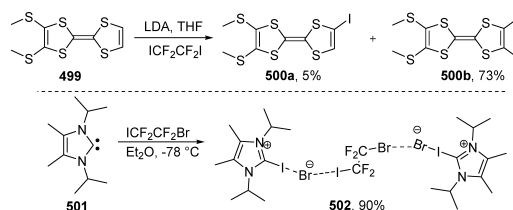
Scheme 265



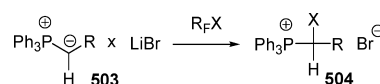
Iodination of dianion, derived from bis(methylthio)-tetrathiafulvalene **499**, using ICF₂CF₂I produced mainly the dihalogenated product **500b** with a small admixture of the monohalogenated compound **500a**.⁵⁷⁶ Imidazol-2-ylidene (stable carbene) **501** reacts with BrCF₂CF₂I to afford stable adduct **502** in 90% yield. The crystal structure of **502** exhibits near linear C–I–Br interactions in addition to short Br–Br and Br–I contacts to the BrCF₂CF₂I fragment (Scheme 266).⁵⁷⁷

In the case of phosphonium ylides **503**, reactions with PFEs led to α -haloalkylphosphonium salts **504** in good to high yields (Scheme 267, Table 41).⁵⁷⁸

Scheme 266



Scheme 267



Silanes can be transformed into bromides by a similar approach. Isomeric bipyridyl derived silanes **505** gave bromides **506** in almost quantitative yields under the treatment with BrCF₂CF₂Br in DMF in the presence of CsF (Scheme 268).⁵⁷⁹

3.8. Complexation of ICF₂CF₂I

The ability of the C–X (X = Cl, Br, I) bond to work as an electron acceptor and to form donor–acceptor complexes C–X...B (X = Cl, Br, I; B = O, S, Se, N) with a neutral basic motif was discovered in the 1950s.⁵⁸⁰ C–X...B noncovalent bonding has been rationalized as a $n \rightarrow \sigma^*$ donation⁵⁸¹ and has been called “halogen bonding”^{580,582} to emphasize the similarity with the hydrogen bonding. ICF₂CF₂I forms complexes **507a–f** with the corresponding bases under simple mixing of reagents in solvent (CHCl₃, acetone) or without solvent (Scheme 269). Complexes thus obtained have 1:1 or 1:2 composition. The attractive intermolecular N...I–RF interaction is specific, directional, and strong enough to drive the self-assembly of PFE and donor modules into 1D networks **507g**, which are solid and stable at room temperature and in the air.⁵⁸³ Structural peculiarities of complexes **507a–g** were established by NMR, IR, Raman spectrometry, and X-ray analysis. The length of noncovalent bonding varies from 2.79 to 2.84 Å. These values are longer than the average N–I covalent bond (2.07 Å). It is approximately 0.8 times of the sum of the van der Waals radius for nitrogen (1.55 Å) and iodine (1.98 Å),⁵⁸⁴ and close to the N–I bond length measured in the complexes formed by I₂ with nitrogen-containing heterocycles (e.g., 2.817, 2.982–3.092, 3.072, 2.724–2.95 Å for pyrazine,⁵⁸⁵ phenazine,⁵⁸⁶ tetramethylpyrazine,⁵⁸⁵ and quinoxaline⁵⁸⁷ complexes with I₂, respectively).

UV-irradiation or heating of complexes **507a–f** in acetonitrile caused their decomposition to form tetrafluoroethylene, iodine, and a molecule of the corresponding donating component **508** (Scheme 270).^{583b,c}

3.9. Polyfluoroethyl Derivatives of Polyvalent Iodo Compounds

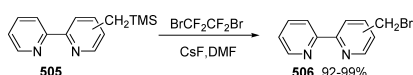
A highly important type of compounds prepared from PFE having iodine atom is polyfluoroethylated iodonium salts. These compounds have rich chemistry and therefore are very significant for organic synthesis. The oxidation of 1,1,1-trifluoro-2-iodoethane with trifluoroperacetic acid⁵⁸⁸ gave the corresponding bis(trifluoroacetate) **510a** in quantitative yield. This iodonium compound can also be prepared in 83% yield by the exchange reaction of silver trifluoroacetate with the corresponding dichloride **509a**, which was synthesized by the chlorination of 1,1,1-trifluoro-2-iodoethane in 66% yield. 1-(Dichloroiodo)-2,2,2-trifluoroethane **509a** behaves as a mild

Table 41. Reaction of Phosphonium Ylides with PFE

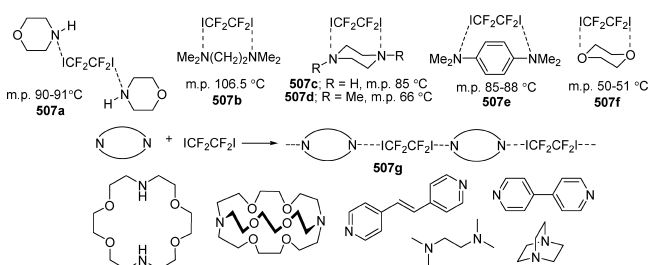
R	R _P X	X	yield of 504 , %	R	R _P X	X	yield of 504 , %
Me	ICF ₂ CF ₂ Cl	I	(93) ^a	<i>n</i> -C ₃ H ₇	CCl ₃ CF ₂ Cl	Cl	90
Me	BrCF ₂ CF ₂ Br	Br	90(72)	<i>n</i> -C ₃ H ₇	BrCF ₂ CF ₂ Br	Br	88
Me	BrCFClCF ₂ Br	Br	(66)	<i>n</i> -C ₅ H ₁₁	ICF ₂ CF ₂ Cl	I	90
Me	CCl ₃ CF ₂ Cl	Cl	71(67)	<i>n</i> -C ₅ H ₁₁	BrCF ₂ CF ₂ Br	Br	89
Et	ICF ₂ CF ₂ Cl	I	93(88)	<i>n</i> -C ₅ H ₁₁	CCl ₃ CF ₂ Cl	Cl	87
Et	BrCF ₂ CF ₂ Br	Br	84(67)	H	ICF ₂ CF ₂ Cl	I	(40)
Et	CCl ₃ CF ₂ Cl	Cl	74(70)	H	BrCF ₂ CF ₂ Br	Br	(18)
<i>n</i> -C ₃ H ₇	ICF ₂ CF ₂ Cl	I	90(80)	H	CCl ₃ CF ₂ Cl	Cl	(40)

^aThe yields were measured by ¹H and ³¹P NMR. The numbers in parentheses are isolated yields.

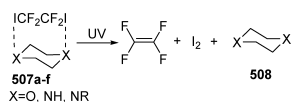
Scheme 268



Scheme 269

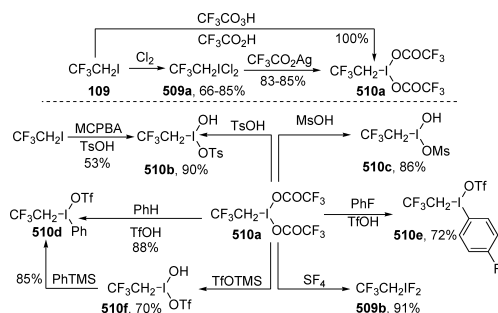


Scheme 270



and efficient chlorinating agent for various alkenes and enols.⁵⁸⁹ Bis(trifluoroacetate) **510a** is a valuable source for the synthesis of other fluorinated iodonium salts **509b-f** as well as 1-(difluoroiodo)-2,2,2-trifluoroethane **509b** (Scheme 271).

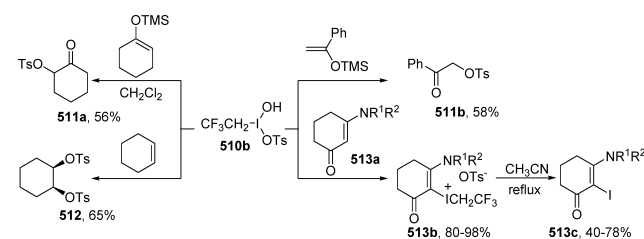
Scheme 271



1-(Hydroxy(sulfonyloxy)iodo)-2,2,2-trifluoroethanes, obtained by the reaction of **510a** with sulfonic acids and trimethylsilyl triflate, are aliphatic analogues of the Koser's reagent (PhI(OH)OTs).⁵⁹⁰ Compound **510b** was found to be a valuable tosylating reagent for silyl enolates and alkenes to give α -tosyloxy ketones **511a,b** and bis-tosylates **512** correspondingly. Using **510b** permits also one to perform the selective iodination of enaminones **513a** via intermediate formation of the corresponding iodonium salts **513b**. Thus, a

series of α -iodoenaminones **513c** was prepared in high yields starting from the corresponding enaminones **513a** (Scheme 272, Table 42).⁵⁹¹

Scheme 272

Table 42. Reactions of Enaminones **513a** with 1-(Hydroxy(sulfonyloxy)iodo)-2,2,2-trifluoroethane **510b**⁵⁹¹

513b , yield	513c , yield	513b , yield	513c , yield

The reaction of trifluoroethylidonium bis(trifluoroacetate) **510a** with benzene and fluorobenzene in the presence of triflic acid leads to the formation of extraordinary interesting iodonium compounds, 2,2,2-trifluoroethyl-substituted arylidonium triflates **510d**, **510e**,⁵⁸⁸ which can be used for transfer of 2,2,2-trifluoroethyl group to various nucleophiles.⁵⁹² The reaction is very general. The O-, N-, S-nucleophiles as well as carbon nucleophiles (Grignard reagents and furan) can be trifluoroethylated by this approach to form compounds **514**. In the case of silyl- and vinylidene ethers **98**, **515**, reaction with phenyliodonium triflate **510d** afforded fluorinated ketones **381** or esters **378** correspondingly (Scheme 273, Table 43).⁵⁹³

Scheme 273

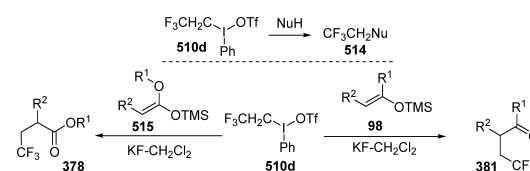
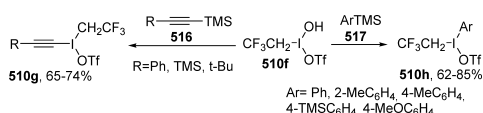


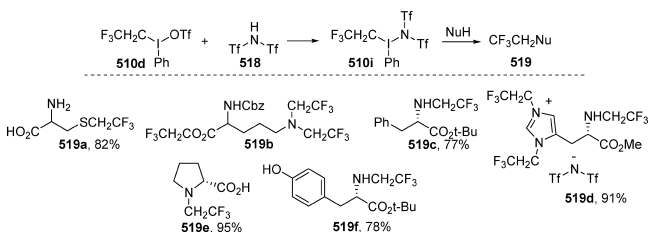
Table 43. Reactions of Aryliodonium Triflate **510d with Nucleophiles⁵⁹⁰**

Product, yield	Product, yield	Product, yield	Product, yield
 NHC ₂ F ₃ 79%	 NHC ₂ F ₃ 88%	 NHC ₂ F ₃ 85%	 NHC ₂ F ₃ 76%
 PhNHCH ₂ CF ₃ 92%	 PhN(Et)CH ₂ CF ₃ 98%	 PhCH ₂ CH ₂ NHC ₂ F ₃ 85%	 PhOCH ₂ CF ₃ 79%
 PhCH ₂ CH ₂ OCH ₂ CF ₃ 61%	 PhCO ₂ CH ₂ CF ₃ 99%	 C ₆ H ₅ SCH ₂ CF ₃ 80%	 EtO ₂ C-CH ₂ -CF ₃ 28%
 PhCH ₂ CH ₂ CH ₂ CF ₃ 36%	 PhCOCH ₂ CH ₂ CF ₃ 30%	 Furan-2-yl-CH ₂ -OTf 52%	 Me ₂ PhN-CH ₂ CF ₃ 89%
 Me ₂ BnN-CH ₂ CF ₃ 89%	 Me-CH ₂ -CH ₂ -CH ₂ -CH ₂ -OTf 83%	 Me-CH ₂ -CH ₂ -CH ₂ -CH ₂ -OTf 84%	 Me-CH ₂ -CH ₂ -CH ₂ -CH ₂ -OTf 83%
 Me-CH ₂ -CH ₂ -CH ₂ -CH ₂ -OTf 77%	 Me-CH ₂ -CH ₂ -CH ₂ -CH ₂ -OTf 63%	 Me-CH ₂ -CH ₂ -CH ₂ -CH ₂ -OTf 93%	 Me-CH ₂ -CH ₂ -CH ₂ -CH ₂ -OTf 87%
 Ph-S-CH ₂ CF ₃ 90%	 Me-S-CH ₂ CF ₃ 92%	 Ph-S-CH ₂ CF ₃ 31%	 PhPPh ₃ -CH ₂ CF ₃ 95%
 8-oxooctyl-OTf 80%	 4-oxopentyl-OTf 42%	 2-oxocyclohexyl-OTf 71%	 2-oxocyclohexyl-OTf 49%
 2,2,2-trifluoroethyl-OTf 53%	 2,2,2-trifluoroethyl-OTf 75%	 2,2,2-trifluoroethyl-OTf 92%	 2,2,2-trifluoroethyl-OTf 76%

Alternatively, 2,2,2-trifluoroethyl-substituted aryliodonium triflates **510h** were prepared by the reaction of TMS-substituted aryls **517** with 1-(hydroxy(trifluoromethanesulfonyloxy)iodo)-2,2,2-trifluoroethane **510f**. Similarly, alkynyl iodonium triflates **510g** were prepared by the reaction with TMS-substituted acetylenes **516** (Scheme 274).⁵⁹¹

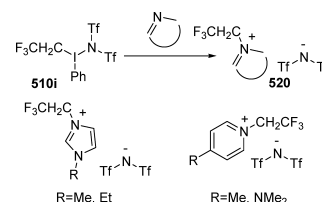
Scheme 274

Another improvement in this chemistry has been done by DesMarteau.⁵⁹⁴ The reaction of **510d** with bis-(trifluoromethylsulfonyl)imide **518** gave a novel type of iodonium salts, (CF₃SO₂)₂Ni(Ph)CH₂CF₃ **510i**. Iodonium salt **510i** is a powerful alkylating reagent, which can be utilized in water. It was demonstrated that this iodonium salt can be used for the trifluoroethylation of various amino acid derivatives, opening a simple and effective route to trifluoroethylated amino acids **519** and their subsequent use in the synthesis of various fluorinated peptides (Scheme 275). The main advantage of this protocol is the possibility to operate

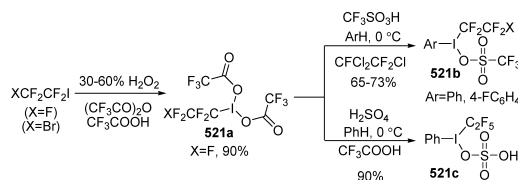
Scheme 275

with the iodonium salt **510i** in aqueous phase, which is impossible for the corresponding triflate. However, low chemoselectivity of alkylation should be noted; for example, in the case of histidine ester three trifluoroethyl groups are incorporated into the molecule (compound **519d**).⁵⁹⁵

Imidazolium and pyridinium salts are a very popular type of ionic liquids that found broad application in modern chemistry. The possibility of direct formation of ionic liquids with bistriflamide counterion has been studied. It was found that the corresponding salts **520** are formed quantitatively from 1-methylimidazole and 1-ethylimidazole, and some pyridines have been alkylated as well (Scheme 276).^{594c,596} Ionic liquids prepared by this manner can be used to form a solution of cellulose.

Scheme 276

Perfluoroethyl iodide can also form hypervalent iodo compounds. The oxidation of C₂F₅I with hydrogen peroxide, performed in a mixture of trifluoroacetic acid with its anhydride, afforded C₂F₅-iodonium bis(trifluoroacetate) **521a** in 90% yield. The treatment of **521a** with benzene in H₂SO₄-CF₃CO₂H mixture led to (perfluoroalkyl)phenyliodonium sulfate **521c**, the so-called FIS-2 reagent. Similarly, reaction with benzene or fluorobenzene in the presence of triflic acid afforded (perfluoroalkyl)phenyliodonium trifluoromethanesulfonates **521b** (FITS-2 reagents). This sequence of transformations was also reported for BrCF₂CF₂I (Scheme 277).⁵⁹⁷

Scheme 277

FITS-2 reagent **521d** was used for perfluoroalkylation of various carbanions. Thus, the reaction of FITS-2 with *p*-xyl magnesium bromide **522** led to fluorinated compound **524**. The reaction of FITS-2 with acetylides **523a**, **523b** yielded the corresponding C₂F₅-acetylenes **297**, **525**.⁵⁹⁸ The reaction of both FITS-2 **521d** and FIS-2 **521c** reagents with vinyl ether **526** in air in pyridine led to oxperfluoroalkylation product **527** in moderate yield (Scheme 278). The mechanism of the transformation was not studied.⁵⁹⁹

In the case of ICF₂CF₂I, the oxidation with peroxytrifluoroacetic⁶⁰⁰ acid or *tert*-perfluorobutyl hypochlorite **529**⁶⁰¹ led to bis(aryliodonio)perfluoroethanes **528a** and **528c**. The reaction of **528a** with benzene in trifluoroacetic acid led to iodonium salt **528b** in 64% yield to link two benzene rings (Scheme 279).

The oxidation of C₂F₅I with TeF₆OCl **530** at -78 °C afforded iodonium salt **531a** in almost quantitative yield. The thermal decomposition of **531a** gave fluorinated tellurium

ArCH_2MgCl **522** $\rightarrow$ $\text{ArCH}_2\text{C}_2\text{F}_5$ **524** $\text{Ar} = 4\text{-CH}_3\text{C}_6\text{H}_4$, 58%
 $\text{Ph}-\text{X}$ **523a** -78°C $\rightarrow$ $\text{Ph}-\text{C}\equiv\text{C}-\text{C}_2\text{F}_5$ **297** $\text{X} = \text{Li}, \text{MgCl}$, 56–58%
 $n\text{-C}_6\text{H}_{13}-\text{X}$ **523b** 0°C $\rightarrow$ $n\text{-C}_6\text{H}_{13}-\text{C}\equiv\text{C}-\text{C}_2\text{F}_5$ **525**, 50%

 $\text{C}_2\text{F}_5\text{I}(\text{Ph})\text{OSO}_2\text{H}$ **521c**
 $\text{Pyridine, air, CH}_3\text{CN, r.t.}$ $\rightarrow$ $\text{C}_2\text{F}_5\text{CH}_2\text{CO}_2\text{CH}_2(\text{CH}_2)_4\text{CH}_3$ **527**, 39%
 $\text{C}_2\text{F}_5\text{I}(\text{Ph})\text{OSO}_2\text{CF}_3$ **521d**
 $\text{Pyridine, air, DMF, r.t.}$ $\rightarrow$ $\text{C}_2\text{F}_5\text{CH}_2\text{CO}_2\text{CH}_2(\text{CH}_2)_4\text{CH}_3$ **527**, 45%

$$\begin{array}{c}
 \text{ICl}_2\text{CF}_2\text{I} \xrightarrow[\text{(CF}_3\text{CO)}_2\text{O}]{\text{CF}_3\text{CO}_2\text{H}} \text{528a, 98\%} \xrightarrow[\text{CF}_3\text{CO}_2\text{H}]{\text{PhH}} \text{528b, 64\%} \\
 \text{529} + \text{ICl}_2\text{CF}_2\text{I} \xrightarrow{0^\circ\text{C}} \text{528c, } \sim 100\%
 \end{array}$$

$$\text{C}_2\text{F}_5\text{I} + \text{TeF}_5\text{OCl} \xrightarrow{-78^\circ\text{C}} \text{C}_2\text{F}_5(\text{OTeF}_5)_2 \xrightarrow{115^\circ\text{C}} \text{C}_2\text{F}_5\text{OTeF}_5 + \text{C}_2\text{F}_5\text{I} + \text{I}(\text{OTeF}_5)_3$$

530 531a, 95% 532, 78% 533

$$\begin{array}{c}
 3\text{C}_2\text{F}_5\text{I} + 2\text{ClF}_3 \xrightarrow[\text{-70 } ^\circ\text{C}]{\text{perfluorohexane}} 3\text{C}_2\text{F}_5\text{IF}_2 + \text{Cl}_2 \\
 \qquad\qquad\qquad \mathbf{534a} \\
 3\text{C}_2\text{F}_5\text{I} + 4\text{ClF}_3 \xrightarrow[\text{-40 } ^\circ\text{C}]{\text{perfluorohexane}} 3\text{C}_2\text{F}_5\text{IF}_4 + 2\text{Cl}_2 \\
 \qquad\qquad\qquad \mathbf{534b} \\
 \hline
 \text{C}_2\text{F}_5\text{IF}_2 \xleftarrow[\text{-78 } ^\circ\text{C}]{\text{F}_2} \text{C}_2\text{F}_5\text{I} \xrightarrow[\text{-40 } ^\circ\text{C}]{\text{F}_2} \text{C}_2\text{F}_5\text{IF}_4 \\
 \mathbf{534a} \qquad\qquad\qquad \mathbf{534b} \\
 \hline
 \text{CF}_3\text{CH}_2\text{IF}_4 \xrightleftharpoons[\text{BF}_3]{\text{XeF}_2} \text{CF}_3\text{CH}_2\text{I} \xrightleftharpoons[\text{CH}_3\text{CN}]{\text{XeF}_2} \text{CF}_3\text{CH}_2\text{IF}_2 \xrightarrow[\text{CH}_3\text{CN}]{\text{Ph}_2\text{Te}} \text{Ph}_2\text{TeF}_2 \\
 \mathbf{509c, 79\%} \qquad\qquad\qquad \mathbf{509b}
 \end{array}$$

Reaction scheme for the synthesis of **537**:

Starting material **535** (a benzodioxole derivative) reacts with $\text{C}_2\text{F}_5\text{TMS}$ in the presence of CsF in MeCN to form intermediate **521e** (a benzodioxole derivative with a CF_2CF_3 group).

Intermediate **521e** then reacts with a substituted piperidine derivative (**536**) in THF at room temperature to form the final product **537** (a benzodioxole derivative with a CF_2CF_3 group and a piperidine ring), in 86% yield.

Reaction scheme for the synthesis of 539:

Starting materials: 538 (a substituted styrene derivative) and 521e (a substituted furan derivative).

Reaction conditions:

- 1.8 equiv 521e
- 14 W-light bulb
- 5 mol% Ru(bpy)₃Cl₂·6H₂O
- MeOH, r.t.

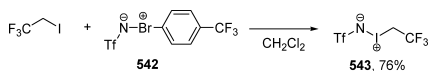
Product: 539 (a substituted styrene derivative).

Reaction conditions and yields:

- R=H; (E/Z=5:6): (E) 53%, er = 89:11
- R=Ac; (E/Z=7:4): (E) 63%, er = 9:1

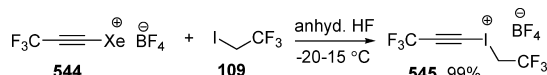
tion of alkenes or in the C–H amidation of alkanes, arenes, and aldehydes using Cu, Rh, Ru, or Ag catalysts (Scheme 285).⁶¹⁰

Scheme 285



Propargyl derivative of $\text{CF}_3\text{CH}_2\text{I}$ was prepared by the reaction with propargyl-substituted xenon monofluoride **544**. The reaction of **544** with $\text{CF}_3\text{CH}_2\text{I}$ was carried out in anhydrous hydrogen fluoride to form polyvalent iodine compound **545** in quantitative yield (Scheme 286).⁶¹¹

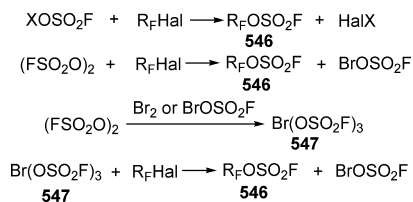
Scheme 286



3.10. Reactions of PFE with HalOSO_2F , $(\text{FSO}_2\text{O})_2$, $\text{CF}_3\text{SO}_3\text{Cl}$, and SO_3

Formal substitution of halide in PFEs by fluorosulfate group was described in a series of reports. Such strong electrophiles as ClOSO_2F , BrOSO_2F , bearing halogen atom in (+1) oxidation state, and $(\text{FSO}_2\text{O})_2$ (peroxydisulfuryl difluoride) react with PFEs to afford the corresponding fluoroethyl fluorosulfates **546** in moderate to high yields (Scheme 287, Table 44). In some

Scheme 287

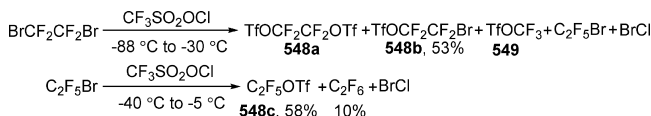


cases, reactivity of ClSO_3F was enhanced by the addition of catalytic amounts of Br_2 , which allowed one to generate in situ more reactive BrOSO_2F . Similarly reactivity of $(\text{FSO}_2\text{O})_2$ was increased due to formation of reactive polyvalent bromine (+3) intermediates. Additives of SbF_5 and HSO_3F increased the

activity of halogen fluorosulfates too. Also, it should be noted that the substitution of chlorine occurs easier than the substitution of bromine.

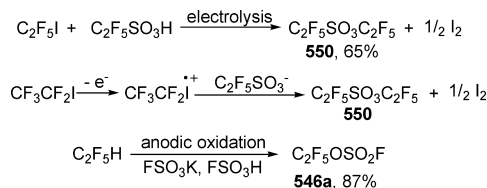
Analogously, the reaction of PFEs with $\text{CF}_3\text{SO}_2\text{OCl}$ resulted in the formation of the corresponding triflates. The treatment of $\text{BrCF}_2\text{CF}_2\text{Br}$ with $\text{CF}_3\text{SO}_2\text{OCl}$ led to a mixture of mono-**548b** and bistriflates **548a** with trifluoromethyltriflate **549**, $\text{C}_2\text{F}_5\text{Br}$, and BrCl . Pentafluoroethyl triflate **548c** was isolated together with hexafluoroethane and BrCl in the case of the reaction with $\text{C}_2\text{F}_5\text{Br}$ (Scheme 288).^{13b}

Scheme 288



Alternatively, products of that kind can be prepared using anodic oxidation of PFE. For example, pentafluoroethyl pentaflate **550** was synthesized by electrolysis of $\text{C}_2\text{F}_5\text{I}$ in solution of $\text{C}_2\text{F}_5\text{SO}_3\text{H}$.⁶¹⁶ Anodic oxidation of perfluoroethane in fluorosulfonic acid in the presence of potassium fluorosulfonate led to $\text{C}_2\text{F}_5\text{SO}_3\text{F}$ **546a** in high yield (Scheme 289).⁶¹⁷

Scheme 289



In contrast, the reaction of PFEs with oleum did not lead to any sulfuric acid derivatives. The treatment of $\text{BrCF}_2\text{CFBrCl}$ and $\text{BrCF}_2\text{CF}_2\text{I}$ with oleum afforded oxalyl fluoride **551a** and bromodifluoroacetyl fluoride **551b**, respectively, by formal hydrolysis of C–Hal bonds (Scheme 290).⁶¹⁸

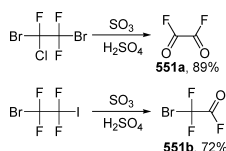
4. APPLICATION OF PFE

In the final part of this Review, some examples of recent application of compounds having fluorinated C-2 fragment are

Table 44. Reactions of PFE with HalOSO_2F , $(\text{FSO}_2\text{O})_2$, $\text{CF}_3\text{SO}_3\text{Cl}$, and SO_3

PFE	conditions	products	yields, %	ref
$\text{BrCF}_2\text{CF}_2\text{Br}$	$(\text{FSO}_2\text{O})_2$ + cat. Br_2	$\text{BrCF}_2\text{CF}_2\text{OSO}_2\text{F}$	75	612
$\text{BrCF}_2\text{CF}_2\text{Br}$	$(\text{FSO}_2\text{O})_2$, HSO_3F , 30–35 °C	$\text{BrCF}_2\text{CF}_2\text{OSO}_2\text{F}$	85	612
$\text{BrCF}_2\text{CF}_2\text{Br}$	ClOSO_2F , 25 °C	$\text{BrCF}_2\text{CF}_2\text{OSO}_2\text{F}$ + $\text{ClCF}_2\text{CF}_2\text{OSO}_2\text{F}$	40	613
			5	
$\text{BrCF}_2\text{CF}_2\text{Br}$	ClOSO_2F , 65 °C	$\text{BrCF}_2\text{CF}_2\text{OSO}_2\text{F}$ + $\text{ClCF}_2\text{CF}_2\text{OSO}_2\text{F}$	64	613
			31	
$\text{BrCF}_2\text{CF}_2\text{Br}$	BrOSO_2F , SbF_5 , HSO_3F , rt	$\text{BrCF}_2\text{CF}_2\text{OSO}_2\text{F}$	86	614
$\text{BrCF}_2\text{CF}_2\text{Br}$	ClOSO_2F , SbF_5 , HSO_3F , rt	$\text{FSO}_2\text{OCF}_2\text{CF}_2\text{OSO}_2\text{F}$	81	614
$\text{ClCF}_2\text{CF}_2\text{Cl}$	ClOSO_2F , 130 °C	$\text{ClCF}_2\text{CF}_2\text{OSO}_2\text{F}$	90	613
$\text{ClCF}_2\text{CF}_2\text{Cl}$	ClOSO_2F , SbF_5 , HSO_3F , rt	$\text{FSO}_2\text{OCF}_2\text{CF}_2\text{OSO}_2\text{F}$	88	614
$\text{ClCF}_2\text{CF}_2\text{Cl}$	ClOSO_2F + cat. Br_2 , 140 °C	$\text{ClCF}_2\text{CF}_2\text{OSO}_2\text{F}$ + $\text{FSO}_2\text{OCF}_2\text{CF}_2\text{OSO}_2\text{F}$	89	613
			6	
$\text{ClCF}_2\text{CF}_2\text{OSO}_2\text{F}$	ClOSO_2F + cat. Br_2 , 140 °C	$\text{FSO}_2\text{OCF}_2\text{CF}_2\text{OSO}_2\text{F}$	31	613
$\text{C}_2\text{F}_5\text{Cl}$	ClOSO_2F + cat. Br_2 , 130 °C	$\text{C}_2\text{F}_5\text{OSO}_2\text{F}$	22	613,615
$\text{C}_2\text{F}_5\text{Br}$	ClOSO_2F , 25 °C	$\text{C}_2\text{F}_5\text{OSO}_2\text{F}$	96	613

Scheme 290



given. These data indicate high attractiveness of incorporation of fluorinated C-2 fragments in molecules having application in medicine and drug design; however, this type of substituents is not so popular as fluorine atom itself and trifluoromethyl group. For example, small fluorinated molecules are widely used in medicine as inhalation anesthetics (Figure 5).

CF ₃ CHClBr Halothane	CF ₃ CH ₂ OCH=CH ₂ Fluoroxene	CF ₃ CF ₂ OCHF ₂ Desflurane
CF ₃ CHClOCHF ₂ Isoflurane		CHFClCF ₂ OCHF ₂ Enflurane

Figure 5.

A novel class of benzimidazole derivatives was prepared as new scaffolds for the development of cannabinoid agonists. These ligands demonstrated good binding affinities with decent potencies toward the cannabinoid CB₂ receptor (nM scale), along with excellent selectivity (>1000-fold) over the CB₁ receptor was prepared.⁶¹⁹ 1,1,1-Trifluoroethyl group is also an important fragment of another type of pharmacologically active bicyclic scaffolds, which are cannabinoid receptor type 1 (CB₁-R) antagonists (Figure 6).⁶²⁰

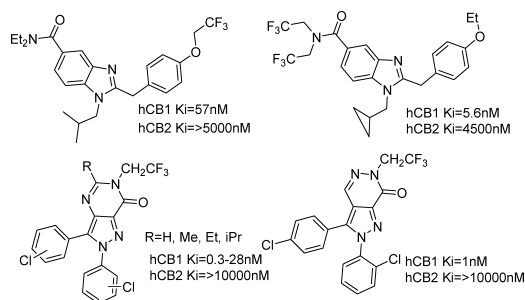


Figure 6.

Benzothiophene, pyridine, benzofuran, and cyclopentene derivatives bearing 1,1,1-trifluoroethyl group (as well as some acyclic analogues) were studied as retinoid X receptors ligands (RXR). The drugs of this type are used for treatment of numerous skin diseases like psoriasis, acne, Kaposi's sarcoma, and CTCL. All of the compounds showed a remarkable selectivity for RXRs over retinoid A receptors RARs. Another type of selective RXR ligands has the tetrahydroquinolinone core also decorated with CH₂CF₃ fragment (Figure 7).^{483,485,621}

A new series of estradiol analogues were synthesized in an attempt to improve on the anticancer activity of 2-methoxyestradiol, a naturally occurring mammalian tubulin polymerization inhibitor. The compounds were evaluated as inhibitors of tubulin polymerization and the binding of [3H]colchicine to tubulin, as well as for in vitro cytotoxicity in human cancer cell cultures. The most potent ligand among

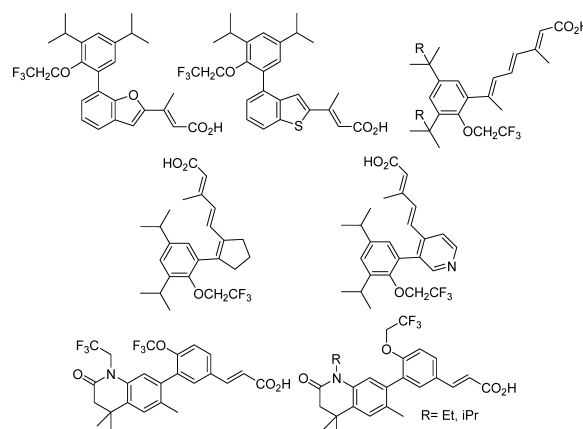


Figure 7.

the new class studied was 2-(2,2,2-trifluoroethoxy)-6-oximinoestradiol (Figure 8).⁶²²

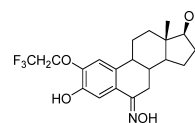


Figure 8.

The platelet P2T receptor plays a major role in platelet aggregation, and its antagonists are predicted to have significant therapeutic potential as antithrombotic agents. Modification of adenosine triphosphate ATP has led to the identification of new scaffolds, which are an important pharmacological tool for the characterization of P2T receptors, and to the discovery of the therapeutically useful analogues (Figure 9).⁴⁴⁶

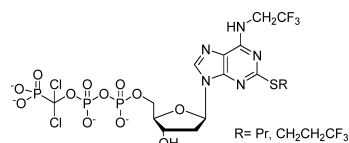


Figure 9.

Phosphodiesterases (PDEs) are a family of enzymes encoded by 21 genes and subdivided into 11 distinct families according to structural and functional properties. These enzymes metabolically inactivate widely occurring intracellular secondary messengers, 3,5-cyclic adenosine monophosphate (cAMP) and 3,5-cyclic guanosine monophosphate (cGMP). A number of new ligands bearing 2,2,2-trifluoroethyl fragment were synthesized. These compounds exhibited high PDE10A inhibitory activity (Figure 10).⁶²³

The peroxisome proliferator-activated receptors (PPARs) are ligand-activated transcription factors in the nuclear hormone receptor superfamily. The physiological role of the PPARs is as

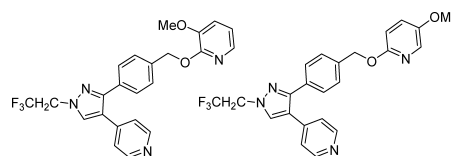


Figure 10.

regulators of lipid and glucose metabolism. The synthesis and structure–activity relationships of novel series of *R*-arylox-*y*-phenylacetic acids as PPAR α/γ dual agonists demonstrated high potency of 2,2,2-trifluoroethyl-substituted derivative (Figure 11).⁶²⁴

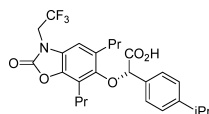


Figure 11.

A series of 4-morpholinopyrrolopyrimidine derivatives were synthesized and evaluated as inhibitors of PI3K α and mTOR, leading to the discovery of PI3KR selective inhibitors and dual PI3K α /mTOR kinase inhibitors. Phosphatidylinositol-3-kinases (PI3Ks) are lipid kinases that catalyze phosphorylation of the 3-hydroxy position of PIP2 (phosphatidylinositol 4,5-diphosphate) to PIP3 (phosphatidylinositol 3,4,5-triphosphate), an important second messenger modulating activity of the PI3K downstream effectors Akt and mTOR. The consequences of biological activation of Akt include tumor progression, proliferation, survival, growth, invasion, angiogenesis, and metastasis. Significant evidence suggests that the PI3K/Akt pathway is deregulated in many human cancers (Figure 12).⁴⁴⁵

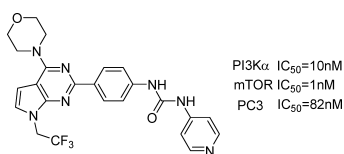


Figure 12.

A series of β -aminoamides bearing triazolopiperazines have been prepared and evaluated as potent, selective, and orally active DPP-4 inhibitors. It was demonstrated that β -amino acids in conjunction with triazolopiperazines with the appropriate substitution (including CF_3CH_2 moiety) provide extremely potent DPP-4 inhibitors showing high selectivity over other related enzymes, good pharmacokinetic profiles, and high in vivo efficacy in an OGTT in lean mice.⁶²⁵ CF_3CH_2 fragment has been introduced via alkylation of the corresponding carbanions with $\text{CF}_3\text{CH}_2\text{Br}$ (Figure 13).

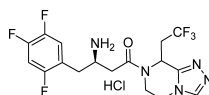


Figure 13.

A number of substituted 2-[(2,2,2-trifluoroethyl)sulfonyl]-1*H*-benzimidazoles and 2-[(1,1,2,2,2-pentafluoroethyl)sulfonyl]-1*H*-benzimidazoles have demonstrated antiinflammatory activity that appears to have a mechanism distinct from typical cyclooxygenase inhibiting nonsteroidal antiinflammatory drugs. These were also found to inhibit release of lysosomal enzymes from neutrophils, raising the possibility that their anti-inflammatory effect is at least partially mediated by an effect on neutrophil function (Figure 14).⁵¹⁷

The peroxisome proliferator-activated receptors (PPARs) are transcription factors belonging to the nuclear receptor super

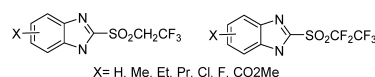


Figure 14.

family. The marketed insulin sensitizers (rosiglitazone and pioglitazone) and the lipid lowering fibrates (e.g., fenofibrate and clofibrate) are PPAR α and PPAR γ agonists, respectively. Computational analysis of the ligand binding pocket of the three PPAR receptor subtypes was utilized in the design of potent PPAR α agonists. Analogues with substituents of different van der Waals volume gave PPAR α agonists with varying potency and selectivity. Optimum PPAR α potency and selectivity were obtained with 1,1,1-trifluoroethyl substituents having van der Waals volume around 260. A key step of the synthesis was alkylation of 3,5-dihydroxybenzaldehyde with 1,1,1-trifluoroethyl iodide (Figure 15).^{482a}

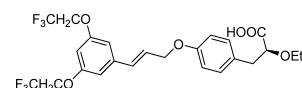


Figure 15.

The ryanodine receptor (RyR) is a nonvoltage-gated calcium channel located in the sarcoplasmic reticulum of muscle cells and endoplasmic reticulum of nonmuscle cells. RyRs regulate the release of intracellular calcium stores critical for muscle contraction. Rynaxypyr is a highly potent and selective activator of insect ryanodine receptors with exceptional activity on a broad range of Lepidoptera. Rynaxypyr is characterized by its high levels of insecticidal activity and low toxicity to mammals attributed to a high selectivity for insect over mammalian ryanodine receptors. Analogues of Rynaxypyr having fluorinated fragment have exhibited high activity. These compounds were synthesized by alkylation of the corresponding pyrazolone with trifluoroethyl iodide (Figure 16).⁶²⁶

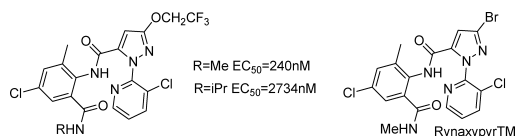


Figure 16.

A simple and flexible pathway to a novel family of fluorinated liquid crystals **552** was developed using 1,2-dibromotetrafluoroethane. Three effective strategies for assembling of molecular core of compounds **552** have been developed (Figure 17). All strategies include three synthetic steps, which are Suzuki coupling (**B**), substitution of bromine in $\text{CF}_2\text{CF}_2\text{Br}$ moiety (**C**), and catalytic olefination reaction (**COR**). Final “tuning” of desired molecule can be done either by choice of appropriate nucleophile (step **C**, strategy 1) or the change of type of linkage by using dehydrofluorination (**D**) at the very last step after the core of **552** is assembled. Various fluorinated $\text{CF}_2\text{CF}_2\text{S}$ -bridged alkanes and CF_2S -, CF_2O -bridged alkenes and alkynes are available by this method. A significant part of the prepared compounds demonstrated attractive LC properties. CF_2O -bridged alkenes revealed the widest range of mesophase existence among the prepared compounds. Simplicity of reaction procedures, high universality, and perspective liquid crystalline properties of compound obtained open broad

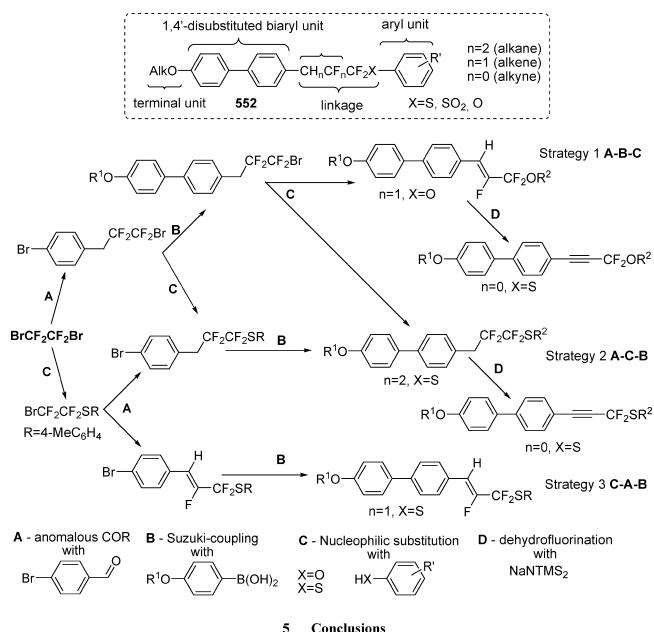


Figure 17.

opportunities in the design and synthesis of new liquid crystals using the method elaborated (Figure 17).⁵⁰²

5. CONCLUSIONS

Fluorine is an element having a really unique position in modern chemistry. First, this is due to fluorinated derivatives having occupied widespread application in various branches of our life from chemistry of materials to drug design. Fluorinated ethanes are really cheap compounds prepared easily from industrially available compounds. It makes these building blocks very attractive for subsequent use in the synthesis of more complex molecules bearing fluorinated C-2 fragment. Recently, Bräse compared prices of common reagents for trifluoromethylation. These data indicated clearly the importance of the principle of “reasonable price”, because the cost of such reagents varies in a huge range. For example, the price per mole for Togni II reagent is 5546 times higher as compared to trifluoroacetic acid.⁶²⁷ In these terms, PFEs are really very competitive reagents for the incorporation of fluorinated fragment into a molecule. Some data (price per one fluorine mole) are given in Figure 18. One can see that the cost of PFEs

KF	2.3 Euro/F mole	$\text{BrCF}_2\text{CF}_2\text{Br}$	9 Euro/F mole
CsF	175 Euro/F mole	$\text{CF}_3\text{CF}_2\text{H}$	14.6 Euro/F mole
CF_3COOH	8.7 Euro/F mole	$\text{ClCF}_2\text{CFCl}_2$	22.3 Euro/F mole
CF_3I	377 Euro/F mole	$\text{CF}_3\text{CF}_2\text{I}$	31 Euro/F mole
	48070 Euro/F mole	$\text{CF}_3\text{CH}_2\text{Br}$	100 Euro/F mole
		$\text{CF}_3\text{CH}_2\text{I}$	189 Euro/F mole

Figure 18.

is comparable with the price of trifluoroacetic acid. Even some very popular inorganic fluorides (cesium fluoride) are more expensive than freons, taking into account incorporation of one fluorine into a molecule.

Looking through a variety of reactions overviewed, one can definitely conclude that PFE have a wide synthetic scope. PFE approved to be very useful building blocks provided plentiful

pathways to fluorinated compounds of different classes. Thus, fluorinated derivatives of alkanes, alkenes, ethers, sulfides, carboxylic acids, sulfinic and sulfonic acids, alcohols and ketones, and aromatic and heteroaromatic compounds were efficiently prepared using cheap and available starting materials. The broad spectrum of PFE reactivity, such as the possibility to participate in both radical and ionic processes, allows one to expect a constant interest of academic and industrial scientists to these readily available and cheap compounds, resulting in sustainable flow of new synthetic pathways to fluorinated molecules elaborated.

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Notes

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